

Measuring discrimination using natural experiments *

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Abstract

Disparities in high-stakes decisions are common, but difficult to interpret as discrimination if unobservable group differences exist. We show how to use natural experiments and a binary instrumental variable (IV) strategy to measure discrimination, adjusted for group differences in unobserved potential outcomes. Our approach does not require random assignment to decision-makers, a prerequisite for existing techniques. We study discrimination in two settings using a different type of empirical strategy in each, and highlight how key assumptions vary by strategy. First, we use a regression discontinuity design to study socio-economic discrimination in the decision to promote Michigan public school students to the next grade. Conditional on unobserved academic ability, economically disadvantaged students at a test score cut-off were promoted at lower rates than non-disadvantaged students. Second, we measure racial discrimination in misdemeanor prosecution using a budget cut that reduced prosecution rates in King County, Washington. Here we use a difference-in-difference (DiD) design and address the challenges in mapping DiD to IV. Before the cut, we find no evidence of discrimination in prosecution conditional on unobserved potential recidivism. After the cut, white defendants became more likely to be prosecuted than minority defendants with similar unobserved potential recidivism. Examining how these prosecution gaps vary by a proxy of case quality suggests that prosecutors may have responded to the budget cut by focusing on high quality cases and offsetting racial disparities in arrests.

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1 Introduction

Disparities between social groups are common in many contexts, from educational institutions to the criminal legal system. Understanding whether such disparities are due to discrimination is usually challenging due to unobservable factors relevant to the decisions at hand (Aigner and Cain, 1977; Charles and Guryan, 2011).¹ The ideal measure of discrimination is often one that compares individuals from two different groups who are similar, even in terms of unobservable characteristics. For example, to assess hiring discrimination, we might want to measure the hiring gap between people from two different groups who would be equally productive if hired. Since productivity is only observed for hired workers, measuring such a gap in practice is difficult. An extensive literature has focused on overcoming challenges like this by operationalizing the Becker (1957) ‘outcome’ test to distinguish taste-based discrimination (or animus) from accurate statistical discrimination (Phelps, 1972; Arrow, 1973), using theoretical restrictions and quasi-experimental tools (Knowles, Persico, and Todd, 2001; Anwar and Fang, 2006; Arnold, Dobbie, and Yang, 2018; Canay, Mogstad, and Mountjoy, 2024; Bharadwaj, Deb, and Renou, 2024). While these tests detect whether animus exists at the margin of a decision, they do not quantify statistical discrimination (Hull, 2021). They also cannot quantify the total magnitude of discrimination away from the margin caused by animus, statistical discrimination, and incorrect beliefs (Bordalo et al., 2016).

Recent work addresses these concerns by using random assignment to decision-makers to extrapolate unobserved outcomes (e.g., productivity of workers not hired) and directly adjusts for these unobservables when measuring discrimination (Arnold, Dobbie, and Hull, 2022). This approach does not provide solutions for contexts where quantifying discrimination is policy-relevant but individuals are not randomly assigned to decision-makers. For example, measuring socio-economic discrimination in the decision to move students from one grade to another would be difficult since students are rarely randomly assigned to teachers and unobserved ability may vary by socio-economic status (Paufler and Amrein-Beardsley, 2014).

In this paper, we broaden the set of contexts where researchers can measure discrimination between groups while adjusting for key unobserved differences. We show how to use natural experiments that yield a binary instrumental variable (IV) framework (Angrist, Imbens, and Rubin, 1996) to measure average discrimination in a treatment decision. We map commonly used quasi-experimental techniques, e.g., regression discontinuity (RD) or difference-in-difference (DiD), to the binary IV framework and use assumptions on selection behavior to recover the distribution of unobserved outcomes for each group. We directly condition on these unobservables when measuring discrimination. Like prior work using random assignment to decision-makers, discrimination estimates from our approach capture animus, statistical discrimination, and incorrect beliefs.

We highlight different aspects of our approach by implementing it to study discrimination in two settings where individuals are not randomly assigned to decision-makers. In one application, we use the local variation generated by an RD design to study socio-economic discrimination in the

¹We are agnostic on the source of such differences. These might be due to multiple factors, including preferences or discrimination prior to the decision of interest, i.e. ‘systemic discrimination’ (Bohren, Hull, and Imas, 2022).

decision to move school students to the next grade (“grade promotion”). In a second application, we study racial discrimination in prosecution using a DiD design and address the challenges inherent in mapping DiD to IV. These analyses emphasize the broad applicability of our approach, especially given the prevalence of DiD-type variation. In what follows, we outline the details of our approach with grade promotion as the guiding example.

Consider a potential outcomes framing of the decision to promote students from one grade to another (Rubin, 1974). Define treatment as $D_i = 1$ if a student is ‘promoted’ to the next grade, while $D_i = 0$ for students ‘retained’ in the same grade. Let the treated potential outcome, $Y_i(1)$, be a measure of a student’s academic success if promoted. We define discrimination by socio-economic status (SES) as the SES gap in promotion rates between students who would have the same academic outcome if promoted, $Y_i(1)$.^{2,3} If the distribution of $Y_i(1)$ varies by SES, then the raw disparity in promotion rates may be due to differences in $Y_i(1)$, and will not align with our definition of discrimination. Let $Y_i(1)$ be an indicator for whether a student would succeed if promoted.⁴ Since $Y_i(1)$ is binary, the distribution of success if promoted is the share of individuals who would succeed if everyone were promoted—the “average promoted outcome”.

To highlight the identification approach, assume for now that we can observe the average promoted outcomes for each group. If the average promoted outcomes differ by SES, we quantify discrimination conditional on the promoted outcome in three steps.⁵ First, we rescale observed SES-specific promotion rates using the ratio between the observed outcome and the average promoted outcome for a given SES group, e.g., the ratio between promoted students’ success rate and the overall SES success rate. This rescaling yields SES-specific promotion rates that are conditional on the promoted outcome (Arnold, Dobbie, and Hull, 2022). Second, we use the resulting promotion rates to construct SES gaps in promotion for each value of the promoted outcome. That is, we construct SES gaps in promotion for people who: a) would succeed if promoted ($Y_i(1) = 1$) and b) would not succeed if promoted ($Y_i(1) = 0$). Third, the weighted average of these two SES gaps yields average discrimination in promotion conditional on the promoted outcome.

Since we typically cannot observe the average promoted outcome for each SES group, we estimate it using an IV framework. Our approach needs a natural experiment that shifts treatment rates – promotion rates in this example – and takes the form of a binary IV. First, we use the binary IV to estimate how potential outcomes (e.g., success in the next grade) vary across students who would always be promoted (‘always takers’) and marginal students promoted due to the IV (‘compliers’) within each SES group (Imbens and Rubin, 1997). Since we do not observe how well the

²Recent work critiques definitions of discrimination that condition on potential outcomes and suggest alternative definitions that may be more or less appropriate depending on the context (Grossman, Nyarko, and Goel, 2024; Bushway et al., 2025). While the focus of this paper is expanding where researchers can measure discrimination conditional on potential outcomes, we discuss these critiques in Section 2.

³We focus on the treated potential outcome in this example since it aligns with the motives of relevant decision-makers in the grade promotion empirical application. We also discuss comparing individuals with the same untreated potential outcome or treatment effect, which may be appropriate in other contexts.

⁴We discuss modifications for multi-valued potential outcomes in Section 3.

⁵These adjustments will be less consequential if group differences are minimal (Goncalves and Mello, 2021; Harrington and Shaffer, 2023; Tuttle, 2023).

retained students (‘never takers’) do, we extrapolate their outcomes using behavioral assumptions from the marginal treatment effects literature (Brinch, Mogstad, and Wiswall, 2017; Mogstad, Santos, and Torgovitsky, 2018; Kowalski, 2023a). Such extrapolations recover the unobserved average promoted outcomes for each SES group, which we use to measure average discrimination conditional on the promoted outcome. Depending on the strength of the extrapolation assumptions, our approach yields either bounds or point estimates of discrimination.

In our first empirical application, we apply this approach to bound SES gaps in the decision to promote 3rd graders to 4th grade (D_i), using administrative records from Michigan public schools. Our measure of the academic outcome if promoted to 4th grade ($Y_i(1)$) is whether students meet state guidelines for being at least partially-proficient in English and Math standardized tests if promoted, although we explore robustness to other definitions. We refer to partial proficiency as “succeeding” in 4th grade, for brevity.

We document SES differences in underlying “success” in 4th grade using an RD design generated by Michigan’s “Read by Grade 3” law. Due to this law, the probability of being promoted discontinuously increased at a cut-off around the 5th percentile of the 3rd grade standardized test score distribution (Westall et al., 2022a,b; Berne et al., 2023). Our bounds imply that 5.7–6.2% of low SES students at the test score cut-off would “succeed” if promoted while 14.0–14.6% of high SES students at the cut-off would do so. These differences imply that discrimination estimates not accounting for potential 4th grade “success”, e.g., covariate-adjusted promotion gaps, will be biased.

We find evidence of discrimination in student grade promotion, even after adjusting for SES differences in underlying 4th grade “success” rates. Our bounds imply that high SES students at the cut-off are 3.3–3.8 p.p. (3.5–4.1%) more likely to be promoted to the 4th grade than similar low SES students; this is driven by students who would not “succeed” in 4th grade. Our bounds imply gaps that are much higher than the covariate-adjusted disparity of 2.8 p.p., which is outside our estimated bounds. While our estimates reflect underlying “success” rates and discrimination among students at the test score cut-off (Angrist and Rokkanen, 2015; Cattaneo et al., 2021), we also discuss the assumptions we would need to measure discrimination away from the test score cut-off. Our empirical tests suggest they are not satisfied in this context, and so we focus on students at the cut-off.

We also implement our IV-based approach with DiD variation to study discrimination in misdemeanor prosecution. We use administrative court records from Washington State to bound racial gaps in prosecution (D_i) among individuals who are equally likely to commit a new offence if prosecuted ($Y_i(1)$).⁶ Analogous to the promotion setting, the first step involves using a natural experiment that shifts prosecution rates and re-offence outcomes to measure the average outcome that would be realized if everyone were prosecuted. We estimate bounds for the race-specific average re-offence outcome if everyone were prosecuted using a budget cut that impacted the King

⁶We also present robustness to conditioning on re-offending if dismissed and changing empirical definitions of re-offence.

County Prosecutors’ Office (serving the Seattle metropolitan area) but did not affect other counties in Washington. We use a DiD strategy comparing King County to adjacent counties to see if the budget cut generated the shifts we need for our approach—indeed, the cut reduced both the prosecution rate and one year recidivism. Reassuringly, we find no evidence that the budget cut affected other key aspects of the criminal legal system or economy.

Mapping the DiD and the IV-based approach to bound average outcomes if prosecuted poses challenges that do not arise in the RD case. With a DiD, time variation in treatment and potential outcomes makes it difficult to disentangle the impacts of the budget cut from the effects of time. We overcome this challenge by making two additional assumptions. First, we assume that time does not shift prosecution status, which ensures that always takers, compliers, and never takers are defined as they would be with a binary IV. Second, we make a parallel trends assumption directly on the potential outcome if prosecuted. This second assumption lets us adjust for the time trends in potential outcomes for always takers, compliers, and never takers. We empirically probe these assumptions and do not find evidence of violations in this setting.

These additional assumptions let us map a traditional 2×2 DiD design to the binary IV method described above, use it to estimate average prosecuted outcomes by race, and estimate discrimination conditional on prosecuted outcomes. We discuss extensions to bounding the average untreated outcome (outcome if dismissed in this application) and to DiDs with staggered treatment adoption.

Using the shifts in prosecution rates and re-offence outcomes generated by the budget cut and the DiD-IV approach, we find racial differences in the unobserved prosecuted outcome. Before the budget cut, 24.5–29.3% of white defendants would commit a new offence if prosecuted, while the same is true for 32.3–37.1% of minority defendants. These estimates imply that discrimination estimates that do not account for differences in unobserved prosecuted outcomes, e.g., covariate-adjusted prosecution gaps, will be biased.

We use the estimates of the average outcomes if everyone were prosecuted to bound the racial gap in prosecution before and after the budget cut, conditional on racial differences in the prosecuted outcome. We cannot reject the null of no racial gap in prosecution rates before the budget cut. This changes after the cut—even though overall prosecution rates fall, white defendants became 1.3–4 p.p. (1.8–5.6%) more likely than minority defendants to be prosecuted. The gap implied by our bounds is lower than and excludes the covariate-adjusted disparity of 4.4.

Next, we investigate potential reasons for the relatively higher white prosecution rate after the budget cut. One explanation might be that: i) resource constraints after the budget cut made prosecutors less likely to pursue low quality cases, and ii) minority cases were supported by weaker evidence due to discrimination in pre-prosecution decisions, e.g., policing (Goncalves and Mello, 2021; Owens and Ba, 2021). If this is the case, we should expect more positive white–minority gaps after the budget cut among a subset of cases that are likely low quality. We define low quality cases as those filed for offences that were relatively less likely to be successfully sentenced before the budget cut. We find that the overall post-cut racial gap is driven by these likely low quality cases. This pattern is suggestive evidence that prosecutors shifted their focus to high quality cases

after the budget cut, offsetting disparities generated in prior stages of the criminal legal system.

This paper makes several methodological and empirical contributions. First, we expand where researchers can measure discrimination conditional on unobserved factors. While existing approaches require random assignment to decision-makers, our approach requires a binary IV, and insights from the marginal treatment effects literature. Our approach is also useful in contexts where researchers do not observe decision-makers in the data, regardless of whether random assignment exists. As a final methodological point, we add to the literature mapping DiD to IV (De Chaisemartin and D’Haultfœuille, 2018) by showing how to use DiD variation to estimate average potential outcomes.

Second, while there is descriptive evidence documenting racial, ethnic, and class disparities in student promotion decisions (Moller et al., 2006; Locke and Sparks, 2019), our analysis in Michigan public schools provides the first evidence of SES discrimination adjusting for unobservable differences. Our results also show that the promotion disparities documented in recent work on the “Read by Grade 3” law (Westall et al., 2022b; Westall, Utter, and Strunk, 2023) are not solely due to SES differences in unobservables.

Third, we add to the understanding of criminal prosecution in multiple ways. There is considerable evidence of disparities by race, ethnicity, and gender in many prosecutorial decisions (Spohn, Gruhl, and Welch, 1987; Davis, 2014; Kutateladze and Andiloro, 2014; Rehavi and Starr, 2014; Tuttle, 2023). Much of this evidence is either descriptive or from contexts where unobservable differences between the comparison groups are minimal. Our analysis in Washington represents the first evidence of racial discrimination in misdemeanor prosecution that documents and directly accounts for unobservable racial differences. Our results also speak to the importance of considering the criminal legal system as a multi-stage system (Harrington and Shaffer, 2024). Consistent with recent work on prosecutorial discretion (Harrington and Shaffer, 2023; Jordan, 2024), we find suggestive evidence of prosecutors offsetting disparities from earlier stages of the criminal legal system. Finally, in validating the budget cut that we use to measure discrimination, we add to the growing evidence showing that non-prosecution for minor offences can reduce future involvement with the criminal legal system (Mueller-Smith and Schnepel, 2021; Agan, Doleac, and Harvey, 2023).

2 Discrimination estimands of interest

We begin with a general potential outcomes framework (Imbens and Angrist, 1994). Individuals are chosen for a binary treatment, D_i , as a function of the unobservable potential outcomes. Individuals realize the potential outcome associated with chosen treatment state, $Y_i(D_i)$. Each potential outcome is only observed if the associated treatment state is realized. That is, the treated potential outcome, $Y_i(D_i = 1)$, is only observed for individuals who are treated. Similarly, the untreated potential outcome, $Y_i(D_i = 0)$, is only observed for individuals who are not treated. These potential outcomes may be continuous, discrete or binary. Individuals belong to one of two groups, denoted

by $R_i \in \{r_1, r_2\}$ and the distribution of potential outcomes may differ across groups.⁷

Our goal is to quantify discrimination, defined as group differences in treatment rates for individuals with the same potential outcome. We begin by estimating differential treatment between individuals who would realize the same outcome **if they were treated**, formally described in Definition 1.⁸ In Section 3, we also discuss conditioning on the outcome if not treated and the treatment effect. The most appropriate factor to condition on will depend on the empirical, theoretical, and possibly normative features of the particular context. The notion of discrimination conditional on potential outcomes maps to classic notions of fairness and discrimination in economics that focus on whether people of two different groups are being treated differently, despite being identical in some objective but latent quality. For example, this notion would consider racial gaps in hiring rates between people who would be equally productive if they were hired as discrimination (Aigner and Cain, 1977). This definition also encompasses multiple sources of discrimination (statistical discrimination, animus, and biased beliefs) and is consistent with the legal interpretation of ‘disparate impact’ (Becker, 1957; Phelps, 1972; Arrow, 1973; Bordalo et al., 2016; Bohren et al., 2019; Arnold, Dobbie, and Hull, 2022).^{9,10}

Definition 1. Differential treatment conditional on treated potential outcome $Y_i(1)$

$$E[D_i|R_i = r_1, Y_i(1) = y] - E[D_i|R_i = r_2, Y_i(1) = y]$$

The discrimination estimand in Definition 1 is generally difficult to estimate empirically. Since treated outcomes are only observed among treated individuals, it is not feasible to directly condition on $Y_i(1)$. One approach to estimate discrimination might be to compute the raw gap in treatment rates by group, $E[D_i|R_i = r_1] - E[D_i|R_i = r_2]$. However, this will differ from the quantity in Definition 1 if treatment decisions are a function of $Y_i(1)$, and if the distribution of $Y_i(1)$ varies by group.

An alternative approach might be to compute the treatment gap conditional on a set of observed covariates. This ‘selection-on-observables’ approach alters the interpretation of the discrimination

⁷This framework is a general version of the framework described in Arnold, Dobbie, and Hull (2022). In their setting of bail reform, individuals vary in a latent unobservable Y^* , which is only observed among treated (released) individuals, while no outcome is observed for untreated (detained) individuals. Similar to Canay, Mogstad, and Mountjoy (2024), we consider a framework that applies to settings where untreated outcomes are also selectively observed among those not treated.

⁸This approach requires the researcher to specify the potential outcomes to condition on, $Y_i(D_i)$. It is possible that the underlying decision-makers also value other factors that are not captured by the chosen $Y_i(D_i)$ (Kleinberg et al., 2018). If the relationship between such omitted factors and our chosen $Y_i(D_i)$ varies by group, this definition would not quantify differential treatment conditional on all unobservable factors. However, such gaps can still be interpreted as unwarranted disparities if conditioning on our chosen $Y_i(D_i)$ maps to a well-defined notion of fairness.

⁹This definition does not require individuals to be identical in terms of all non-race characteristics, as in Canay, Mogstad, and Mountjoy (2024). We think of differences in observed covariates, even conditional on potential outcomes, as possible drivers of discrimination, and investigate them in our empirical applications.

¹⁰Using the pretrial detention context as a motivating example, Grossman, Nyarko, and Goel (2024) and Bushway et al. (2025) argue that comparisons like Definition 1 are unsuitable for measuring discrimination, preferring definitions that compare individuals with similar predicted misconduct. While such definitions may be attractive from a legal standpoint in certain contexts, they may not always compare unobservably similar individuals, especially if prediction quality varies by group (e.g., race).

test. Instead of measuring the extent of differential treatment between people who have the same potential outcome but different group membership, it moves towards a narrower test measuring how much two individuals with identical **observables** are treated differently **because** of their group identity, generating ‘included variables bias’ (Ayres, 2010). In general, even if such bias is small, controlling for covariates will typically not recover the measure of differential treatment, conditional on potential outcome, in Definition 1 unless the covariate used is perfectly correlated with $Y_i(1)$.

In practice, groups often differ in terms of unobserved potential outcomes in many settings where measuring discrimination is of interest. For example, relevant to our application studying racial discrimination in misdemeanor prosecution, discrimination in policing could generate cross-race differences in the underlying potential outcome distributions. In the context of our application studying socio-economic discrimination in students’ grade promotion decisions, differential access to educational inputs by socio-economic status might generate group differences in skills (and thus potential outcomes). The estimand described in Definition 1 measures discrimination holding such upstream differences fixed. This measures the magnitude of unwarranted disparities arising in the decision process of interest.

We next demonstrate how to use a natural experiment to identify the estimand in Definition 1. Consider a quasi-random intervention that generates a binary instrument, Z . Assume Z satisfies the usual instrumental variables (IV) assumptions of relevance, independence, exclusion and monotonicity. In one of our empirical applications, we use a difference-in-difference (DiD) approach to isolate quasi-random shifts in treatment. Time trends in potential outcomes and treatment status that are inherent in a DiD approach can act as confounders and make it difficult to use DiD variation in an IV set up. For ease of exposition, we abstract away from time trends for now and let Z be a binary instrument, where $Z \in \{0, 1\}$ denotes periods before and after an intervention. We first discuss identification and measurement of the discrimination estimand in Definition 1 using this binary instrument. We then address the adjustments required for the identification using a DiD approach in Section 3.

Definition 2 describes a time period-specific version of Definition 1: Δ_{zy} is a group treatment gap that is specific to a given period and value of the treated outcome. Each such gap is composed of treatment rates that are conditional on time period, group and potential outcome (π_{zry}).

Definition 2. Differential treatment within a given period, conditional on $Y_i(1)$

$$\begin{aligned}\Delta_{zy} &= (E[D_i|Z = z, R_i = r_1, Y_i(1) = y] - E[D_i|Z = z, R_i = r_2, Y_i(1) = y]) \\ &= (\pi_{zr_1y} - \pi_{zr_2y}) \\ \Delta_z &= \sum_{y \in \text{supp}(Y_i(1))} Pr(Y_i(1) = y) \Delta_{zy}\end{aligned}\tag{1}$$

Our objects of interest are the period-specific estimates of discrimination that are conditional on having the same outcome if treated (Δ_z). These are averages of the period- and treated outcome-specific gaps, weighted by the population prevalence of each value of the treated potential outcome ($Pr(Y_i(1) = y)$). These can also be used to measure changes in discrimination **due to the inter-**

vention, $\Delta_{z=1} - \Delta_{z=0}$, the difference in discrimination before versus after some intervention.¹¹

To understand how to estimate our main object of interest, note that the building blocks of Δ_z are treatment rates that are conditional on period, group, and treated outcome (π_{zry}). Since treated outcomes are not always observed, we cannot directly condition on it to compute each π_{zry} . However, following Arnold, Dobbie, and Hull (2022), we re-write π_{zry} in Equation 2 using: 1) the definition of conditional expectations, and 2) the IV assumptions. The second line follows from the definition of conditional expectations, while the third line follows from the fact that $Y_i(1) \perp Z$.

$$\begin{aligned}\pi_{zry} &\equiv E[D_i|Z = z, R_i = r, Y_i(1) = y] \\ &= \frac{E[Y_i(1) = y|Z = z, R_i = r, D_i = 1] \times E[D_i|Z = z, R_i = r]}{E[Y_i(1) = y|Z = z, R_i = r]} \\ &= \frac{E[Y_i(1) = y|Z = z, R_i = r, D_i = 1] \times E[D_i|Z = z, R_i = r]}{E[Y_i(1) = y|R_i = r]}\end{aligned}\tag{2}$$

Equation 2 highlights how to quantify the period-, group-, and potential-outcome specific treatment rates used to estimate discrimination conditional on having the same outcome if treated (Δ_z). We need the following objects:

1. $E[D_i|Z = z, R_i = r]$:
Share of individuals of each group r treated in each period z
2. $E[Y_i(1) = y|Z = z, R_i = r, D_i = 1]$:
Share of treated individuals with treated outcome y , for each group r and period z
3. $E[Y_i(1) = y|R_i = r]$:
Prevalence of treated potential outcome y in each group's population

Each period-, group-, and potential-outcome specific treatment rate π_{zry} is a function of two moments directly observable in data, and one typically unobserved moment. Objects 1 and 2 are observed in data—we see the share of individuals of each group who are treated, and the outcomes realized for treated individuals. Object 3 represents the underlying share of individuals of group r who would experience a given value of the treated potential outcome. This object would only be observed in a counterfactual where everyone of that group was treated. This share is an especially crucial element because it also provides a test of whether the distribution of potential outcomes varies by group. If the distributions are the same across race, we could interpret the raw observed group differences in treatment as discrimination. We also use $E[Y_i(1) = y|R_i = r]$, along with each group's shares of the population (p_r), to compute the underlying prevalence of treated potential outcome y in population: $E[Y_i(1) = y] = p_{r_1}E[Y_i(1) = y|R_i = r_1] + p_{r_2}E[Y_i(1) = y|R_i = r_2]$.

¹¹Even though Z represents quasi-experimental variation, the cross-group gap in the impact of Z on D will not generally recover $\Delta_{z=1} - \Delta_{z=0}$ unless 1) potential outcomes are similar across groups or 2) the impact of Z on D is uncorrelated with potential outcomes (see Appendix C.1). Such cross-group comparisons can also suffer from the pitfalls of conducting marginal outcome tests with discrete instruments (Canay, Mogstad, and Mountjoy, 2024).

We use this to aggregate each π_{zry} to estimate the discrimination conditional on having the same outcome if treated in a given period (Δ_z) in Definition 2.

However, $E[Y_i(1) = y | R_i = r]$ is not typically directly observable in the data, unless there are unique institutional features such as random assignment to supremely lenient decision-makers who treat (almost) everyone (Arnold, Dobbie, and Hull, 2022; Baron et al., 2023; Reeves, 2023). There are many settings where measuring discrimination is of interest, but individuals are not randomly assigned to decision-makers. For instance, criminal misdemeanor defendants in King County are not randomly assigned to prosecutors, and students in Michigan public schools are not randomly assigned to teachers. Measuring discrimination in prosecution or students’ promotion decisions in these contexts is thus difficult if the distribution of potential outcomes differ by group.

Next, we incorporate insights from the IV and marginal treatment effects literatures to estimate $E[Y_i(1) = y | R_i = r]$, which we need to understand if the groups to be compared differ in terms of the potential outcomes if treated. If so, we adjust for such differences following Equation 2, and then measure discrimination that accounts for differences in potential outcomes.

3 Estimating discrimination using a natural experiment

In this section we discuss using a binary instrumental variable (IV) to understand if the distribution of potential outcomes differs by social group using insights from the IV and marginal treatment effects literatures. First, we describe estimating the share of each group that would experience a given value of the **treated** potential outcome if everyone of that group were treated. This is a key input to estimating discrimination that is conditional on treated potential outcomes. We then address complications that arise when using difference-in-difference (DiD) variation for this purpose. Finally, we discuss conditions required to account for group differences in **untreated** potential outcomes or **treatment effects** instead, which may be appropriate in contexts that have treated and untreated potential outcomes that are meaningful. We use the misdemeanor prosecution setting as the guiding example in this section since key elements of the prosecution decision process and our empirical application let us highlight the generality of the approach.

3.1 Implementation with a binary instrument

We describe the framework in the context of racial discrimination in misdemeanor prosecution. Individuals belong to either ‘White’ or ‘Minority’ groups, denoted by $R_i \in \{w, m\}$, and the treatment decision is prosecution. This definition of racial groups, rather than a White–Black comparison, is motivated by the context of Washington State, which we describe in detail in Section 5. We start with a discussion of estimating racial differences in prosecution rates for individuals who would have the same outcome if prosecuted. Later, we discuss conditioning on the outcome if dismissed (i.e., the untreated potential outcome), or the treatment effect of prosecution.

Individuals, indexed by i , are chosen for treatment, D_i . In the context of prosecution, this decision is made by a bundle of multiple agents who influence case outcomes, including prosecuting

attorneys and judges, rather than by the individual i . Let $D_i = 1$ if an individual’s case is prosecuted, and $D_i = 0$ if an individual’s case is dismissed. Let the potential outcomes be binary indicators for whether an individual commits a new offence in the future, after prosecution or dismissal, i.e., $Y_i(D_i) \in \{0, 1\}$. While we use binary potential outcomes in this section for simplicity, the expressions in Definition 2 and Equation 2 accommodate multi-valued potential outcomes.

Finally, let Z be a binary instrument that shifts the rate of prosecution. Let Z represent periods before and after an unanticipated budget cut that sharply **reduced** prosecution rates. As mentioned in the previous section, our prosecution application isolates the quasi-experimental variation using a difference-in-difference (DiD) strategy in which one county adopts a reform and the others do not. The time trends inherent in this approach make directly using an IV challenging. For ease of exposition, we first abstract away from time trends and discuss how a binary IV identifies the required moments. We then make adjustments for the DiD in the following subsection.

Recall that we need to estimate each period-, group-, and potential outcome-specific treatment rate (π_{zry}) in Equation 2 to quantify the discrimination estimands in Definition 2. The key challenge is that the denominator, $E[Y_i(1) = y | R_i = r]$, is unobserved. Overcoming this challenge requires estimating the proportion of each group r that would realize treated outcome $Y_i(1) = y$ if everyone in that group were treated. Since $Y_i(1)$ is binary, $E[Y_i(1) = 1 | R_i = r]$ is the average outcome that people in group r would realize if everyone in that group were treated. We next show how to use the binary instrument Z to estimate bounds and point estimates of $E[Y_i(1) = 1 | R_i = r]$.

Under the IV assumptions listed in Section 2, the variation from the binary instrument Z partitions the population into three “compliance groups” (always takers, compliers, and never takers) and identifies their associated proportions and certain average potential outcomes (Angrist, Imbens, and Rubin, 1996). Assuming that Z shifts treatment for both racial groups, these quantities are identified separately by race group. For each race, we directly observe the proportions of always takers (p_A) and compliers (p_C) in the data by examining the share of the population that would receive treatment regardless of the reform and would only receive treatment because of the reform, respectively. In the application that we use as the example here, the treatment is prosecution, and the reform decreases prosecution rates. Hence, p_A is the share of people who are prosecuted after the budget cut, and p_C is the change in prosecution rates due to the budget cut. Since always takers, compliers and never takers partition the population, the share of never takers is $p_N = 1 - p_A - p_C$.

The variation from the binary IV also provides estimates of average potential outcomes for some of these groups. Since our initial focus is on treated potential outcomes, we concentrate on that potential outcome here. In the prosecution application, the average treated outcome for always takers is the average outcome of people treated (i.e., prosecuted) after the budget cut. Note that the group of people treated before the budget cut consist only of compliers and always takers. Hence, the average outcome of people treated before the reform is a weighted average of treated outcomes for compliers and always takers. Using these two averages, along with the population shares of always takers and compliers, we estimate the average treated outcomes of compliers (Imbens and Rubin, 1997). This recovers average treated outcomes for two of the three

“compliance groups” that partition the population: always takers and compliers. However, we do not observe the average treated outcomes of never takers, since they are never treated. This is the final piece to estimate the average outcome that would be realized if everyone of each racial group were treated, $E[Y_i(1) = 1|R_i = r]$.¹²

We estimate bounds (or point estimates) for the average treated outcomes of never takers by placing restrictions on the relationship between treatment propensity and average treated outcomes. Each “compliance group” is defined by its propensity to be treated: always takers are more likely to be treated than compliers, who are in turn more likely to be treated than never takers. [Figure 1](#) depicts a hypothetical example where always takers and compliers are roughly 70% and 20% of the population respectively. In this example, compliers have greater treated outcomes than always takers. In the context of prosecution, this might be the case if prosecutors were more likely to prosecute individuals who were unlikely to commit a new offence if prosecuted. We use this estimated relationship to infer the treated outcomes of never takers.

In Panel a) we assume that **average** treated outcomes are weakly monotonic in the treatment propensity of “compliance groups”, following Mogstad, Santos, and Torgovitsky (2018) and Kowalski (2023a). This assumption extends the relationship between always takers’ and compliers’ average treated outcomes to never takers’ average treated outcomes. In this example, the assumption implies that the average treated outcomes for never takers must be weakly greater than the average treated outcomes for compliers, which pins down the lower bound for never takers’ average outcomes.¹³ Since $Y_i(1) \in \{0, 1\}$, the average treated outcomes for never takers is bounded above by 1. Combining the bounds on the treated outcomes for never takers with the point estimates for the average treated outcomes for compliers and always takers yields bounds on the average treated outcomes for each racial group, $E[Y_i(1) = 1|R_i = r]$.¹⁴

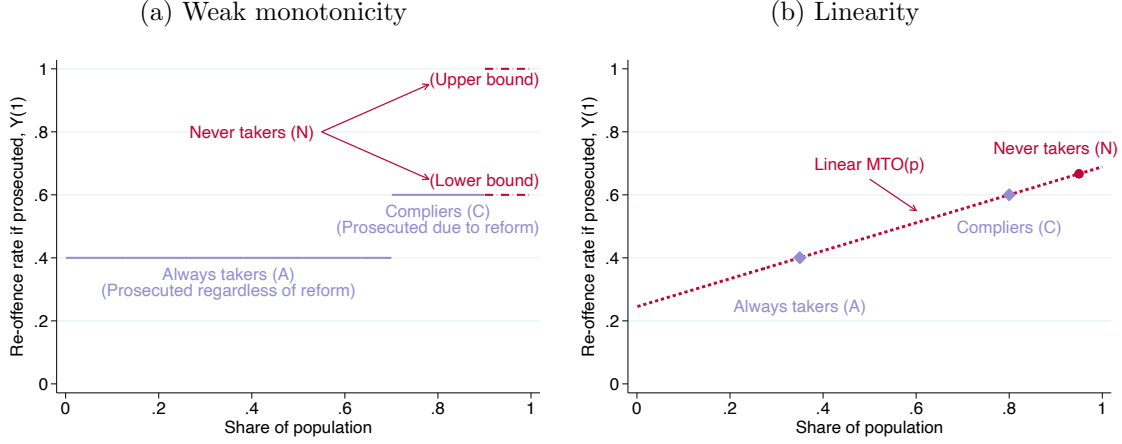
Note that the restrictions that we place to bound average treated outcomes are restrictions on the natural experiment and do not require assuming the underlying decision-makers focus on a single narrow objective. Rather, this approach allows decision-makers to adopt a wide range of complex and multi-dimensional models as long as their decisions generate patterns that imply that average potential outcomes for each of the compliance groups is weakly monotonic in their likelihood of being treated. In the context of prosecution, the extrapolation in Panel a) of [Figure 1](#) assumes that never takers, who are least likely to be prosecuted, are at least as likely as compliers to re-offend if prosecuted. This assumption might be violated if other inputs into prosecution decisions generate contradictory patterns. For example, say that prosecutors are also less likely to

¹²When both a natural experiment and random assignment to decision-makers are present, the approach that identifies outcomes using information for a larger proportion of treated (or untreated, if conditioning on the untreated potential outcome) individuals will require less extrapolation and involve less extrapolation error.

¹³This restriction is mathematically similar to the “Performance Bound” in Jordan (2024), who studies racial discrimination in felony review in a context where prosecutors are randomly assigned cases. However, those bounds arise from modelling the underlying objectives of prosecutors, while ours place restrictions on the policy reform.

¹⁴If potential outcomes are multi-valued, we can still use this approach. In such a case, we require an estimate of the prevalence of each possible value of the potential outcome, $E[Y_i(1) = y|R_i = r]$, to identify the discrimination estimand in Definition 2. The logic underlying the bounding approach will still hold, since each expectation $E[Y_i(1) = y|R_i = r]$ represents a proportion and is hence also bounded between $[0, 1]$.

Figure 1: Identifying the average treated outcome with a binary IV



Note: This figure uses simulated data. Lower values of the x-axis denote individuals who are more likely to be treated. $Y(1)$ denotes the treated potential outcome. The diamonds and dots in Panel b) reflect outcomes of the median individual in that group. Solid lines and diamonds represent moments observed in the data, and dashed lines and circles represent objects that are extrapolated.

pursue cases with poor quality evidence, such that all the never takers' cases are poor quality. If individuals whose cases are poor quality never re-offend if prosecuted, this could violate our weak monotonicity assumption. In such situations, an alternative approach is to bound never takers' treated outcomes between 0 and 1—the widest logically possible bounds (Manski, 1989).

We obtain point estimates instead if we restrict the relationship between the underlying treatment propensity and treated outcomes to be linear. Panel b) of Figure 1 demonstrates this, where the diamonds plot the treated outcomes for the median always taker and complier against their respective treatment propensities.¹⁵ Assuming this relationship is linear lets us extrapolate the treated outcomes across the support of the treatment propensity and point identify the treated outcomes of never takers. This restriction in turn identifies the marginal treated outcome function ($MTO(p)$) which we would integrate to estimate $E[Y_i(1) = 1 | R_i = r]$ (Heckman and Vytlacil, 2000; Brinch, Mogstad, and Wiswall, 2017; Mogstad, Santos, and Torgovitsky, 2018; Kowalski, 2023b).¹⁶ Since this approach involves assuming all of the marginal treatment response functions are linear, it places stronger restrictions on behavior than the previous partial identification approach.

Implementing this separately by race group r provides either bounds or point estimates for the average outcome that would be realized if everyone of each group were treated, $E[Y_i(1) = 1 | R_i = r]$. This is the final object that we need to estimate discrimination that is conditional on treated potential outcomes, reproduced below for the case of binary treated outcomes in Equation 3.

¹⁵Linearity and the uniformity of the underlying latent index determining treatment implies that the median outcome of each compliance group has the average treated outcome of that compliance group (Kowalski, 2023b).

¹⁶Linearity assumptions identify all the marginal treatment response functions. We identify the marginal untreated outcome function by using the average untreated outcomes for compliers & never takers to extrapolate the untreated outcome for always takers. Along with $MTO(p)$, this identifies the marginal treatment effect function. Appendix C.2 sketches a simple model of selection, describes these assumptions, and illustrates this point-identification approach with a brief empirical example studying discrimination in incarceration decisions in Texas.

$$\begin{aligned}
\pi_{zr1} &= \frac{\overbrace{E[Y_i|Z = z, R_i = r, D_i = 1] \times E[D_i|Z = z, R_i = r]}^{\text{Observed in data}}}{\underbrace{E[Y_i(1) = 1|R_i = r]}_{\text{Extrapolated}}} \\
\pi_{zr0} &= \frac{\overbrace{E[(1 - Y_i)|Z = z, R_i = r, D_i = 1] \times E[D_i|Z = z, R_i = r]}^{\text{Observed in data}}}{\underbrace{1 - E[Y_i(1) = 1|R_i = r]}_{\text{Extrapolated}}} \tag{3}
\end{aligned}$$

$$\begin{aligned}
\Delta_{zy} &= \pi_{zwy} - \pi_{zby} \\
\Delta_z &= \sum_{y \in \{0,1\}} Pr(Y_i(1) = y) \Delta_{zy}
\end{aligned}$$

We observe the re-offence rate among prosecuted individuals in each period and race in the data: $E[Y_i|Z = z, R_i = r, D_i = 1]$. We also observe the prosecution rate for each period and race: $E[D_i|Z = z, R_i = r]$. Plugging the bounds/point estimates for $E[Y_i(1) = 1|R_i = r]$ into the first two lines of Equation 3 generates bounds/point estimates for each period-, race-, and potential outcome-specific treatment rate, π_{zry} . Using π_{zry} in the third line yields bounds/point estimates for the period- and outcome-specific discrimination Δ_{zy} . We then construct the period-specific discrimination estimates, Δ_z , as a weighted average of the period- and potential outcome-specific discrimination, where the weights are defined by the prevalence of the treated outcome in the population, $Pr(Y_i(1) = 1)$ (fourth line of Equation 3).

There are important practical differences between our approach using natural experiments to estimate discrimination conditional on potential outcomes and prior work using random assignment to decision-makers (Arnold, Dobbie, and Hull, 2022). By using random assignment to decision-makers, prior work exploits continuous variation in treatment rates. As a result, exercises that examine discrimination within subsamples of the data, e.g., by time period, can suffer from lack of power or a first stage within the subsamples. In contrast, such exercises will typically be more feasible using our binary IV approach. Additionally, random assignment to decision-makers in such settings is usually only conditional on certain covariates, e.g., courts, offence type, days of the week. Conditioning on covariates to ensure random assignment can introduce ‘included variables bias’ if the covariates themselves are generated due to some discriminatory behavior (Ayres, 2010). While our approach can accommodate such conditioning, it does not necessarily require it.

3.2 Accommodating difference-in-difference into the approach

So far, we have discussed using a binary IV to assess if treated outcomes differ by group (e.g., race) and estimate discrimination, conditional on treated outcomes. As discussed earlier, not all natural experiments easily map to the IV framework that is crucial for our approach. This is especially true for our application with a DiD approach using the timing of a budget reform adopted in one

county but not others. Unlike typical DiD implementations, individuals are treated (rather than counties) and there is treatment non-compliance. That is, some individuals in counties that do not adopt a reform are still treated, and not all individuals in the county that adopts the reform are treated. Using a binary IV, e.g., before vs after the reform, in such a setting is complicated by time-variation, which does not let us disentangle changes in outcomes due to the reform from the effects of time. This in turn makes it difficult to identify proportions and average outcomes for each of the compliance groups.

We overcome this complication by using the change in treated outcomes in counties that *did not adopt a reform* as an estimate of the change in treated outcomes the county that *adopted the reform* would have experienced if it had not adopted the reform. This assumes that 1) time alone does not influence treatment status, and 2) that the time trends in average treated outcomes are the same for always takers and compliers, and are independent of county. These assumptions need to hold within racial group. Next, we introduce additional notation to formalize the assumptions and describe the adjustment. In keeping with the previous subsection, we follow the same example of measuring discrimination in prosecution, conditional on the prosecuted outcome, using a budget cut that reduces prosecution rates. Appendix C.3 discusses the adjustment in further detail in a more general framework.

Let $T \in \{0, 1\}$ denote periods before/after the reform, and let $G \in \{0, 1\}$ be an indicator for the county that adopted the reform. Let $Z \equiv T \times G$ be an indicator for after the reform and in the county that adopted the reform. $D_i(g, z) \in \{0, 1\}$ denotes whether an **individual** takes up treatment or not (i.e., is prosecuted or not). This is a key feature of the IV framework that differs from typical DiD implementations. Here, it is not the case that an entire county is “treated” by the budget reform. Rather, the budget reform shifts individuals into or out of treatment. As a result, individuals in either county can be treated both before or after the policy. The lack of a time subscript in the treatment indicator implicitly makes the first assumption—time does not influence treatment status. Similar to the monotonicity assumption in IV, we allow the reform to shift individuals into or out of treatment in only one direction.

This assumption lets us partition the population of individuals in the county that adopts the reform ($G = 1$) into always takers (A), never takers (N), and compliers (C), since their proportions are constant over time. Equation 4 demonstrates how to estimate each of these in the data, following our example where the reform reduces treatment rates.

$$\begin{aligned} p_A &= E[D_i | G = 1, T = 1] \\ p_N &= 1 - (E[D_i | G = 1, T = 0]) \\ p_C &= 1 - (p_A + p_N) \end{aligned} \tag{4}$$

Since time trends are still allowed to affect potential outcomes, the treated potential outcomes for each of these groups is not directly observed in both periods. In a setting where a reform reduces treatment take-up, the outcomes of individuals who are treated after the reform identifies

the treated outcomes for always takers in $G = 1$. Equation 5 demonstrates that the treated outcomes for always takers in the post-period and in the pre-period differ by the trend in treated potential outcomes, θ_1 (second line of Equation 5), which is unobserved.

$$\begin{aligned} E[Y_i|D_i = 1, G = 1, T = 1] &= E[Y_{i1}(1, 1)|A, G = 1] \\ \underbrace{E[Y_{i1}(1, 1)|A, G = 1]}_{\text{Observed}} &= \underbrace{E[Y_{i0}(1, 1)|A, G = 1]}_{\text{Unobserved}} + \underbrace{\theta_1}_{\text{Unobserved}} \end{aligned} \quad (5)$$

We identify θ_1 by making an assumption in the spirit of parallel trends. Equation 6 formally describes this, where $Y_{it}(1, g)$ is an individual’s treated potential outcome.¹⁷ We assume that the average change in treated outcomes is the same for always takers and compliers, and is independent of county. This restricts the effects of time on treated potential outcomes to be constant across these two compliance groups, but not all of them, and does not force the effects of time to be identical across individuals.¹⁸

$$E[Y_{i1}(1, g) - Y_{i0}(1, g)|g, \text{Always taker}] = E[Y_{i1}(1, g) - Y_{i0}(1, g)|g, \text{Complier}] \text{ and } \perp g \quad (6)$$

Under these two additional assumptions, the change in outcomes among treated individuals in the counties that *did not adopt the reform* provides an estimate of the trend in treated outcomes in counties that *adopted the reform*: $\theta_1 = E[Y_{i1}(1, 0) - Y_{i0}(1, 0)|G = 0]$. We estimate θ_1 and use it to separate the reform’s impact on treated outcomes from the effects of time. This allows us to estimate average treated outcomes for always takers and compliers before and after the reform, in the county that adopted the reform. We implement these adjustments within race, and use the resulting moments to bound or point identify average outcomes for never takers before and after the reform, which lets us estimate the average outcomes if everyone of each racial group were treated.

The only difference between the DiD approach and the approach with a simple binary IV is that the average treated outcomes for each compliance group and the average outcomes if everyone were treated vary over time. As a result, bounds on average discrimination conditional on treated outcomes also vary with time.

The logic of this approach also extends to settings with staggered policy adoption. We can pair each county that adopts a policy (‘adopter’) with a set of counties that never adopted or

¹⁷This contrasts with the usual DiD implementations where individuals in counties where a policy occurs are considered ‘treated’ after the policy takes effect, while the rest are considered ‘untreated’ by the policy. There, the standard parallel trends assumption to identify impacts of the policy requires assuming parallel trends in the average untreated potential outcomes between the two counties.

¹⁸This is similar to the assumption underlying the “time-corrected” Wald estimand in De Chaisemartin and D’Haultfœuille (2018). There, the treated (untreated) potential outcomes for those treated (not treated) in the pre-period are the same across group. This identifies the LATE, but does not allow us to identify the average potential outcomes of each compliance group separately. That is because their assumption pins down time trends in a) treated outcomes for always takers and b) an average of untreated outcomes for both never takers and compliers. This does not pin down time trends for compliers specifically.

have not yet adopted the policy (‘not-yet-adopter’) (Cengiz et al., 2019). Then, the assumption described in Equation 6 needs to hold between each ‘adopter’ and its associated set of ‘not-yet-adopters’ in a window around policy adoption. Combining the IV extrapolation approach described above with these DiD adjustments for each ‘adopter’ provides estimates of discrimination for each ‘adopter’. These estimates can be aggregated across ‘adopters’ to construct an average measure of discrimination across all ‘adopters’, both before and after the policy.

3.3 Discrimination conditional on other functions of potential outcomes

We have discussed estimating discrimination conditional on **treated** potential outcomes. However, it might be more appropriate in certain contexts to measure discrimination as differential treatment among individuals with the same **untreated** potential outcomes, or even the same **treatment effect**. Each of these may also map to different normative notions of fairness. For example, say we wanted to study discrimination in the decision to nominate students for advanced educational programs. Conditioning on the treated/untreated potential outcomes and the treatment effect in this example would provide an understanding of group differences in educational program nominations between individuals who would: i) do equally well in the program, ii) do equally well without the program, and iii) have equal gains from the program. We next discuss estimating these other discrimination estimands, and the additional assumptions that may be required, again assuming that potential outcomes are binary for simplicity.

Conditioning on untreated potential outcomes, $Y_i(0)$

Differential treatment among individuals with the same untreated potential outcome is a function of period-, race-, and potential outcome-specific treatment rates, as shown in Equation 7. This diverges from the treatment rates conditional on treated potential outcomes from Equation 3 in two ways. 1) The denominator is now the average outcome that would be realized if **no one was treated**. 2) The first term in the numerator, the average **untreated** outcome among those who were **treated**, is no longer directly observed in the data, since always takers are always treated.

$$E[D_i|Z = z, R_i = r, Y_i(0) = 1] = \frac{E[Y_i(0) = 1|Z = z, R_i = r, D_i = 1] \times E[D_i|Z = z, R_i = r]}{E[Y_i(0) = 1|R_i = r]} \quad (7)$$

Assumptions on the relationship between treatment propensity and average **untreated** potential outcomes, analogous to those described for treated outcomes, deal with both instances of divergence. These assumptions bound or point-identify the average outcomes if no one were treated, $E[Y_i(0) = y|R_i = r]$. Extrapolating $E[Y_i(0) = y|R_i = r]$ involves extrapolating the average untreated outcomes of always takers (since they are always treated) using estimates of the untreated outcomes of compliers and never takers. Note that always takers’ untreated outcomes are a component of $E[Y_i(0) = y|Z = z, R_i = r, D_i = 1]$. Plugging in bounds/point estimates from each of these steps into Equation 7 yields bounds/point estimates for $E[D_i|Z = z, R_i = r, Y_i(0) = y]$. These

treatment rates would then be aggregated up, in a way analogous to [Equation 3](#), to estimate group differences in treatment among those who would have identical outcomes if not treated.

Alternatively, since treatment is binary here, the **prosecution rate** conditional on outcome if dismissed is equivalent to subtracting the **dismissal rate** for those who would have that outcome if dismissed from 1. [Appendix C.4](#) shows this formally. Given this mapping, if one was interested in estimating discrimination conditional on untreated outcomes, an attractive natural experiment is one that generates a small share of always takers. While this discussion has focused on a binary IV, [Appendix C.3](#) discusses the assumptions required to estimate discrimination conditional on untreated potential outcomes with DiD variation.

Conditioning on treatment effects, $Y_i(1) - Y_i(0)$

[Equation 8](#) describes treatment rates that condition directly on the treatment effect (τ_i). This object departs from [Equation 3](#) in two ways. 1) The denominator is the share of individuals in the population who would realize a given treatment effect value. 2) The first term in the numerator is the share of treated individuals who would realize a given treatment effect value. We cannot estimate these quantities in the same way that we have done when conditioning on potential outcomes because τ_i is never observed for any individual and can take multiple values. For example, even if both $Y_i(1)$ and $Y_i(0)$ are binary, $\tau_i \in \{-1, 0, 1\}$. Thus even though the IV assumptions identify the local average treatment effect for compliers, we will not know the prevalence of a given value of τ_i .

$$E[D_i|Z = z, R_i = r, \tau_i = y] = \frac{E[\tau_i = y|Z = z, R_i = r, D_i = 1] \times E[D_i|Z = z, R_i = r]}{E[\tau_i = y|R_i = r]} \quad (8)$$

$$\tau_i \equiv Y_i(1) - Y_i(0)$$

Conditioning on the treatment effect requires restricting the support of the treatment effect. If we assume τ_i is binary, say due to a theoretical or empirical justification, we can overcome the two challenges above. For example, we may be in a context where the treatment (prosecution) is unlikely to reduce future re-offending. We then might assume $\tau_i \in \{0, 1\}$, which implies that the average treatment effect (ATE), $E[\tau_i|R_i = r]$, coincides with the denominator of [Equation 8](#). This solves the first challenge. Similarly, $\tau_i \in \{0, 1\}$ implies that $E[\tau_i = y|Z = z, R_i = r, D_i = 1]$ is the average treatment effect on the treated (ATT). Recall that we directly observe average treated outcomes for the treated in the data and that the discussion regarding conditioning on untreated outcomes provided an estimate of the average untreated outcomes for the treated. The difference between these two averages is the ATT. Under the restriction that $\tau_i \in \{0, 1\}$, this identifies the proportion of treated individuals with a treatment effect of $\tau_i = 1$: $E[\tau_i = 1|Z = z, R_i = r, D_i = 1]$.

Plugging in bounds/point estimates from each of these steps into [Equation 8](#) yields bounds/point estimates for $E[D_i|Z = z, R_i = r, Y_i(0) = y]$.¹⁹ These treatment effect-specific treatment rates

¹⁹Bounding this requires bounding a non-linear function of point identified and partially identified objects. Ap-

would then be aggregated up in a way analogous to [Equation 3](#), to estimate group differences in treatment among those who would have the same treatment effect, assuming binary treatment effects.

3.4 Recap: Assumptions and step-by-step guide

In this section we have discussed how to use a natural experiment to estimate discrimination between two groups, accounting for any group differences in the distribution of potential outcomes.²⁰ Implementing this requires the following:

1. A natural experiment that can be mapped to an IV framework
 - If using DiD variation, need time to not shift treatment status, and a parallel trends assumption for potential outcomes
2. A potential outcome, $Y_i(D_i)$, that corresponds to a notion of fairness. That is, the choice of $Y_i(D_i)$ should be such that group differences in treatment between people with the same $Y_i(D_i)$ can be interpreted as an unwarranted disparity.
3. Use the natural experiment to estimate whether the underlying distribution of potential outcomes differ between the groups. For example, if $Y_i(D_i)$ is binary, this amounts to comparing the average outcomes if everyone of each group were treated (or not treated).
4. If the potential outcomes differ by group, estimate average discrimination conditional on potential outcomes following [Equation 2](#).

We apply this approach to two empirical applications. The first studies socio-economic discrimination in student grade promotion, using a regression discontinuity approach. The second studies racial discrimination in prosecution, using a DiD approach. In both applications, we adopt the partial identification approach, since the data are inconsistent with the linearity restriction required for point identification.

In each application, we first bound the average potential outcomes separately by group, and then plug these bounds directly into the expression for our object of interest: the average period-specific discrimination, Δ_z , in [Equation 3](#). We compute the bounds for Δ_z by searching over a grid defined by the combinations of the average potential outcomes for each group. Our

pendix [C.4](#) discusses how we compute the bounds in such a case.

²⁰So far we have discussed estimating aggregate measures of discrimination. One can also use our approach to quantify decision-maker-specific discrimination estimates with large enough samples and first stages within the subsample for each decision-maker, j . Repeating our approach within each j -subsample yields estimates of the average potential outcomes in each j 's subsample, $E[Y_i(D_i)|j]$. Decision-makers with similar $E[Y_i(D_i)|j]$ observe subsamples with similar potential outcomes. j -specific discrimination estimates are comparable, in a 'selection-on-unobservables' strategy, among such a subset of decision-makers.

approach also provides bounds for the underlying group- and potential outcome-specific treatment rates, which we report in appendices.²¹

We generate 95% bootstrapped confidence intervals for these bounds using a Bayesian bootstrapping procedure (Rubin, 1981). We use weights randomly drawn from $\Gamma(1, 1)$ to compute the moments in the estimation procedure, enforcing the weak monotonicity restriction within each re-weighted bootstrap sample. We then report the confidence intervals for the single true underlying parameter using the resulting bootstrap distribution (Imbens and Manski, 2004).

4 Socio-economic discrimination in student grade promotion

In this section we use the approach described in Section 3 to measure discrimination by socio-economic status (SES) in the decision whether to promote Michigan public school students to the next grade. This is a key decision in students’ lives that can have large academic and non-academic impacts (Jacob and Lefgren, 2004, 2009; Eren, Lovenheim, and Mocan, 2022). We test for SES differences in potential outcomes if promoted using a policy that made promotion decisions a function of standardized test scores, and generated a regression discontinuity (RD) design. Using the RD, we find large SES differences in the underlying chances of potential success if promoted to the next grade. We find significant SES gaps in promotion rates, even after accounting for SES differences in potential success in the next grade. We also highlight the relative advantages and disadvantages of implementing our approach with an RD design, which has a close link to the instrumental variables framework, rather than a difference-in-difference design.

4.1 Natural experiment: Michigan’s ‘Read by Grade 3’ law

In 2016, the State of Michigan passed legislation regarding the retention and promotion of 3rd graders in public schools (Public Act 306 of 2016). The new bill, also known as the ‘Read by Grade 3’ (RBG3) law, intended to improve public school students’ reading skills. One component of the law introduced a formulaic rule to determine when a student should be retained instead of being promoted. Prior to the bill, promotion decisions were at the discretion of the school or district staff. The bill stipulated that 3rd graders scoring 1252 or lower (approximately the 5th percentile) on the standardized reading test (English Language Arts Michigan Student Test of Educational Progress, or ELA M-STEP) were to be retained while the rest were to be promoted to 4th grade.²² While the promotion component of the policy was meant to come into effect in the 2019-20 school year, its implementation was delayed by a year due to the COVID-19 pandemic, and was ultimately

²¹Note that one could also bound average period-specific discrimination by 1) first constructing gaps using the bounds on the group- and potential outcome-specific treatment rates, 2) computing the average, and then 3) taking the minimum and maximum. This will differ from our approach of plugging in the bounds on average potential outcomes directly into the equation for average period-specific discrimination because minimum/maximum are not linear functions. We prefer directly plugging bounds on average potential outcomes into the expression for Δ_z in Equation 3 since our main goal is estimating discrimination conditional on potential outcomes.

²²The policy also required schools to provide students below the cut-off with additional interventions. Appendix A.1 discusses this further, along with other details on the policy.

repealed due to its wide unpopularity (Donahue, 2023; Povich, 2023). As a result, formula-based promotion and retention decisions affected students who were in 3rd grade during the 2020-21 and 2021-22 school years.

The formula-based system would present an opportunity to study SES discrimination in student grade promotion decisions if the formula induced an exogenous shift in the probability of being promoted – i.e., the treatment rate – at the specified 3rd grade ELA-M-STEP cut-off score. We need this shift to quantify whether there are differences by SES in the underlying potential outcomes of promotion. We follow the approach in Section 3, but use a regression discontinuity (RD) design to estimate discrimination between students who would have the same potential outcome. Alternative quasi-experimental strategies that rely on the variation in the timing of these policies, e.g., difference-in-difference, would be challenging, given how close this policy was to the start of the COVID-19 pandemic.

We use the same data as in recent work studying the impacts of multiple aspects of the RBG3 policy with an RD design at this test score cut-off. (Westall et al., 2022a,b; Berne et al., 2023; Westall, Utter, and Strunk, 2023). For each public school 3rd grader in the affected cohorts, we observe: their 3rd grade test scores, whether they were promoted to 4th grade, their performance in the following academic year (regardless of the promotion decision), and a host of baseline characteristics. We include all first-time 3rd graders during the two years that the formula-based retention rule was active. In our analysis sample, we only include students who scored within 15 points of the ELA M-STEP cut-off, which is the baseline sample in Berne et al. (2023).²³

Column 1 of Table 1 compares the general 3rd grade student body in the two affected cohorts to our analysis sample (Column 2). Michigan public school 3rd graders are diverse in terms of race, ethnicity, and economic status. Overall, 32% of students are non-white and 54% of students are designated as facing some economic disadvantage.²⁴ We refer to economically-disadvantaged students as ‘low SES’ and the rest of the students as ‘high SES’.

Our analysis sample limits to students with ELA M-STEP scores in a small window around the 5th percentile cut-off, and students in this region are relatively more disadvantaged than the average 3rd grader in Michigan’s public schools. 80% are economically-disadvantaged, and 52% are non-white. Students in our analysis sample are more likely to attend a charter school, demonstrate limited English proficiency, and participate in special education programming. Their baseline 3rd grade ELA M-STEP scores are mechanically lower—1257 on average (compared to 1293 among all 3rd graders). This places the average student in our sample at the lowest 3rd grade proficiency level designated by Michigan Department of Education (‘Level 1: Not proficient’).

Assessing the first stage & identifying assumptions

While this variation has been validated by prior work, we present validation results separately by SES since we implement our approach within SES group. Figure A1 finds no evidence of manipulation in the running variable, using the Cattaneo, Jansson, and Ma (2018) density test.

²³Berne et al. (2023) demonstrate robustness to many alternate bandwidths.

²⁴A student is designated as economically-disadvantaged if the student: was eligible for free/reduced-price lunch, received SNAP/TANF, was homeless, was a migrant, or was in foster care.

Table 1: Characteristics of Michigan 3rd grade students (2020–22)

	Overall	Estimation sample
	(1)	(2)
<i>N</i>	162,355	31,285
<i>Demographics</i>		
White (Non-Hispanic)	0.68	0.48
Black	0.19	0.38
Hispanic	0.08	0.11
Male	0.51	0.55
Economic disadvantage	0.54	0.80
<i>Academic</i>		
3rd Grade M-STEP ELA score	1292.9	1257.4
Attends charter school	0.11	0.19
Limited English Proficiency (LEP)	0.08	0.12
Special Education	0.14	0.27

Note: Column 1 includes all first-time Michigan public school 3rd graders during academic years 2020-21 and 2021-22. Column 2 limits to those 3rd graders who scored between [1238, 1267] on the 3rd grade ELA M-STEP.

Figures A2–A3 show that almost all demographics and academic characteristics are smooth around the cut-off. A4 presents a summary test of whether 4th grade ELA performance predicted using these covariates is smooth around the cut-off. While the covariates would predict that low SES students above the cut-off would score lower than low SES students below the cut-off, this is a small and marginally significant estimate—the estimated discontinuity represents 0.005 of the standard deviation of the 4th grade ELA test score.

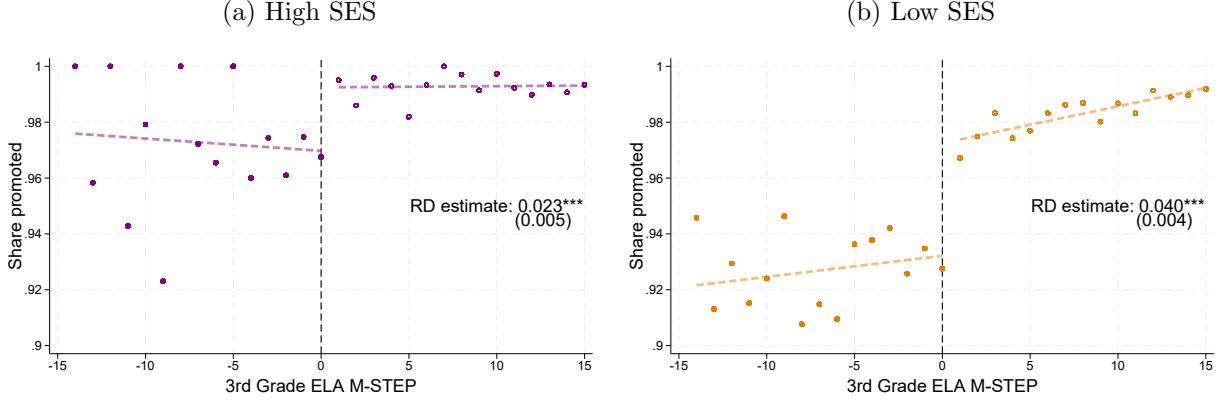
Next, we turn our attention to whether the policy rule actually resulted in discontinuous changes in the probability of promotion. If so, we can use this to estimate average potential outcomes for always takers, compliers, and never takers, and use that information to estimate average outcomes if all students of both SES groups were promoted. We start by estimating Equation 9:

$$\text{Outcome}_i = \alpha + \beta \mathbb{I}(\text{Score}_i > 1252) + \delta_1 \text{Score}_i + \delta_2 \mathbb{I}(\text{Score}_i > 1252) \times \text{Score}_i + \varepsilon_i \quad (9)$$

Figure 2 plots the relationship between the 3rd grade ELA M-STEP and the probability of being promoted, along with the estimated β from Equation 9, separately for high and low SES students. We see significant changes in the probability of being promoted around the cut-off for both groups of students. High SES students just above the cut-off are 2.3 p.p.(2.4% of the average promotion rate within 3 points of the cut-off) more likely to be promoted than high SES students below the cut-off. Low SES students just above the cut-off are 4.0 p.p.(4.3% of the average promotion rate within 3 points of the cut-off) more likely to be promoted than their counterparts below the cut-off.

Notably, while the RBG3 law stipulated a formulaic approach to promotion and retention policy, these figures suggest that a large amount of discretion was still used in making promotion decisions, consistent with prior work studying the law’s implementation (Westall et al., 2022a,b). The reason we do not see promotion shares jump from zero below the cut-off to one above the cut-off is due to

Figure 2: Effect of test score cut-off on promotion rates



Note: Each Panel presents RD estimates investigating the impact of the RBG3 test score-based promotion policy on promotion rates, using a local linear specification. The x-axis represents the running variable, the 3rd grade ELA M-STEP, re-centred by the cut-off of 1252. The sample includes first-time 3rd graders during academic years 2020-21 and 2021-22 who scored within 15 points of the ELA M-STEP cut-off. ‘RD estimate’ presents β from Equation 9. Standard errors are clustered at the level of the running variable.

a section of the law that allowed students below the cut-off (who were supposed to be retained) to be promoted if they met specific criteria.²⁵ However, despite how common these exemptions seem to be, the test score cut-off still caused a modest but meaningful share of students who would have otherwise been retained to be promoted.

4.2 Estimating discrimination in grade promotion using the test score cut-off

We start by mapping the objects from the empirical discussion to the potential outcomes framework in Section 3. Students are considered treated if they are promoted to 4th grade for the upcoming school year ($D_i = 1$) and ‘untreated’ if retained in 3rd grade. The treated potential outcome, $Y_i(1)$, is how well a student would perform if promoted to the 4th grade for the upcoming school year. For students who are retained, we do not observe how well they do in the 4th grade, since they are still in the 3rd grade.²⁶ As a result, we do not observe $Y_i(0)$.

The instrument, Z , is an indicator for being above or below the RD cut-off (we discuss complications due to the local nature of the RD below) and we use the RD to assess if there are SES differences in how ready students are for the 4th grade. We then adjust for such SES differences following Section 3, and estimate SES promotion gaps for students who would have performed equally well if promoted. We define whether students are high or low SES by whether they are flagged as being economically-disadvantaged, and denote this by $R_i = r \in \{h, l\}$.

Estimating shifts in the outcome if promoted

²⁵Common exemptions include students who: are English language learners, have disabilities, and whose parents submit an exemption request. See Appendix A.1 for additional details.

²⁶These students will eventually reach the 4th grade, but their outcomes at that point will be a function of being retained as well as being older for their grade. Our focus is on the upcoming school year, and we do not observe outcomes in 4th grade in the upcoming school year for retained students.

We construct our empirical analog of $Y_i(1)$ using students’ test scores in the next school year, if they are promoted. Since Michigan public school students in 4th grade are required to undergo standardized testing, we observe an objective measure of student performance among the promoted students when they take the 4th grade test. This definition of $Y_i(1)$ has a direct mapping to fairness norms—between two students equally likely to do well in 4th grade, it would be unfair to promote one student and not the other. This definition also maps well to the underlying behavior of teachers and other educational staff. In fact, there were concerns that some students were advancing to 4th grade without the skills to cope. The reform was conceived to improve skills that educators viewed as key inputs to students’ success in later grades (French, 2019; Povich, 2023). This anecdotal evidence suggests that around this time period, teachers and other school authorities ideally wished to only promote students who were ready for the next grade.

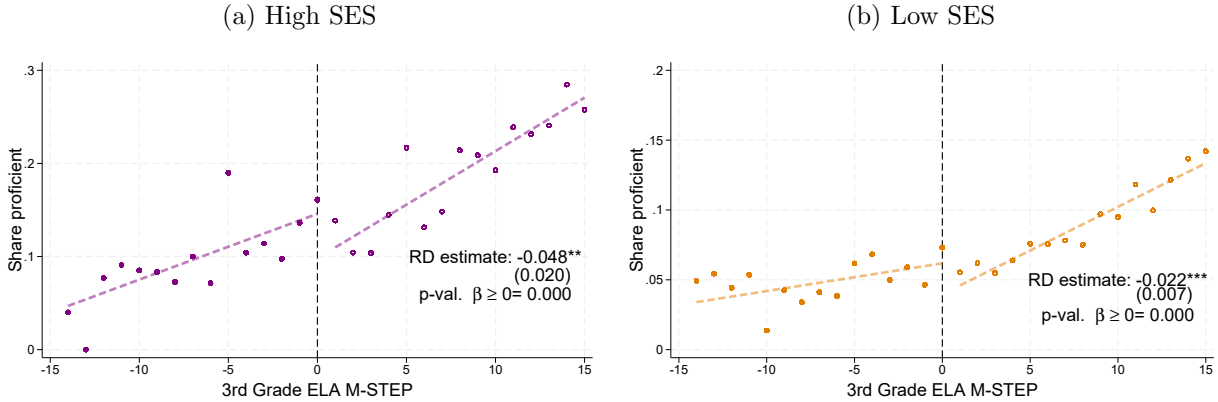
We define $Y_i(1) = 1$ if a student demonstrates “any proficiency” in both the Math and ELA M-STEP tests in the 4th grade, if promoted. A student is considered to demonstrate “any proficiency” if they receive a score of Level 2 or above according to the Department of Education guidelines in [Table A1](#). We consider this binary outcome to be a proxy for whether a promoted student was ready for the 4th grade. Returning to the potential outcomes framework, this means that $Y_i(1) = 1$ if a student is ready for 4th grade and $Y_i(1) = 0$ if a student is not ready.²⁷

As before, we use quasi-experimental variation in promoted outcomes to estimate how average promoted outcomes vary across always takers, compliers, and never takers. We use that information to bound the average test score outcomes if everyone in the analysis sample were to be promoted. [Figure 3](#) plots the relationship between the 3rd grade ELA M-STEP score and 4th grade outcomes, along with the estimated β from [Equation 9](#). We limit the sample here to only students who were promoted, since we are trying to understand how promoted outcomes vary around the cut-off. Marginal high SES students who are promoted due to the RBG3 policy are 4.8 p.p.(35.9% of the average share proficient within 3 points of the cut-off) less likely to demonstrate ‘any proficiency’ in 4th grade than promoted students below the cut-off. Marginal low SES students are 2.2 p.p.(36.7% of the average share proficient within 3 points of the cut-off) less likely to demonstrate ‘any proficiency’ in the 4th grade than promoted students below the cut-off.

In an RD setting, the proportions and outcomes of always takers, never takers, and compliers are identified by the intercepts of the local linear lines of best fit at the cut-off. Here, the proportion of always takers is the y-intercept in [Figure 2](#) as the line approaches the cut-off from below zero. The proportion of compliers is the difference between i) the y-intercept in [Figure 2](#) as the line approaches the cut-off from above zero and ii) the proportion of always takers. Never takers are the remaining share of the relevant population. Similarly, the average treated outcome of always takers is the y-intercept in [Figure 3](#) as the line approaches the cut-off from below zero. The average treated outcomes of always takers and compliers together is the y-intercept in [Figure 3](#) as the line approaches the cut-off from above zero. As described in [Section 3](#), this is the information we need

²⁷We focus on a binary proficiency measure since our sample consists of students with 3rd grade ELA M-STEP scores around the 5th percentile of the score distribution. As a result, most of the variation in outcomes is between Levels 1 and 2.

Figure 3: Impact of test score cut-off on 4th grade outcomes (Only promoted students)



Note: Each Panel presents RD estimates investigating the impact of the RBG3 test score-based promotion policy on 4th grade proficiency rates, using a local linear specification. ‘Share proficient’ represents the share of individuals who demonstrated any proficiency on both the Math and ELA M-STEP in 4th grade, as defined in Table A1. The x-axis represents the running variable, the 3rd grade ELA M-STEP, re-centred by the cut-off of 1252. The sample includes first-time 3rd graders during academic years 2020-21 and 2021-22 who scored within 15 points of the ELA M-STEP cut-off, and were promoted to 4th grade. ‘RD estimate’ presents β from Equation 9. The p -value in the second line is a one-sided test of whether the ‘RD estimate’ is weakly positive. Standard errors are clustered at the level of the running variable.

to estimate the average promoted outcomes for compliers, and bound average outcomes for never takers. Combining the estimates of average promoted outcomes for always takers, compliers, and never takers for each SES group, we have the information to bound the average promoted outcomes if everyone **at the cut-off** from both SES groups were to be promoted.

Bounding the socio-economic group-specific average outcomes if promoted

Before presenting our SES-specific estimates of average outcomes if everyone were promoted, we first address the fact that the estimated discontinuities in outcomes if promoted are large relative to the first stage. For example, taken at face value, the estimates for high SES students suggest that a 2.3 p.p. increase in the probability of being promoted causes a 4.8 p.p. reduction in the probability of demonstrating any proficiency in 4th grade. The large magnitude of the discontinuity in outcomes could be due to noise or treatments other than promotion shifting around the cut-off.²⁸ We take two approaches to deal with these issues. The first approach takes the sign of the discontinuity seriously, but is agnostic about the magnitude. The second approach is also agnostic about the sign of the discontinuity. We discuss the implementation details of the first approach next, and present the resulting SES promotion gaps. We discuss the second approach along with other robustness tests.

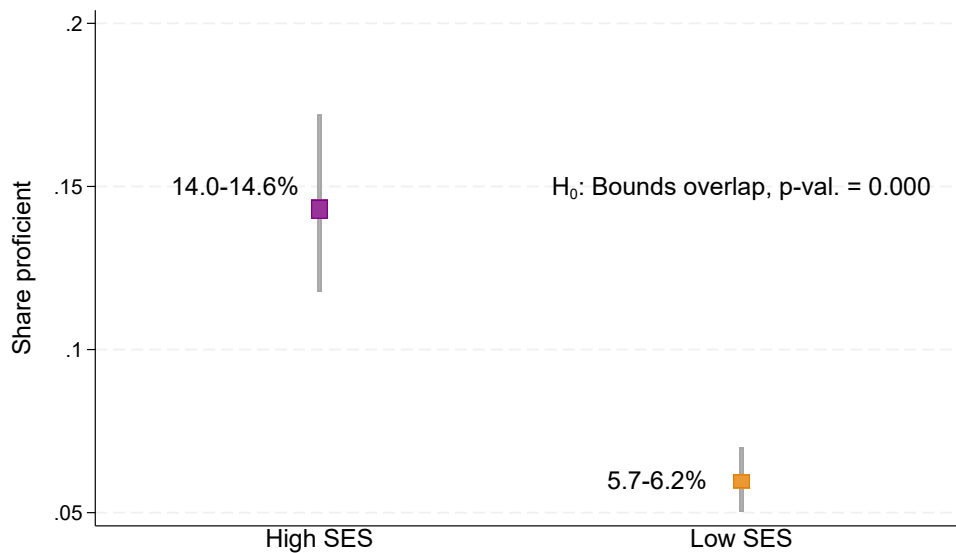
The first approach assumes that marginally promoted students at the cut-off are weakly less likely to be ready for 4th grade than students who would have been promoted anyway. This is consistent with the literature finding that some 3rd grade students might benefit from retention in

²⁸The policy mandated that students below the cut-off receive additional interventions and encouraged (but did not mandate) this for students above the cut-off. Berne et al. (2023) find evidence that some school districts followed this aspect of the policy. Appendix A.1 formally discusses the conditions under which multiple treatments would bias our analysis, and the resulting implications.

terms of academics (Jacob and Lefgren, 2004; Hwang and Koedel, 2022). The p -values presented in each Panel of Figure 3 suggest that our estimate of the sign of the relationship is credible—in both cases we reject the null that the relationship is weakly positive.²⁹

We use this estimated **sign** of the discontinuity, rather than the **magnitude**, to adjust the bounding approach described in Section 3. Figure 3 implies that, for both SES groups, always takers for promotion are more likely to demonstrate ‘any proficiency’ in 4th grade than compliers. Given these estimates, using the **sign and magnitude** would involve: i) backing out the proficiency rate if promoted for compliers and ii) assuming never takers’ proficiency rate if promoted is bounded above by that of compliers and below by zero. We modify this to assume that the average proficiency rate for both **compliers and never takers** together is bounded above by that of always takers, and below by zero.³⁰ Under this adjustment, we still assume weak monotonicity of the relationship between compliance groups’ treatment propensity and the average treated outcomes, using the **estimated direction** of the relationship between treated outcomes for always takers and compliers.

Figure 4: Average outcomes if promoted, $E[Y_i(1)|R_i = r]$



Note: This figure presents bounds on the average treated outcome obtained using the approach described in Section 3. The treatment is promotion and the treated outcome, $Y_i(1)$, is whether a student demonstrated any proficiency on both the Math and ELA M-STEP in 4th grade, as defined in Table A1. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap. The p -value is from a formal bootstrapped test of whether the identified sets overlap, described in Appendix C.5.

²⁹Figure A5 presents additional results validating the sign of the discontinuity in treated outcomes. We conduct a placebo exercise that re-estimates the RD specification in Figure 3, using every other possible test score in our analysis sample as the cut-off, while keeping the bandwidth the same. Each panel displays the distribution of resulting placebo RD estimates, and the vertical lines plot the observed estimates shown in Figure 3. Only 1.2% and 3.5% of placebo estimates for the high and low SES sample respectively are larger than the ones we observe.

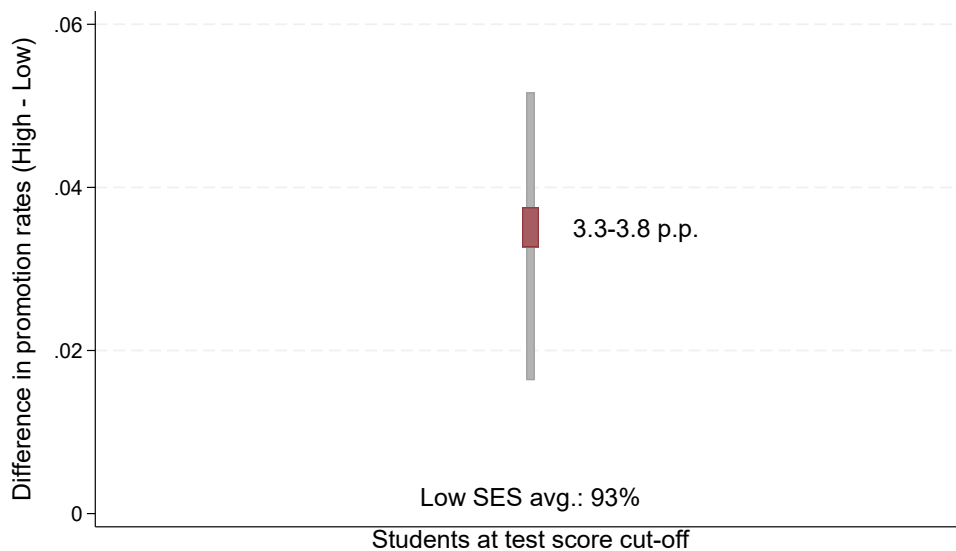
³⁰Figure A6 displays the resulting estimates of average outcomes if promoted by compliance group and SES. Compliers and never takers, the portion of the population whose treated outcomes we bound, make up 3% and 6.8% of the population of high and low SES students respectively.

With these adjustments, Figure 4 shows that there are indeed large SES differences in the underlying 4th grade outcomes that would be realized if all students in the analysis sample were promoted. 14.0–14.6% of high SES students would demonstrate ‘any proficiency’ if promoted to 4th grade, while this is true for only 5.7–6.2% of low SES students. Using a bootstrapped inference procedure described in Appendix C.5, we reject the null that these bounds overlap ($p = 0.000$). Given that there are meaningful cross-SES differences in the underlying readiness for 4th grade, observed promotion gaps by SES are likely contaminated by differences in this unobservable factor.

Estimating the socio-economic promotion gap

We now estimate bounds on the SES differences in promotion rates that condition on the outcome if promoted. Since each of the average outcomes if promoted in Figure 4 is estimated only at the cut-off, our baseline estimates will only use the average outcomes of promoted students **at the cut-off** and the promotion rate of students **at the cut-off** when following Equation 3. Such estimates are informative of SES discrimination in promotion rates **at the cut-off**.

Figure 5: SES promotion gap conditional on promoted outcome



Note: This figure presents bounds on the average difference in promotion rates conditional on treated potential outcomes, using the approach described in Section 3. The treatment is promotion and the treated outcome, denoted by $Y_i(1)$, is whether a student demonstrated any proficiency on both the Math and ELA M-STEP in 4th grade, as defined in Table A1. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Figure 5 presents our estimates of the high–low SES differences in promotion rates for students who are at the test score cut-off. Our results indicate that high SES students are 3.3–3.8 p.p. (3.5–4.1%) more likely to be promoted relative to low SES students, even after accounting for SES differences in how prepared students are for the 4th grade.³¹ Without our approach, an alternative

³¹Disaggregating gaps from Figure 5 into promotion rates for students at the cut-off who would and would not be ready for 4th grade if promoted suggests that the SES gaps we find are driven by students who are not ready for the 4th grade (see Figure A7). Figure A7 also suggests that promotion rates for students who would be ready for 4th

strategy might be to measure promotion disparities that control for observable characteristics. ‘Selection-on-observables’ approaches that control for gender, special education status, English language learner status, race, school district characteristics, and neighborhood share with at least a BA would estimate high–low SES promotion gaps of 2.8 p.p., which is 15–26% smaller than the one that we find.

Extrapolating away from the cut-off

So far, we have focused on estimating discrimination at the cut-off, but the approach can extend to measuring discrimination away from the cut-off (in a small window around the cut-off) under two additional assumptions. First, we would need to assume that the running variable alone does not influence promotion rates. Second, we would need to assume that the average outcome if everyone at the cut-off were promoted would be the same as the average outcome if everyone above or below the cut-off were promoted.³²

Table A2 tests these two assumptions in our context. Columns 1 and 3 provide suggestive evidence that the first assumption is satisfied, failing to reject that promotion status does not vary with 3rd grade ELA M-STEP scores for either SES group within the small window around the cut-off. On the other hand, Columns 2 and 4 of Table A2 show that promoted outcomes clearly increase with 3rd grade ELA M-STEP scores, violating the second assumption. Given that the assumptions needed to extrapolate our discrimination estimates away from the cut-off are not satisfied in this setting, we focus on discrimination in promotion at the cut-off.

While these assumptions may seem similar to those described in Section 3 to estimate discrimination using difference-in-difference designs, they are stronger and less likely to be satisfied in general. In the difference-in-difference case, the assumptions restrict the impact of **time** on treatment and potential outcomes. Here, the assumptions restrict the impact of the **running variable** on treatment and potential outcomes. The running variable is likely to have a tight relationship with both of these factors in many settings, making such assumptions typically unlikely to hold.

Robustness checks

The analysis so far has assumed that even if there are shifts in multiple treatments that introduce bias, the estimated sign of the discontinuity in promoted outcomes is still correct. Our results are robust to weakening this assumption by allowing outcomes if promoted for compliers and never takers to lie between 0 and 1, instead of restricting the bounds using the sign of the discontinuity. This allows for a wide range of outcomes, allowing none or all of the compliers and never takers to be ready for 4th grade. Given the weaker assumption, we estimate wider bounds for how well all students at the cut-off would do if promoted and the adjusted promotion gaps.

As Figure A8 shows, we still find that students from high SES backgrounds are much more likely to be ready for 4th grade than students from low SES backgrounds. Our bounds suggest that

grade are weakly higher than that for students who would not be, consistent with school decision-makers targeting promotion towards students who are ready for the 4th grade.

³²These assumptions, described in detail in Appendix C.6, are stronger than those in typical approaches to extrapolate away from RD cut-offs because our focus is on extrapolating average potential outcomes rather than treatment effects (Angrist and Rokkanen, 2015; Cattaneo et al., 2021; Ricks, 2022).

14–18% of high SES students and 5.7%–12.7% of low SES students would be ready for 4th grade (Figure A9). Adjusting for these underlying differences, we find evidence that high SES students are 3.3–6.0 p.p. more likely to be promoted than low SES students. These patterns are in line with our baseline estimates, suggesting that our results are not driven by the assumption on the sign of the discontinuity in promoted outcomes.

We also explore robustness to making different empirical decisions. First, we see similar patterns if we change our definition of the promoted outcome. We observe similar patterns when we condition on whether students would be at least partially proficient in the 4th grade ELA M-STEP (rather than ELA and Math), which is the academic subject directly linked to the RBG3 policy (see Figure A10 for the estimated discontinuity in 4th grade ELA proficiency). Using this measure, we still estimate that students from high SES backgrounds are almost twice as likely as students from low SES backgrounds to be ready for 4th grade (Figure A11), and find nearly identical SES promotion gaps as in our baseline (left panel of Figure A13).

Second, we also show that our findings are not driven by omitted factors that may be correlated with SES. We rule out that the gaps we estimate are driven by SES differences in how likely parents are to request retention exemptions for their children. We observe similar patterns in SES differences in average promoted outcomes (Figure A12) and SES promotion gaps that condition on these differences if we exclude students whose parents requested that they be promoted instead of being retained (right panel of Figure A13). Figure A14 displays patterns similar to our baseline when we simultaneously adjust for promoted outcomes and baseline covariates (gender, special education status, English language learner status, race, school district characteristics, and neighborhood share with at least a BA), suggesting that the unwarranted disparities that we estimate are not being driven entirely by the covariates that we have access to.

4.3 Summary

Our analysis suggests that despite the intended formulaic nature of the RBG3 policy rules, the discretion that decision-makers exercised resulted in unwarranted disparities in promotion decisions by SES. We also find that these gaps are concentrated among students who are not ready for 4th grade. These results make clear that the promotion disparities documented in recent work on the RBG3 law are not solely driven by differences in underlying unobservables (Westall et al., 2022a,b; Westall, Utter, and Strunk, 2023).

5 Racial discrimination in misdemeanor prosecution

In this section we use the approach described in Section 3 to measure racial discrimination in the decision to prosecute misdemeanor defendants. This is important decision can have long-lasting impacts on individuals’ lives, given the adverse effects of involvement with the criminal legal system (Leasure, 2019; Mueller-Smith and Schnepel, 2021; Agan, Doleac, and Harvey, 2023). We find evidence that potential re-offence outcomes if prosecuted differ by race, using difference-in-

difference variation to isolate the quasi-experimental effects of a cut to the King County prosecutors’ budget in Washington State. Accounting for these racial differences in potential outcomes, we find no evidence of discrimination in prosecution before the budget cut. Even though prosecution rates fall due to the cut, we find that white defendants were more likely to be prosecuted than minority defendants after the cut. We find suggestive evidence that this is driven by prosecutors dropping low quality cases, which are more likely to involve minority defendants.

5.1 Natural experiment: King County budget reform

We study racial discrimination in misdemeanor prosecution in King County, Washington (Seattle metropolitan area and suburban areas) using administrative records on all criminal cases from 2004–2014 from the Washington Administrative Office of the Courts. Crucially, misdemeanor cases are typically not randomly assigned to prosecutors in King County, ruling out existing random assignment-based tools to measure discrimination here.³³ We consider an individual as having their case prosecuted if their case **did not** meet the following condition: dismissed without requiring any punishments.³⁴ Our primary definition of a punishment excludes fines, but we assess robustness to including them. Our sample consists only of individuals in the court records, and so all of our estimates are representative of individuals who have been arrested and whose cases have been sent by police to the prosecutors’ office. As a result, our data may not contain cases where police arrested individuals for minor offences, but did not forward the cases to prosecutors.

We focus on differences in prosecution between white (non-Hispanic) and ‘minority’ defendants. This comparison is motivated by the fact that the population that has contact with the Washington criminal legal system is quite diverse. A large proportion of non-white defendants are of Native Hawaiian and Pacific Islander descent—these groups often face disadvantage of various forms and are over-represented in the criminal legal system in the Western United States (Hu and Esthappan, 2017; Malott, 2024). Nevertheless, we later demonstrate that our results are robust to comparing prosecution rates between white (non-Hispanic) and Black & Hispanic defendants.

As discussed in Section 3, we need a natural experiment that shifts prosecution rates to assess if unobserved potential outcomes vary by race, and then estimate discrimination conditional on the unobserved potential outcomes. We use the fact that in September 2010, King County announced a large and unanticipated cut to the Prosecutor’s Office budget. Facing a \$60 million budget shortfall, the County cut the Prosecutors’ budget by approximately \$3.9 million, the equivalent of 33 full-time employees (Constantine, 2010). The Prosecutors’ Office warned that this would reduce their ability to prosecute challenging and time-consuming cases, and that they would have

³³Correspondence with the King County Prosecuting Attorney’s Office suggests most cases submitted by police are assigned to an attorney (or a team) based on offence type, experience, and workload, who decide whether to file the case. The filing decision is then approved by supervising attorneys. While some subsequent proceedings for filed cases may be quasi-randomly assigned (e.g., arraignment based on a stipulated calendar), prosecutors for many steps are often assigned at the discretion of supervising attorneys.

³⁴Our data do not allow us to accurately distinguish between situations where prosecution was pursued and: individuals were convicted, prosecution failed, and charges were dismissed upon successful completion of a sentence. Our definition considers all of the above scenarios as ‘prosecution’.

to focus resources on offences they deemed to be high-priority (Ervin, 2010). The County tried to mitigate the budget cuts’ impact on the criminal legal system by raising additional funds through a referendum to increase the sales tax. However, the referendum failed in November 2010, consigning the King County Prosecutors’ Office to the new budget realities (Ballotpedia, 2010).³⁵

The circumstances created by the budget reform present an opportunity to use our approach to study racial discrimination in prosecution. The sharp contraction to prosecutorial resources should result in many cases being dropped, especially given the Prosecutors’ Office’s public statements. A shift in prosecution rates would let us partition each racial group into always takers, compliers, and never takers for prosecution. We would then examine how outcomes if prosecuted vary across these groups, and then account for any racial differences in outcomes if prosecuted.

We isolate the quasi-experimental variation using a difference-in-difference strategy that compares changes in prosecution and recidivism rates around the budget reform in King County, relative to changes in the adjacent counties unaffected by the reform (Chelan, Kitsap, Kittitas, Pierce, and Snohomish). We construct our analysis sample using criminal cases disposed in the District Courts of these counties. District Courts are one type of court in which the County prosecutors work, and typically hear criminal misdemeanor cases of varying severity. Given the messaging from the Prosecutors’ Office regarding the types of cases they will find difficult to pursue, cases in District Courts are most likely to be affected by the prosecutorial budget cuts. We limit to misdemeanor cases disposed of in the relevant District Courts between October 2008 and September 2012, a two-year span on either side of the budget cut announcement.³⁶

We construct re-offence outcomes and measure criminal history using information on cases filed by law enforcement in any Washington courts, including felony offences (filed in Superior Courts) and offences filed outside of the October 2008 and September 2012 interval. We measure re-offending by tracking whether a defendant appears again in these data after disposition. This measure of re-offending will not capture undetected criminal activity or arrests that police did not forward to prosecutors. However, it will capture new arrests that police forward to prosecutors, including cases that prosecutors choose not to pursue.

The final sample, described in [Table 2](#), consists of around 120,000 unique cases. 30% of the sample consists of non-white defendants, and this group is quite diverse—43% are Black, 32% are Hispanic and 18% are Asian American or Pacific Islander (AAPI). We refer to the group of non-white defendants as ‘minority’ defendants. Women make up almost a quarter of the sample, and the sample consists of a wide range of individuals in terms of age and criminal background. The

³⁵We rule out that the change in the Seattle City Attorney, who pledged to reduce racial disparities and prosecution of minor offences, on January 1, 2010 poses a confounding threat. Since the City Attorney has jurisdiction over Seattle’s local municipal courts (and county prosecutors do not) we exclude local municipal courts from our sample. Hence, direct impacts of the City Attorney change are unlikely to be present in our analysis. Since this change occurred before the county budget cut, any impacts of the City Attorney’s reforms on the broader courts in our sample should show up as differential pre-trends in the year leading up to the budget reform. We do not find strong evidence of differential pre-trends during this period in our event studies examining prosecution, recidivism, or caseload composition (discussed below).

³⁶If an individual has cases filed on multiple dates within this time frame, we only include the first case to ensure that the probability of multiple appearances is not a function of prosecution decision.

average defendant is almost 35 years old, and 47% of individuals have had at least 1 conviction in the past (averaging 3.5 prior convictions). Prosecution rates are quite high, but individuals are unlikely to face jail time if they are prosecuted. Only 9% of individuals prosecuted in King County (and 6% overall) in our sample were sentenced jail time. Among those who serve any time in jail, the average sentence is around 40 days long. By statute, the longest jail sentence for misdemeanors in Washington is one year, a sentence which only occurs for 0.5% of prosecuted cases.

Table 2: Characteristics of Washington District Court sample

	Overall	King County	Adjacent Counties
<i>N</i>	122,156	51,242	70,914
Demographics			
White (Non-Hispanic)	0.72	0.65	0.78
Black	0.12	0.16	0.09
Hispanic	0.09	0.09	0.09
AAPI	0.05	0.08	0.03
Age at disposition	34.6	34.8	34.4
Male	0.74	0.73	0.74
Criminal history			
Any prior convictions	0.47	0.42	0.50
# prior Any	3.8	3.5	3.9
Case outcomes			
Case prosecuted	0.86	0.82	0.88
<i>White</i>	0.86	0.83	0.87
<i>Minority</i>	0.85	0.81	0.89
Jail sentence Prosecuted	0.06	0.09	0.05
Sentence length (Days) Jail sentence	40.4	39.6	41.3

Note: Sample includes all criminal cases disposed of in the District Courts in Chelan, King, Kitsap, Kittitas, Pierce and Snohomish counties in Washington State between October 2008 and September 2012. For defendants with multiple dispositions in this time frame, we include only the first case. “AAPI” stands for Asian American or Pacific Islander.

Estimating first stage & shifts in recidivism outcomes

We first compare the changes in prosecution rates before and after the King County budget reform to changes in prosecution rates in adjacent counties that were unaffected by the budget reform. This involves estimating the specification in Equation 10, where D_{itg} denotes whether defendant i was prosecuted in quarter t and g denotes whether the case was disposed in King County or the adjacent counties. We investigate this separately for white and minority defendants to ensure that we have enough variation in prosecution rates in each racial subgroup.

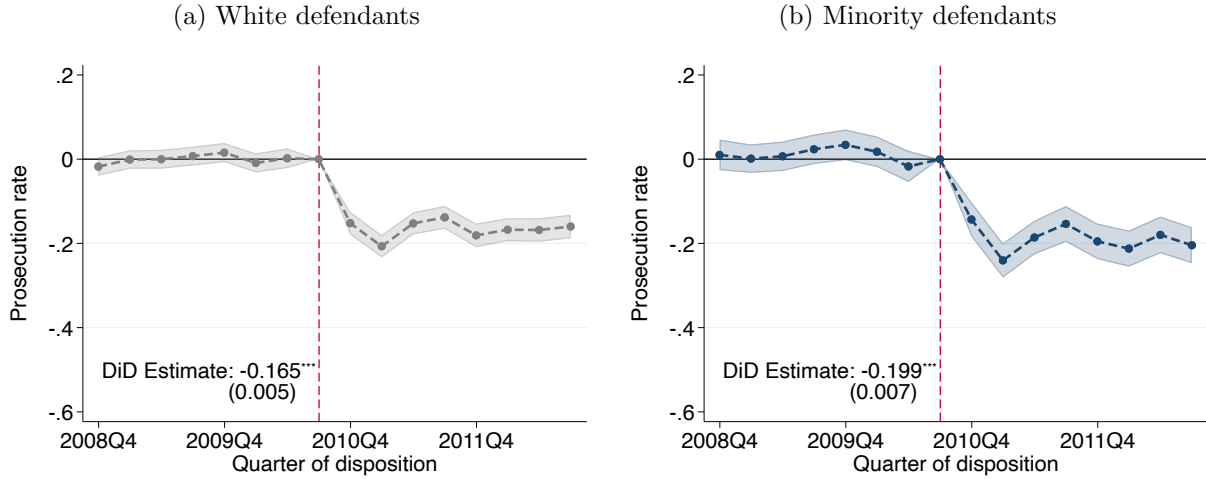
$$D_{itg} = \theta_t + \delta \mathbb{I}[\text{King County}] + \sum_{k \neq -1} \beta_k (\mathbb{I}[t - \text{Budget Reform} = k] \times \mathbb{I}[\text{King County}]) + \varepsilon_{itg} \quad (10)$$

Figure 6 documents large drops in prosecution rates due to the King County budget reforms. Prosecution rates fall by 16.5 p.p. and 19.9 p.p. for white and minority defendants respectively (17.8% and 21.7% of pre-reform average prosecution rates in King County).³⁷ We do not see

³⁷Figure B1 shows that the reduction in prosecution due to the reform is robust to using a more expansive definition of prosecution that considers cases listed as ‘Dismissed’ but with fines or fees as ‘Prosecuted’. Using that definition,

any evidence of pre-trends in prosecution rates, which provides some evidence that this natural experiment is credible (we discuss additional validity tests below). The magnitude of these shifts in prosecution rates pin down the complier share of white and minority defendants. Given the overall high rate of prosecution, these large shifts in prosecution suggest that we will have to bound outcomes for a relatively small share of the population.

Figure 6: Impact of King County budget reform on prosecution rates



Note: Each Panel presents event study estimates investigating the impact of the King County budget reform. Sample includes all misdemeanor defendants as described in Table 2. ‘DiD Estimate’ pools the coefficients on relative time indicators and estimates $D_{igt} = \alpha + \delta_1 \mathbb{I}[\text{King County}] + \delta_2 Post_i + \beta^{DD} \mathbb{I}[\text{King County}] \times Post_i + \epsilon_{igt}$, where $Post_i = 1$ if the case is disposed of on or after September 28, 2010, when the budget reform was announced. 95% confidence intervals are constructed using heteroscedasticity-robust standard errors.

We next examine whether these shifts in prosecution rates influenced individuals’ outcomes, which will reduce the width of the bounds for never takers’ prosecuted outcomes. Figure 7 repeats the event study approach, assessing the impact of the budget reform on the probability that a defendant re-offends one year after disposition. We see that not being prosecuted reduces one year re-offence rates by 3.1 p.p. for white defendants and 4.7 p.p. for minority defendants (13.1% and 15% of pre-reform average re-offence rates in King County).³⁸ Again, we see no evidence of pre-trends in re-offence rates. The direction and magnitude of these estimates are consistent with recent work demonstrating that avoiding prosecution reduces future interactions with the criminal legal system (Mueller-Smith and Schnepel, 2021; Agan, Doleac, and Harvey, 2023).^{39,40}

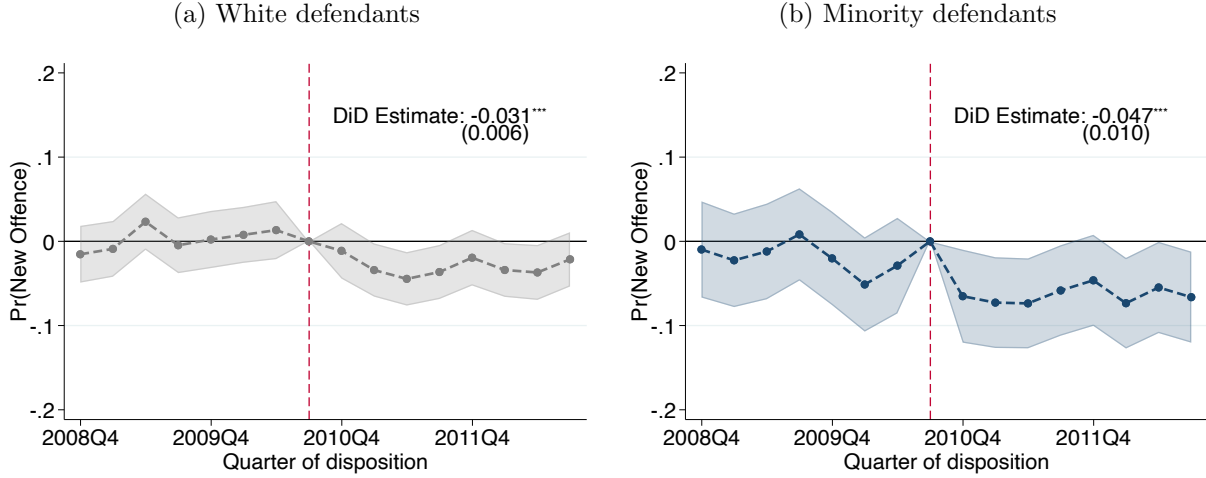
we see that prosecution falls by 15.9 p.p. and 18.8 p.p. for white and minority defendants, which is very similar to what we see with our main definition.

³⁸These reductions are unlikely to be driven by incapacitation, given that only 9% of sentences for prosecuted defendants in King County involved jail. Among those with jail sentences, the average length was 40 days, much shorter than the one year horizon used to compute the outcome (see Table 2). Finally, as discussed above, even though the maximum possible jail sentence for a misdemeanor in Washington is one year, this sentence occurs for 0.5% of prosecuted cases in our sample. As a result, potential outcomes will not be mechanically censored by jail spells.

³⁹These impacts are stable over time and persist even after two years, see Figure B2.

⁴⁰These reduced form estimates are not driven by prosecutors systematically changing what cases they accept from law enforcement, e.g., by refusing all low priority cases. Figure B3 presents results from an alternative specification

Figure 7: Impact of King County budget reform on re-offence within one year



Note: Each Panel presents event study estimates investigating the impact of the King County budget reform. The outcome includes any new misdemeanor or felony offences committed anywhere in Washington State. Sample includes all misdemeanor defendants as described in Table 2. ‘DiD Estimate’ pools the coefficients on relative time indicators and estimates $Y_{igt} = \alpha + \delta_1 \mathbb{I}[\text{King County}] + \delta_2 Post_i + \beta^{DD} \mathbb{I}[\text{King County}] \times Post_i + \epsilon_{igt}$, where $Post_i = 1$ if the case is disposed of on or after September 28, 2010, when the budget reform was announced. 95% confidence intervals are constructed using heteroscedasticity-robust standard errors.

Assessing identifying assumptions

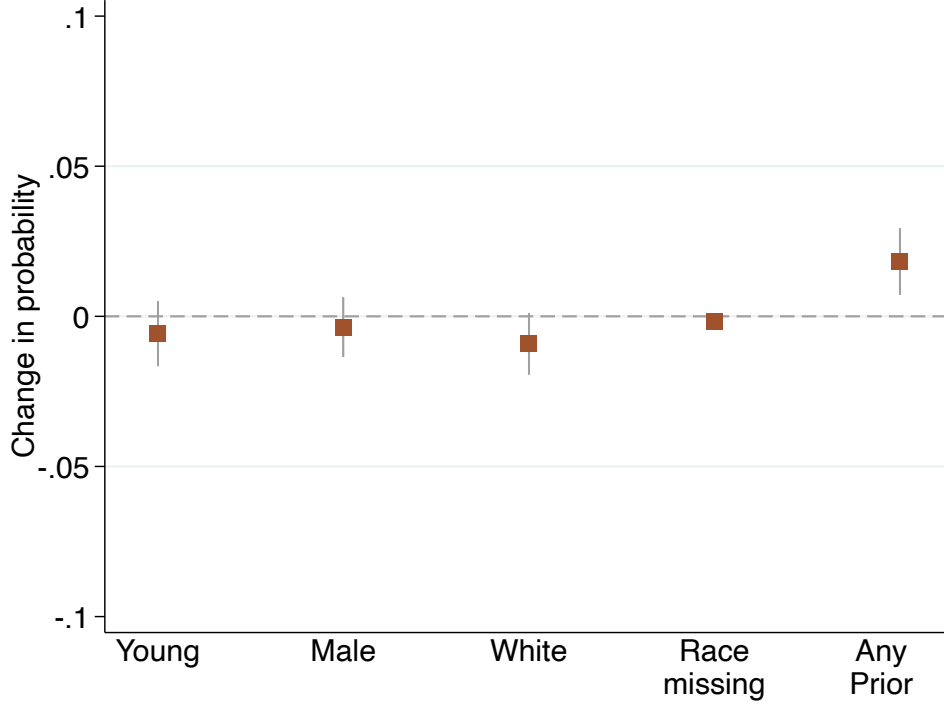
Next, we rule out different threats to the natural experiment’s validity. We should be concerned if there are any concurrent policy or behavioral changes that could influence the determinants of crime and confound our estimates of the budget reform’s effects. These might occur if other aspects of King County institutions, e.g., police or social services, were affected by the budget reform, or if prosecutors altered their behavior in ways other than prosecuting fewer cases. We assess how likely these concerns are through multiple exercises.

First, we test whether the composition of cases that prosecutors choose to accept from law enforcement changes due to the budget reform. We estimate a series of DiD regressions, where we compare the change in the share of individuals with a given baseline covariate before versus after the reform in King County to that same change in the adjacent counties. Figure 8 presents each of these coefficients and shows no evidence of compositional shifts in terms of most baseline characteristics (Figure B4 presents the underlying event study patterns for each covariate). We see some evidence that individuals whose cases were accepted after the reform were 1.8 p.p. more likely to have been previously charged with an offence but this is a small shift in composition (3.5% of the pre-period mean in King County).⁴¹

that excludes charges for offences commonly dropped right after the budget reform announcement (resisting arrest, criminal trespass, driving with a suspended licence, minor marijuana possession, reckless driving and DUI). Assuming this contains information about charges that the prosecutor’s office considers low-priority, we designate other charges as ‘high-priority’. We see reductions in the probability of ‘high-priority’ re-offending that are similar to our baseline estimates in terms of their proportion of the relevant pre-reform means.

⁴¹This minor compositional shift might be due to the end of the federal investigation of Seattle Police Department (SPD) which resulted in lower SPD stops and arrests (Campbell, 2023). We would see this compositional shift if police

Figure 8: Testing for changes in observable characteristics of cases filed



Note: Each square is β^{DD} from $X_{igt} = \alpha + \delta_1 \mathbb{I}[\text{King County}] + \delta_2 Post_i + \beta^{DD} \mathbb{I}[\text{King County}] \times Post_i + \epsilon_{igt}$, where $Post_i = 1$ if the case is filed on or after September 28, 2010, when the budget reform was announced. X_{igt} is the relevant baseline characteristic. ‘Young’ defendants are those who ≤ 28 years old at disposition and ‘Any Prior’ is an indicator for whether an individual has ever been previously charged with an offence in Washington State. 95% confidence intervals are constructed using heteroscedasticity-robust standard errors.

We also test whether law enforcement agents in King County were laid off or reduced their arrest effort (perhaps anticipating that prosecutors would stop prosecuting certain cases) due to the reform, and whether the reform affected other economic determinants of crime. [Figure B6](#) & [Figure B7](#) show no significant reductions in either officer employment or different categories of arrests. [Figure B8](#) similarly finds no differential changes in house prices, unemployment rates, or population, suggesting that the reform did not impact other economic factors that might influence criminal behavior.

The results of these exercises suggest that the drop in prosecution rates and resulting reduction in recidivism are driven by the unanticipated budget reform, and not by concurrent policy or behavioral changes. While the King County budget reform is a valid natural experiment, we need to make the adjustments described in [Section 3](#) to accommodate the DiD variation and use the reform to estimate discrimination. Next, we describe the adjustment and associated assumptions in the context of this setting. We then use this natural experiment to estimate racial differences in

focused on more serious offences. Campbell (2023) finds the largest reductions in SPD activity after the investigation ended (December, 2011), which corresponds to when we see changes in defendants with a prior charge (see Panel c) of [Figure B4](#)). Our first stage and reduced form estimates are nearly identical if we exclude this time period from our sample (see [Figure B5](#)), and so we proceed with the full sample for our baseline analysis.

prosecution among individuals who would have the same outcomes if prosecuted.

5.2 Estimating discrimination in prosecution using the budget reform

We begin by mapping the objects in the empirical discussion to the potential outcomes framework discussed in Section 3. Individuals are considered ‘treated’ if they are prosecuted ($D_i = 1$) and considered ‘untreated’ if dismissed. Potential outcomes in each treatment state are the binary indicators of recidivism used in the event study estimates in Figure 7. That is, $Y_{it}(d) = 1$ if an individual assigned to treatment state $D_i = d$ would re-offend within one year of disposition. We start by considering $Y_i(1)$, the outcome if prosecuted. Z indicates periods around the King County budget reform. As described in Section 3, the first step is to use the quasi-experimental variation from the budget reform to estimate the average re-offence outcome that would be realized if all defendants of each race were prosecuted. This would help us understand whether the unobserved potential outcomes vary by race, in which case we adjust the discrimination estimates for them.

Given the time-varying nature of potential outcomes in a DiD setting, we need to adjust for these trends. Following the DiD adjustments discussed in Section 3.2, we use the trend in re-offence outcomes among prosecuted individuals in adjacent counties as an estimate of the time trend in outcomes that prosecuted individuals in King County would have experienced had the budget reform not occurred. Using this, we isolate the change in re-offence outcomes in King County only due to the reform.

Next, we validate this adjustment, which lets us use the budget reform DiD to estimate discrimination, and then apply our approach to estimate racial discrimination in prosecution among individuals who would have the same re-offence outcome if prosecuted. We then discuss estimating discrimination conditional on having the same re-offence outcome if dismissed.

5.2.1 Empirically validating the difference-in-difference adjustments

In this context, we can estimate discrimination using the DiD if we assume the following conditions hold within each race group:

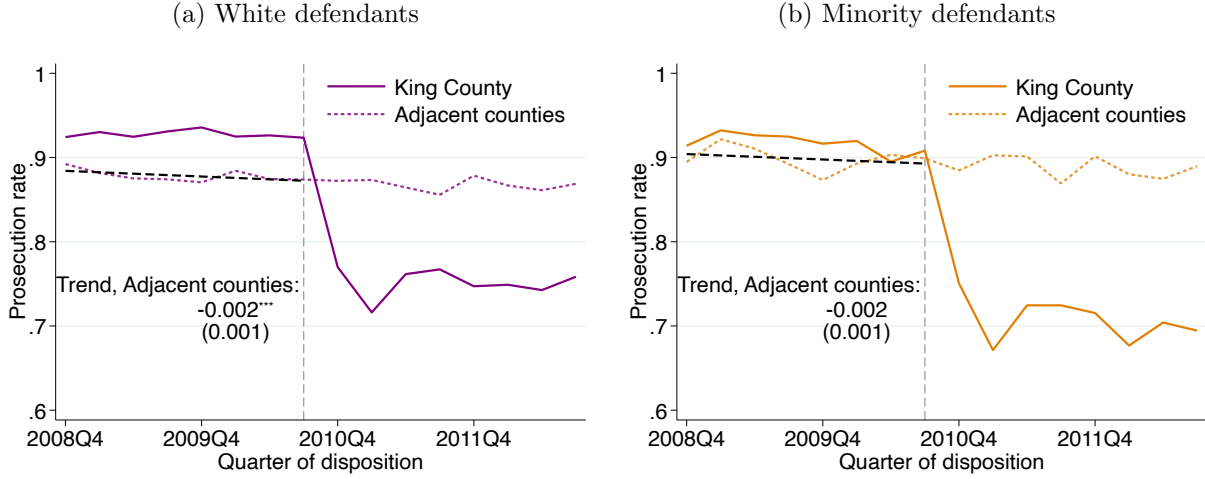
A1 Time does not shift individuals’ prosecution status.

A2 Re-offence outcomes if prosecuted, $Y_{it}(1)$, would trend similarly for always takers & compliers absent the budget reform and is independent of county.

While these assumptions are fundamentally untestable, we provide suggestive evidence that they are not violated here in this setting. Figure 9 examines **A1**, testing for time trends in prosecution using pre-period data from the adjacent counties, separately for white and minority defendants. These shifts in prosecution rates are very small. Even though the estimated trend for white defendants is significant, it represents a 0.2 p.p. reduction in prosecution rates every quarter, which is approximately 1% of the reductions in prosecution that we see in Figure 6. To ensure that these aggregate trends are not masking shifts into and out of prosecution for different types of people, we

repeat this exercise separately by covariate subgroups (gender, criminal history, age). [Figure B23](#) shows that trends in prosecution rates are similarly very small even within these subgroups. This builds confidence that individuals are not shifting into or out of treatment over time.

Figure 9: Testing trends in prosecution rates in adjacent counties



Note: The displayed coefficients are from estimating a linear regression of prosecution on a linear quarterly trend using pre-period data in the counties adjacent to King County. Standard errors on pre-period trend estimates are heteroscedasticity-robust.

Next, we test **A2**, which requires parallel trends to hold between always takers and compliers. This is because we need to account for time trends in the average prosecuted outcomes for compliers and always takers so we can use that information to construct bounds for never takers. Since we cannot identify always takers and compliers in the adjacent counties, we test for differential pre-trends by subgroup (gender, criminal history, age), which might be correlated with being an always taker or complier. Using pre-period data, [Figure B24](#) shows that there is no evidence of differential trends in $Y_{it}(1)$ across counties within the various subgroups. [Figure B25](#) conducts a similar exercise, finding limited evidence of differential trends in $Y_{it}(1)$ within counties but across subgroups. While this is not definitive evidence that these assumptions are satisfied, they build credibility that they are reasonable in this setting.

5.2.2 Baseline estimates of discrimination

We now use the variation from the budget reform DiD to 1) estimate if there are racial differences in the average re-offence outcome that we would see if all defendants of each race were prosecuted and 2) measure racial gaps in prosecution that account for such differences. We denote Z as indicating periods around the King County budget reform. Since Y_{it} varies with time, we estimate average prosecuted outcomes in each period t : $E[Y_{it}(1)|R_i = r]$. [Figure B9](#) presents event study estimates showing how the reform impacts prosecuted outcomes—we use this shift in prosecuted outcomes to estimate average prosecuted outcomes for always takers and compliers.

Bounding the race-specific average outcomes if prosecuted

Figure 10 displays how average outcomes if prosecuted vary across always takers, compliers, and never takers for white and minority defendants using the IV approach and weak monotonicity assumptions discussed in Section 3.⁴² Panels a) and c) present estimates from before the reform, and Panels b) and d) present estimates from after the reform. Among white defendants, average outcomes for always takers and compliers are uniformly lower than the corresponding averages for minority defendants. This suggests that there likely are racial differences in the average re-offence outcomes that we would see if all defendants were prosecuted. Additionally, we see that never takers comprise 7-8% of the population of each racial group, implying that we have to bound outcomes for a relatively small portion of the population. As a result, the estimated bounds on the average outcomes if everyone were prosecuted are likely to be relatively tight. Since the bounds on the average outcomes if everyone were prosecuted are inputs into estimating the discrimination estimands (see Equation 3), we should expect relatively tight bounds on discrimination as well.

As discussed earlier, the next steps assumes that the average potential outcome if prosecuted is weakly monotonic across always takers, compliers, and never takers, which implicitly places assumptions on the underlying decision-maker behavior. Here, we assume that the defendants who prosecutors are least likely to prosecute (never takers) are at least as likely to re-offend if prosecuted as marginal defendants.⁴³

Figure 11 estimates bounds on the average prosecuted outcome by computing a weighted average of the average outcomes for always takers, compliers, and never takers from Figure 10. We see meaningful and significant cross-race differences in the average outcomes that would be realized if all defendants were prosecuted. Before the budget reform, approximately 25–29% of white defendants would have re-offended if prosecuted, while approximately 32–37% of minority defendants would have done so. Using a bootstrapped inference procedure described in Appendix C.5, we reject the null that these bounds overlap ($p = 0.004$). After the reform, the bounds on average prosecuted outcomes are still meaningfully different (approximately 20–25% vs 27–32%), although testing the probability that they overlap is less precise ($p = 0.105$).⁴⁴ Given that outcomes if prosecuted differ meaningfully by racial group, not accounting for unobservable cross-race differences would yield incorrect estimates of discrimination.

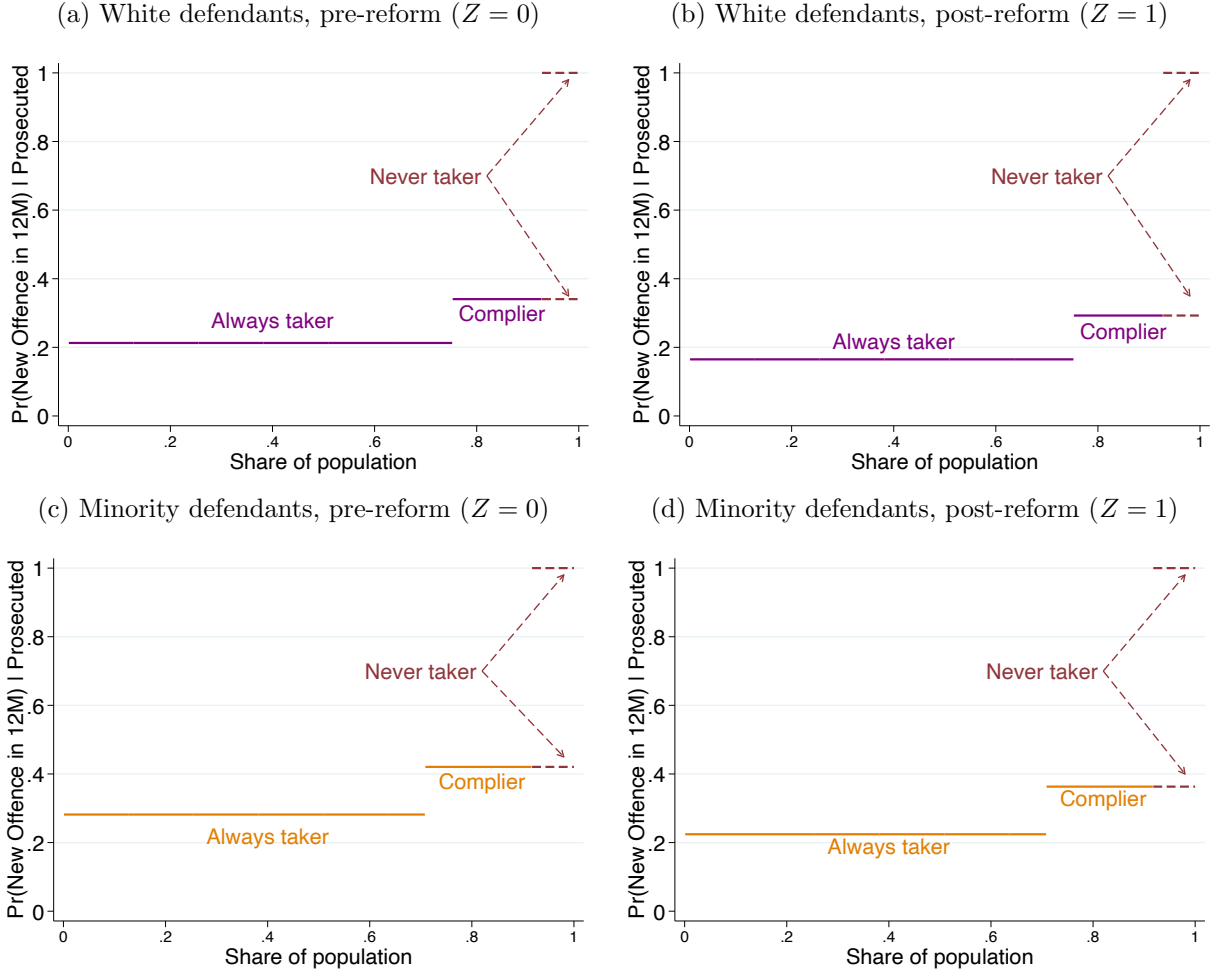
Estimating the racial gap in prosecution

⁴²As discussed in Section 2, estimating average potential outcomes by ‘compliance group’ is valid under IV monotonicity, which rules out the existence of ‘defiers’. Here, ‘defiers’ would be individuals who would not be prosecuted before the reform but would be after the reform. We assess the likelihood of ‘defiers’ in this context by re-estimating the first-stage separately by race and baseline covariate (age bins, gender, criminal history). We find large and significant reductions in prosecution similar to the estimates in Figure 6 across all race $\times$ covariate cells, suggesting that ‘defiers’ are unlikely to be present here (Figure B10).

⁴³This assumption could be violated if other inputs into prosecution decisions are correlated with re-offence outcomes in specific ways. E.g., if i) never takers are not prosecuted since their cases are backed up by low quality evidence, and ii) individuals whose cases are low quality are systematically unlikely to re-offend if prosecuted. While we do not have evidence that such violations occur in our setting, we test robustness to not assuming weak monotonicity and find similar results.

⁴⁴These patterns are not due to our decision to define a broad minority subsample. Figure B11 disaggregates the average re-offence outcomes if prosecuted for minority defendants separately by the largest race/ethnicity subcategories. While there is variation across these subcategories, the average re-offence rates for non-Black and non-Hispanic defendants are not large outliers.

Figure 10: Average outcomes if prosecuted ($Y_i(1)$) by compliance group

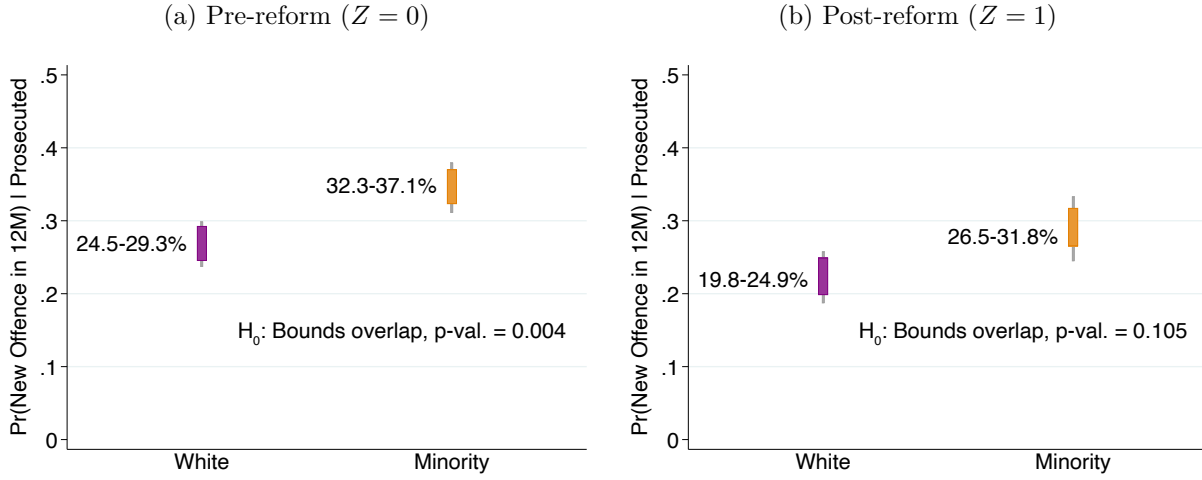


Note: This figure shows the average treated outcomes for always takers, compliers, and never takers for each time period. The treatment is prosecution and the treated outcome, $Y_i(1)$, is whether an individual re-offends one year after disposition, if prosecuted. The bounds for the treated outcomes for never takers come from the assumption of weak monotonicity of average treated outcomes across compliance groups, and that $Y_i(1) \in \{0, 1\}$.

Next, we estimate bounds on the racial differences in prosecution rates that condition on outcomes if prosecuted. Following Equation 3, we need, for each racial group and time period, 1) the average re-offence outcomes among prosecuted defendants, 2) the prosecution rate, and 3) the average re-offence outcomes if everyone were prosecuted. 1) and 2) are directly observed in the data, and we use bounds for 3) from Figure 11.

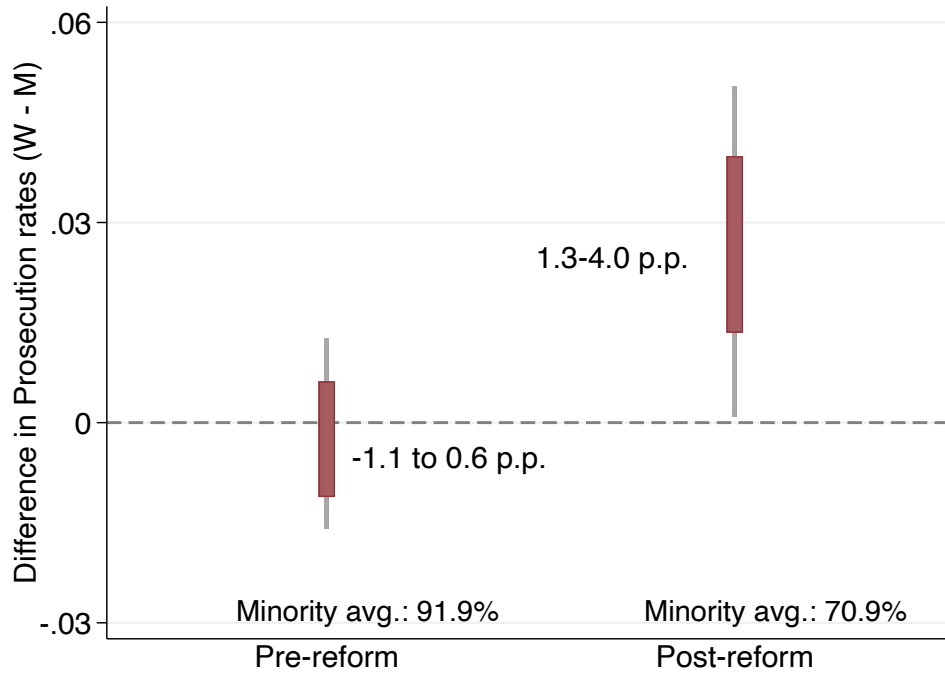
Figure 12 displays bounds on the average white–minority gap in prosecution rates conditional on outcomes if prosecuted. After accounting for racial differences in the outcomes if prosecuted, we cannot reject that white and minority defendants were prosecuted at similar rates before the budget reform. Our bounds suggest that white defendants before the reform were between 0.6 p.p. **more likely** to 1.1 p.p. **less likely** to be prosecuted compared to their minority counterparts. While this suggests that prosecution in this context may not have been discriminatory before the reform,

Figure 11: Average outcomes if prosecuted, $E[Y_{it}(1)|R_i = r]$



Note: This figure presents bounds on the average treated outcome obtained using the approach described in Section 3, separately by race and time period. The treatment is prosecution and the treated outcome, $Y_i(1)$, is whether an individual re-offends one year after disposition, if prosecuted. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap. The p-value is from a formal bootstrapped test of whether the identified sets overlap, described in Appendix C.5.

Figure 12: Racial prosecution gap conditional on prosecuted outcome



Note: This figure presents bounds on the average difference in prosecution rates in each time period, conditional on prosecuted outcomes, $Y_i(1)$, using the approach described in Section 3. $Y_i(1)$ is whether an individual re-offends one year after disposition, if prosecuted. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

this does not speak to discrimination in other aspects of the criminal legal system or society.

This pattern changes dramatically after the reform—white defendants were 1.3–4 p.p. (1.8–5.6%) more likely than minority defendants to be prosecuted, after accounting for racial differences in the outcomes if prosecuted. That is, even though the budget reform results in prosecution rates falling overall, they fall by a greater amount for minority defendants than for white defendants, even after conditioning on outcomes if prosecuted.

Alternative approaches to estimate discrimination here yield estimates that are outside of the bounds that we estimate. ‘Selection-on-observable’ estimates that control for age, gender, criminal history, and court fixed effects would estimate white–minority prosecution gaps of 1.2 p.p. before the budget reform and 4.4 p.p. after the reform. This suggests that alternative approaches to estimate of discrimination in prosecution before the reform might be incorrectly signed, and that estimates of discrimination after the reform would be biased upward by 10%–238%.

Robustness checks

These findings are robust to changing our empirical definitions of racial groups, prosecution, and re-offence outcomes. First we show that these patterns are not driven by our definition of the ‘minority’ group. [Figure B12](#) and [Figure B13](#) present the average re-offence outcomes if prosecuted and discrimination estimates that include only Black and Hispanic defendants in the minority group. These estimates are qualitatively and quantitatively similar to our baseline. We also show that our findings are robust to a broader definition of prosecution that includes fine-only punishments. [Figure B14](#) shows racial gaps that are qualitatively similar and overlap considerably with the baseline bounds. The racial gaps in prosecution that we estimate remain stable even if we reduce or expand the time horizon used to measure re-offences (see Panel (a) of [Figure B15](#)).

Our results are also robust to weakening key identifying assumptions. The patterns observed in [Figure 12](#) are not driven by the weak monotonicity assumption imposed in [Figure 10](#). [Figure B16](#) relaxes the weak monotonicity assumption by allowing never takers’ outcomes to lie between 0 and 1—i.e., allowing none or all of the never takers to re-offend if prosecuted. This exercise yields nearly identical discrimination estimates as our baseline.

Finally, [Figure B17](#) displays patterns similar to our baseline when we simultaneously adjust for potential re-offence outcomes and baseline covariates (age, gender, criminal history, court fixed effects). This implies that covariates in this context do not provide additional information that is not already captured by the potential re-offence outcomes. This also suggests that the unwarranted disparities that we observe are not being mediated by the covariates that we have access to.

5.3 Understanding drivers of the racial gap in prosecution after the reform

Our results so far document that in King County: 1) there was little evidence of discrimination in prosecution before the budget reform, and that 2) even though overall prosecution rates fell after the reform, white defendants were more likely to be prosecuted than minority defendants who

would experience identical outcomes if prosecuted.⁴⁵ Next, we investigate potential factors that could be driving this result.

Since the budget reform posed significant strain on resources, the relatively higher prosecution rate for white defendants (even conditional on prosecuted outcome) could be due to cases involving minority defendants requiring more resources to prosecute. Discrimination in pre-prosecution stages of the criminal legal system, such as policing, could lead cases involving minority defendants to be backed up by weaker evidence (Goncalves and Mello, 2021; Owens and Ba, 2021; Jordan, 2024). Since cases with weak evidence require greater resources to prosecute successfully, such cases may have been less likely to be pursued after the budget reform when resources became scarce. Instead, prosecutors might have shifted their resources to cases that they were more likely to win. As described in Section 5.1, statements from the King County Prosecutors’ Office suggested they were concerned about not being able to prosecute resource-intensive and time-consuming cases.

We investigate this explanation by using our approach to estimate discrimination in two subsamples that may vary in terms of average case quality. We classify offence categories into two bins based on the share of charges that ended up being successfully punished in King County, using (pre-reform) data from September 2004 to September 2010. The logic is that for offences where the average case is typically high quality, we should see a high conversion rate of charges into punishments. Offences for which a relatively high share of charges end up being punished are: drug, DUI, property, prostitution, and weapons violations. We refer to these as “high quality” cases. On the other hand, arrests for traffic, ‘other’, and violent offences have a relatively low share of charges that are punished, and we refer to these types of cases as “low quality”. As Table 3 shows, before the budget reform, cases involving minority defendants were relatively more likely to be lower “quality” than cases involving white defendants.⁴⁶

While other factors may also vary between these two categories (e.g., resource intensity), if the budget cut forced prosecutors to re-allocate resources based on case quality, and minority defendants’ cases tended to be lower quality, we should expect to see a more positive post-budget cut gap within the “low quality” subset.

Figure 13 displays our discrimination estimates separately for “high quality” and “low quality” cases. We see that our findings of discrimination after the reform are driven by the low quality cases. In this subsample, white defendants after the reform were 1.6–4.6 p.p. more likely than minority defendants to be prosecuted, after accounting for racial differences in the outcomes if prosecuted. In contrast, we see a gap of only 0.2 to 1.1 p.p. among the low quality subset of cases, which is much smaller and lies outside of the post-period high quality bounds. However, we cannot

⁴⁵Appendix B.3 examines how prosecution rates vary by functions of potential outcomes. We see suggestive evidence that the post-reform racial gaps are more pronounced among the set of individuals who would not re-offend if prosecuted. We also find that prosecution is less likely for defendants who would commit a new offence if prosecuted. Under additional assumptions restricting the evolution of dismissed outcomes over time and restricting the support of treatment effects of prosecution, we also find suggestive evidence that prosecution is less likely in cases where it would be harmful, i.e., by generating new crime.

⁴⁶Some cases in the data are dismissed without charges ever being recorded and we also consider these cases as “low quality”. However, we see similar shares of “low quality” cases for white defendants (76%) and minority defendants (79%) if we exclude such cases from the exercise.

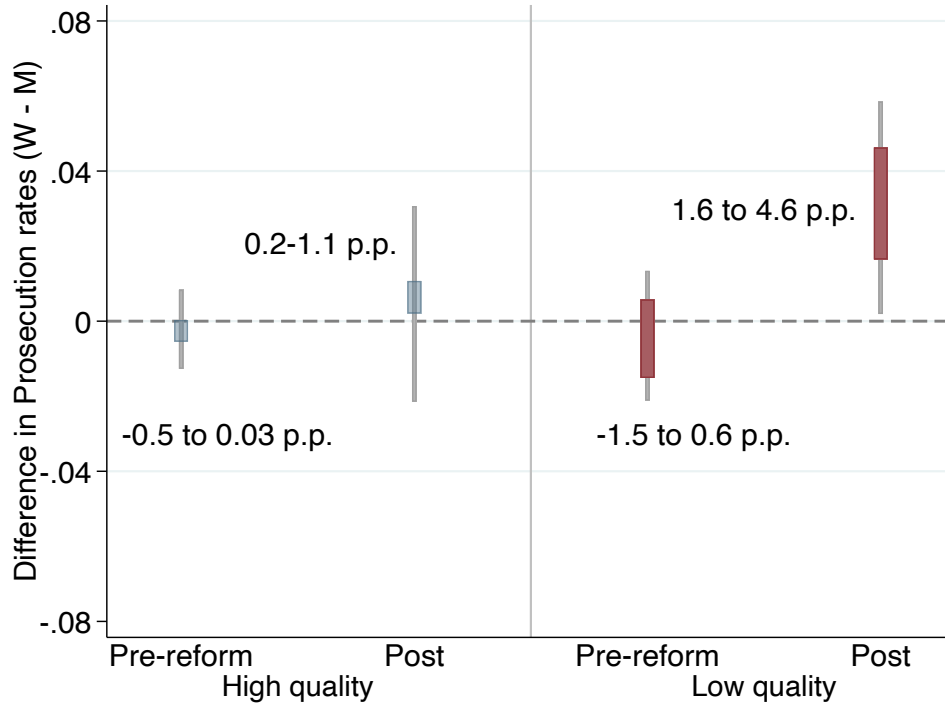
Table 3: Distribution of case “quality” by defendant racial group

	Share of cases that are:	
	High quality	Low quality
White	0.24	0.76
Minority	0.20	0.80

Note: Sample includes all criminal cases disposed of in the King County District Courts between October 2008 and September 2010 (i.e., in the analysis sample but before the budget reform). ‘High quality’/‘low quality’ offences are those with an above/below median share of charges that result in any punishment using pre-reform data.

reject that these bounds overlap due to the reduction in power from splitting our data into two subsamples.^{47,48}

Figure 13: Racial prosecution gap, by proxy for case quality



Note: This figure presents bounds on the average difference in prosecution rates in each time period, conditional on prosecuted outcomes, $Y_i(1)$, using the approach described in Section 3. $Y_i(1)$ is whether an individual re-offends one year after disposition, if prosecuted. ‘High quality’/‘low quality’ offences are those with an above/below median share of charges that result in any punishment using pre-reform data. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

⁴⁷An alternative exercise is to condition on whether the case was successfully sentenced to any punishment. However, such gaps would be biased by the exclusion of potential re-offence outcomes. On the other hand, the exercises above condition on re-offence outcomes as well as an admittedly imperfect measure of case quality. Nevertheless, Figure B18 conditions on the potential punishment outcome if a case was prosecuted. Racial gaps from this exercise are similar to the baseline analysis in Figure 12, and we cannot reject that the bounds between the two cases overlap.

⁴⁸These findings are similar but less precise if we exclude cases with missing charge information (Figure B19).

While this exercise only provides suggestive evidence, the divergence in discrimination patterns between the “high quality” and “low quality” subsamples and the patterns in Table 3 suggest that minority defendants’ cases are lower quality, and that such cases are dropped due to the budget reform. Taken seriously, this implies that prosecutors were pursuing most cases before the reform, and passing through any pre-existing disparities from prior stages in the criminal legal system (Harrington and Shaffer, 2024). However, in shifting their focus to high quality cases, prosecutors may have been offsetting such pre-existing disparities. While this behavior is consistent with prior work showing how prosecutors can use their discretion to attenuate prior discrimination (Harrington and Shaffer, 2023; Jordan, 2024), it contrasts with how other agents in the criminal legal system react when fiscally-constrained. For example, police alter their search behavior in ways that increase racial disparities when budget deficits bind (Makowsky, Stratmann, and Tabarrok, 2019).

5.4 Conditioning on outcome if dismissed, $Y_i(0)$

So far, we have defined discrimination as racial differences in prosecution conditional on the outcome if prosecuted. We consider an alternate definition of discrimination that conditions on the outcome if **dismissed**. This definition can be interpreted as an estimate of discrimination that holds fixed a notion of the baseline “risk” that an arrested individual might re-offend. As discussed in Section 3 (Equation 7), this requires bounding outcomes if dismissed for always takers (who are always prosecuted). Given that always takers are approximately 80% of the population in this setting (see Figure 10), we should expect the bounds on average outcomes and discrimination to be wider here.

Again, given the time-varying nature of the potential outcomes, we need to purge the time trends in potential re-offence outcomes if dismissed ($Y_{it}(0)$) in King County using trends in $Y_{it}(0)$ in adjacent counties. We make a similar parallel trends assumption as we previously did, but now for the re-offence outcomes if **dismissed**. As described in A3, we require parallel trends to hold between never takers and compliers (and not for always takers). We use this assumption to account for time trends in the average dismissed outcomes for compliers and never takers, and construct bounds for always takers’ dismissed outcomes.⁴⁹

A3. Re-offence outcomes if dismissed, $Y_{it}(0)$, would trend similarly for never takers & compliers absent the budget reform and is independent of county.

Using this assumption, we isolate the shift in re-offence outcomes if dismissed that is due to the budget reform. Figure B20 presents estimates of the average re-offence rates if all individuals were dismissed, separately by race and time period. We see suggestions that minority defendants are more likely to commit a new offence if dismissed, but we cannot reject that the bounds overlap. For example, our estimates suggest that before the reform, 4.3%–15.6% of white defendants would

⁴⁹Similar to the previous validation exercises, we test for differential pre-trends in the outcomes of dismissed individuals across county and covariate subgroups. Figure B26 shows no evidence of differential pre-trends in re-offence outcomes if dismissed across counties but within various demographic subgroups. Figure B27 finds limited evidence of differential pre-trends in re-offence outcomes if dismissed within counties but across subgroups. Almost all estimates are statistically indistinguishable from zero, which we interpret as evidence that A3 is not grossly violated.

commit a new offence if all were dismissed, while 6.4%–19.7% of minority defendants would commit a new offence if dismissed.⁵⁰

We use the bounds on the race-specific average outcomes if dismissed to compute racial gaps in prosecution, conditional on the outcome if dismissed. [Figure B21](#) shows that similar to our baseline results, we cannot reject that there is no discrimination in prosecution prior to the reform, and that white defendants are 0.9–8.2 p.p. (1.3–11.6%) more likely to be prosecuted after the reform. These findings are robust to altering the time horizon used to measure re-offending (see Panel (b) of [Figure B15](#)) and simultaneously controlling for baseline covariates (see Panel (b) of [Figure B17](#)).

We also redo the exercise estimating racial gaps in prosecution separately for potentially “high quality” and “low quality” cases, this time conditioning on re-offence outcomes if dismissed. We see patterns that are qualitatively similar to what we observe when conditioning on re-offence outcomes if prosecuted, but with wider and less precise bounds. In [Figure B22](#), the white–minority gap after the reform lies in [1.4 p.p., 8.5 p.p.] for “low quality” cases and lies in [−1.4 p.p., 6.8 p.p.] for “high quality” cases. While these patterns are less clear-cut than in [Figure 13](#), the bounds on post-period gaps for “low quality” cases do not contain zero, while the post-reform bounds among “high quality” cases do contain zero and are quite imprecise.

5.5 Summary

Our analysis presents the first evidence of racial discrimination in misdemeanor prosecution that directly accounts for meaningful unobservable differences across groups. We find that after a budget reform which reduced overall prosecution rates, white defendants were more likely to be prosecuted than minority defendants with similar potential re-offence outcomes. Digging deeper, our evidence suggests that prosecutors seemed to be dismissing low quality resource-intensive cases, attenuating discrimination in prior stages of the criminal legal system. Finally, our findings that not being prosecuted due to the budget reform reduces future involvement with the criminal legal system adds to the literature on the impacts of non-prosecution.

6 Conclusion

This paper shows how to use a natural experiment that generates a binary instrumental variable (IV) to estimate discrimination conditional on potential outcomes or treatment effects. We combine the shift in treatment rates and outcomes induced by the binary IV with behavioral assumptions on the relationship between selection into treatment and average potential outcomes. Depending on the strength of the assumptions, we obtain bounds or point estimates. While existing methods require random assignment to decision-makers, our approach does not. Thus, we expand the set

⁵⁰The similarity of these bounds across race contrasts to other work that finds minority defendants have higher likelihood of re-offending (Arnold, Dobbie, and Hull, 2022). One potential reason for our findings is discrimination in policing, which could push the race-specific re-offence distributions closer together if the marginal minority defendant arrested has a lower likelihood of re-offending.

of places where researchers can study discrimination while accounting for unobserved differences in potential outcomes.

We study discrimination by socio-economic status (SES) in the decision to promote Michigan public school 3rd graders to 4th grade, using a test score RD design. We find significant SES differences in the underlying probability of succeeding in 4th grade among students at a test score cut-off. Even after accounting for these underlying differences, high SES students were more likely to be promoted than low SES students, suggesting that disparities documented by prior work in this context are not solely driven by unobserved 4th grade outcomes (Westall et al., 2022a,b).

We also study racial discrimination in misdemeanor prosecution, using a budget cut in King County, Washington that sharply reduced prosecution rates. Using a DiD strategy, we find significant racial differences in the unobserved re-offence outcomes. Adjusting for these differences, we find no evidence of discrimination in prosecution before the budget reform. After the reform, white defendants were more likely to be prosecuted than minority defendants with identical potential re-offence outcomes. We find suggestive evidence of prosecutors responding to fiscal constraints by focusing on easy cases and offsetting disparities from prior stages of the criminal legal system.

Our implementation with a difference-in-difference (DiD) strategy also adds to the DiD-IV literature. We show how to use selection into being an always taker, complier, or never taker to estimate average potential outcomes among units that *experienced an intervention* using information on units who *did not experience the intervention*.

While our analysis conditions on a single potential outcome at a time, future methodological steps might incorporate insights from work estimating average population outcomes while accounting for two dimensions of unobservable heterogeneity (Dutz et al., 2021). Such an approach could quantify discrimination between individuals similar on more than one dimension, e.g., discrimination in prosecution conditional on potential re-offence and employment outcomes.

Fruitful next steps for the empirical analyses might assess the mechanisms underlying how discrimination is affected by changes in discretion (in the case of student grade promotion) and resource constraints (in the case of prosecution). Focusing on the prosecutorial context, data that can track individual events within each court case would allow us to better understand which prosecutorial actions amplify versus offset any pre-existing disparities.

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Appendix A Additional results & details: Grade promotion

A.1 ‘Read by Grade 3’ policy details

We focus on the aspect of the ‘Read by Grade 3’ policy that required students who scored below 1252 (approximately the 5th percentile) on the Michigan standard reading test, the English Language Arts Michigan Student Test of Educational Progress (ELA M-STEP), to be retained. The exact mapping of test scores to the policy guidance is below:

- **Score ≤ 1252 :** These students were to be retained unless they had a ‘good cause exemption’. These students are also supposed to receive additional supports over the next school year to improve their reading skills, e.g., extra instructional time, improved instructional quality.
 - ‘Good cause exemptions’ applied to the following students: English language learners with ≤ 3 years of English instruction, students with disabilities (i.e., with a Section 504 Plan or an Individualized Education Plan), students who have been retained before and have received supports for ≥ 2 years, students who have been enrolled in the district for ≤ 2 years and were not previously designated as having reading issues, students who later demonstrate proficiency through alternate assessments and students whose parents submit an exemption request (subject to superintendent approval) (Westall et al., 2022b).
- **Score in $[1253, 1271]$:** The policy did not mandate that these students were to be retained. These students were eligible for the additional supports that students who scored below 1252 were mandated to receive. However, this provision was not mandatory
- **Score ≥ 1272 :** These students are to be promoted, and there is no direct mandate or encouragement for them to receive additional academic supports.

Given that the RBG3 policy mandate includes the provision of additional academic interventions, and that this may change around the 1252 cut-off, there is concern that moving from one side of the cut-off to another may shift students across multiple treatments, biasing our estimates of average outcomes if promoted. We investigate potential bias by decomposing the reduced form treatment effect of promotion estimated with this variation into margin-specific treatment effects using the logic of Humphries et al. (2023).

Define the following treatment states: retention & supports (rs), promotion & no supports (pn) and promotion & supports (ps).⁵¹ Let $D_i = j$ if individual i received treatment j . For simplicity, consider the RBG3 test score cut-off rule as a binary instrument, where $Z = \mathbb{I}(\text{Score} > 1252)$. We then have that $Pr(\text{Promoted} | Z = z) = Pr(D_i = pn \text{ or } D_i = ps | Z = z)$ is increasing in Z .

⁵¹We rule out retention & no supports since a survey of school principals conducted by Michigan State University researchers suggests that most of the retained students receive some additional intervention (Strunk et al., 2022; Berne et al., 2023).

Adopting the conditional pairwise monotonicity (CPM) assumption from Humphries et al. (2023), we permit the following set of treatment flows that occur when moving from $Z = 0$ to $Z = 1$:

- Retention & supports to promotion & no supports: $rs \rightarrow pn$
- Retention & supports to promotion & supports: $rs \rightarrow ps$
- Promotion & supports to promotion & no supports (or the reverse): $ps \rightarrow pn$ (or $pn \rightarrow ps$)

This assumption ensures that flows between treatments are weakly one way and forces individuals to only flow out of retention, not into it.

Let $\omega_{p \rightarrow q}$ denote the proportion of individuals flowing in direction $p \rightarrow q$ and let $\Delta_{p \rightarrow q}^{j-k}$ denote the treatment effect between treatments j and k for people shifted along the p and q margin. Finally, let the third flow be $ps \rightarrow pn$ for simplicity.

We then rewrite the reduced form estimate from a regression of the form $Y_i = \alpha + \beta Z_i + \varepsilon_i$ as the following expression:

$$\beta = E[Y(Z=1) - Y(Z=0)] = \omega_{rs \rightarrow pn} \Delta_{rs \rightarrow pn}^{pn-rs} + \omega_{rs \rightarrow ps} \Delta_{rs \rightarrow ps}^{ps-rs} + \underbrace{\omega_{ps \rightarrow pn} \Delta_{ps \rightarrow pn}^{pn-ps}}_{\text{Bias}}$$

The above expression shows that the reduced form estimates of the effect of the RBG3 test score cut-off on outcomes is a weighted average of the treatment effect of 1) promotion without supports relative to retention (Δ^{pn-rs}), 2) promotion & supports relative to retention (Δ^{ps-rs}) and 3) additional supports with promotion relative to promotion without supports. A weighted average of the first two treatment effects can be interpreted as a total effect of promotion relative to retention and does not necessarily represent bias. However, term 3) contaminates the reduced form estimates with the effect of supports if $\omega_{ps \rightarrow pn} > 0$, i.e. if some promoted students are shifted between additional supports and no additional supports around the cut-off.⁵² If additional supports help students, then $\Delta_{ps \rightarrow pn}^{pn-ps} < 0$. Since our estimate of the effect of promotion is negative, bias under these conditions would imply that the true effect of promotion is less negative than our estimate.

⁵²This bias term arises under the CPM assumption, but not under more restrictive monotonicity assumptions. For example, if we were to assume unordered partial monotonicity instead (Mountjoy, 2022), the only permissible flows would be $rs \rightarrow pn$ and $rs \rightarrow ps \implies \omega_{ps \rightarrow pn} = 0$, implying that the bias term is 0.

A.2 Additional results

Table A1: M-STEP proficiency levels

		Level 1	Level 2	Level 3	Level 4
	Grade	Not proficient	Partially proficient	Proficient	Advanced
ELA	3	1203-1279	1280-1299	1300-1316	1317-1357
Math	3	1217-1280	1281-1299	1300-1320	1321-1361
ELA	4	1301-1382	1383-1399	1400-1416	1417-1454
Math	4	1310-1375	1376-1399	1400-1419	1420-1455

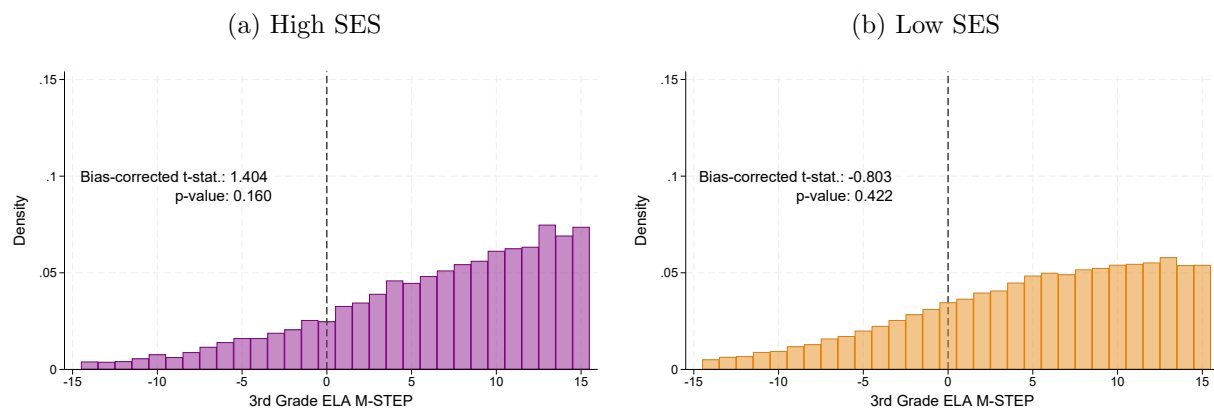
Note: From page 15 of Spring 2023 Interpretive Guide to M-STEP reports (Michigan Department of Education, 2023).

Table A2: Examining effect of test score on promotion and promoted outcomes

	High SES		Low SES	
	Promoted	Proficient	Promoted	Proficient
Above cut-off	0.0227*** (0.0048)	-0.0477** (0.0196)	0.0404*** (0.0041)	-0.0221*** (0.0071)
Score	-0.0004 (0.0010)	0.0071*** (0.0013)	0.0008 (0.0006)	0.0020** (0.0009)
Above cut-off $\times$ Score	0.0005 (0.0010)	0.0044** (0.0019)	0.0006 (0.0006)	0.0043*** (0.0010)
Mean Outcome	0.9697*** (0.0040)	0.1459*** (0.0110)	0.9322*** (0.0033)	0.0618*** (0.0058)
N	6201	6132	25084	24320

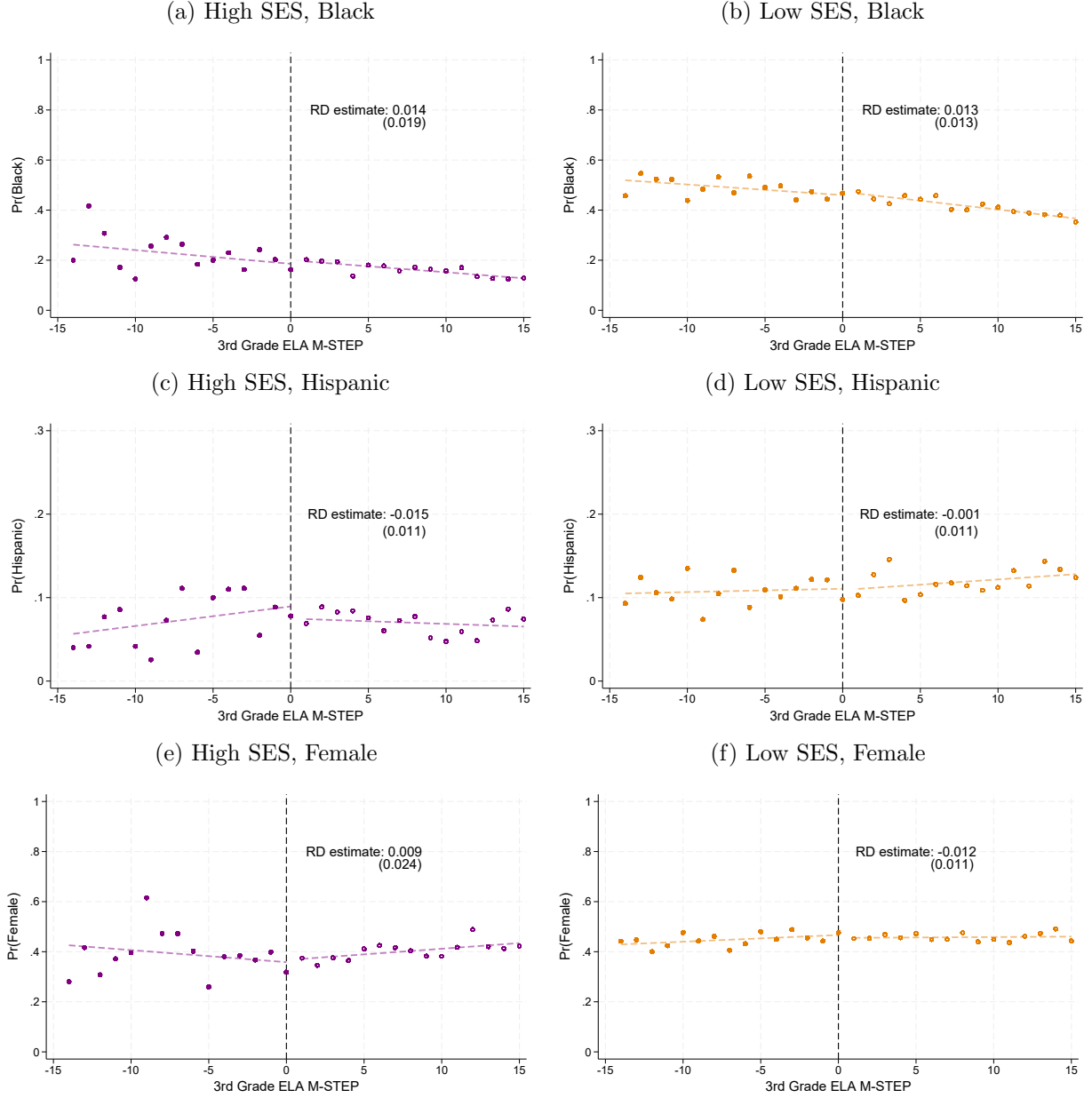
Note: This presents estimates of $X_i = \alpha + \beta \mathbb{I}(\text{Score}_i > 1252) + \delta_1 \text{Score}_i + \delta_2 \mathbb{I}(\text{Score}_i > 1252) \times \text{Score}_i + \varepsilon_i$, where X_i is either an indicator for promotion or the proficiency outcome if promoted. The ‘Proficient’ columns only include promoted students. Standard errors are clustered at the level of the running variable.

Figure A1: Smoothness of test score distribution density around cut-off



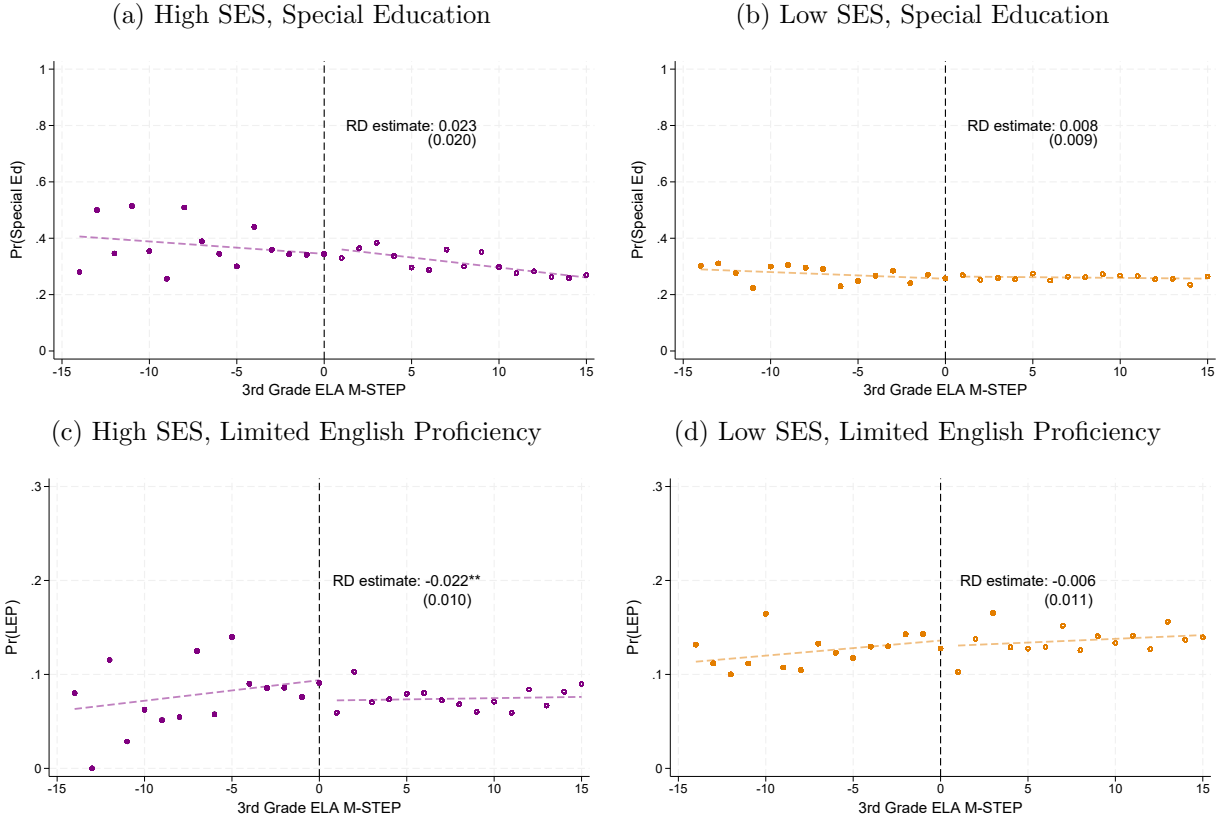
Note: These figures present results of the Cattaneo, Jansson, and Ma (2018) density test for the smoothness of the running variable. The x-axis represents the running variable, the 3rd grade ELA M-STEP, re-centred by the cut-off of 1252. The sample includes first-time 3rd graders during academic years 2020-21 and 2021-22 who scored within 15 points of the ELA M-STEP cut-off.

Figure A2: Smoothness of demographics around test score cut-off



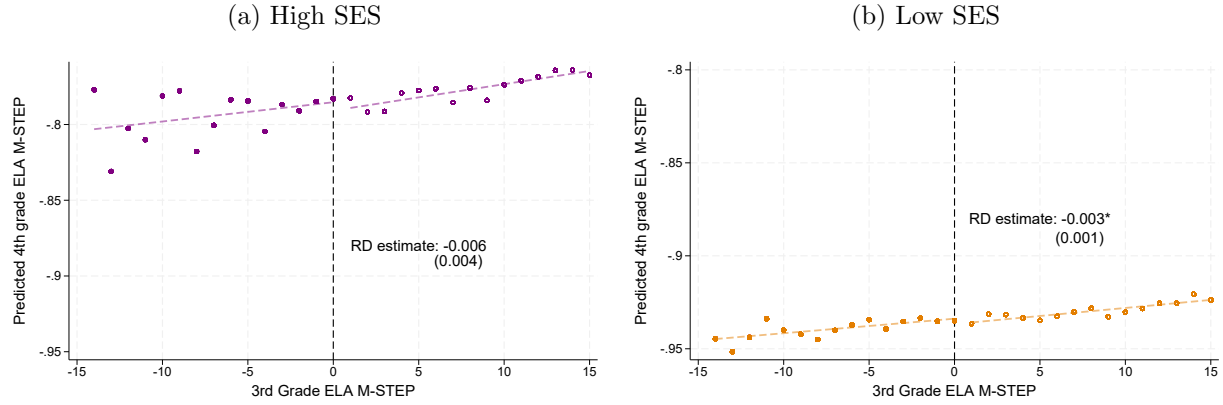
Note: Each Panel presents RD estimates investigating the impact of the RBG3 test score-based promotion policy on the listed baseline covariate, using a local linear specification. The x-axis represents the running variable, the 3rd grade ELA M-STEP, re-centred by the cut-off of 1252. The sample includes first-time 3rd graders during academic years 2020-21 and 2021-22 who scored within 15 points of the ELA M-STEP cut-off. 'RD estimate' presents β from $X_i = \alpha + \beta \mathbb{I}(\text{Score}_i > 1252) + \delta_1 \text{Score}_i + \delta_2 \mathbb{I}(\text{Score}_i > 1252) \times \text{Score}_i + \varepsilon_i$. Standard errors are clustered at the level of the running variable.

Figure A3: Smoothness of additional academic programming around test score cut-off



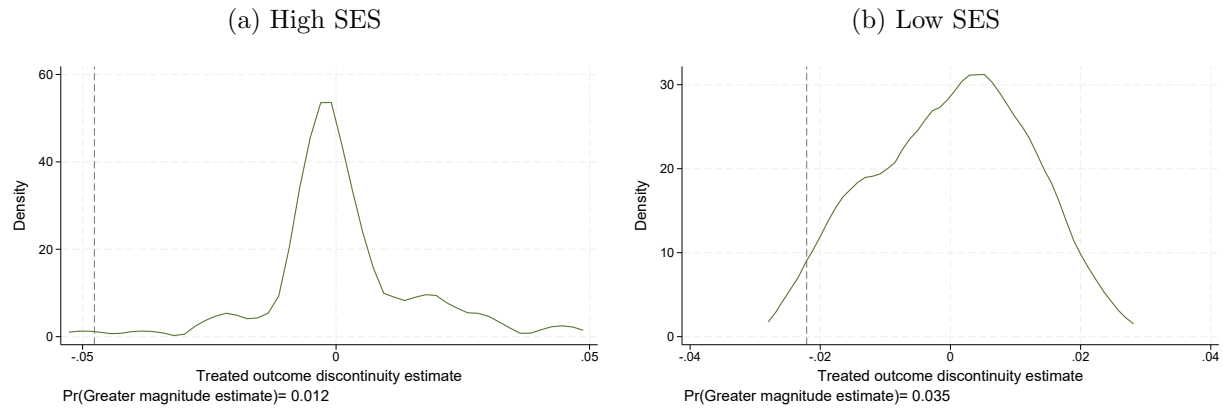
Note: Each Panel presents RD estimates investigating the impact of the RBG3 test score-based promotion policy on the listed baseline covariate, using a local linear specification. The x-axis represents the running variable, the 3rd grade ELA M-STEP, re-centred by the cut-off of 1252. The sample includes first-time 3rd graders during academic years 2020-21 and 2021-22 who scored within 15 points of the ELA M-STEP cut-off. ‘RD estimate’ presents β from $X_i = \alpha + \beta \mathbb{I}(\text{Score}_i > 1252) + \delta_1 \text{Score}_i + \delta_2 \mathbb{I}(\text{Score}_i > 1252) \times \text{Score}_i + \varepsilon_i$. Standard errors are clustered at the level of the running variable.

Figure A4: Smoothness of predicted ELA M-STEP next year around test score cut-off



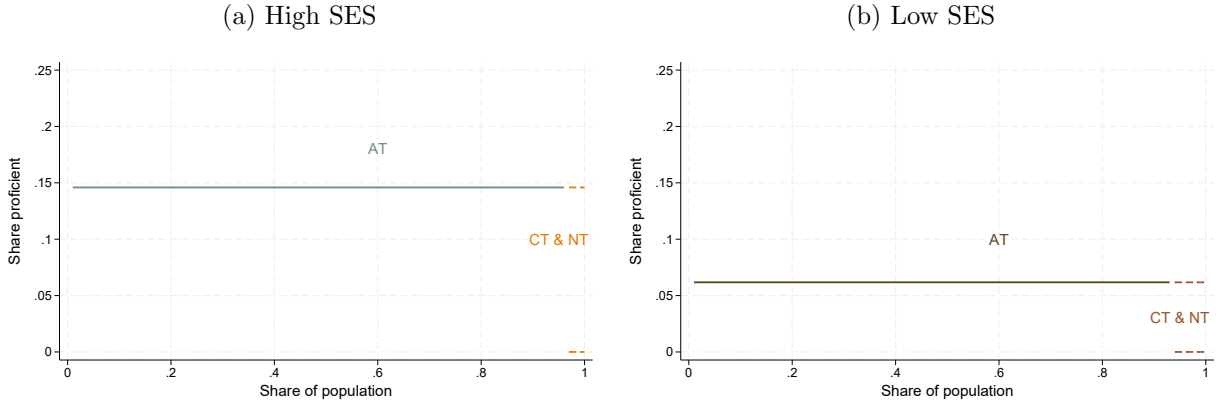
Note: These figures present RD estimates investigating the impact of the RBG3 test score-based promotion policy on the predicted ELA M-STEP score taken in the following school year (regardless of actual promotion status), using demographics, Limited English Proficiency and special education status, whether the student was previously retained, whether the student is new to the district, and school fixed effects. ‘RD estimate’ presents β from $X_i = \alpha + \beta \mathbb{I}(\text{Score}_i > 1252) + \delta_1 \text{Score}_i + \delta_2 \mathbb{I}(\text{Score}_i > 1252) \times \text{Score}_i + \varepsilon_i$. Standard errors are clustered at the level of the running variable. The x-axis represents the running variable, the 3rd grade ELA M-STEP, re-centred by the cut-off of 1252. The sample includes first-time 3rd graders during academic years 2020-21 and 2021-22 who scored within 15 points of the ELA M-STEP cut-off.

Figure A5: Distribution of placebo estimates of discontinuity in treated outcomes



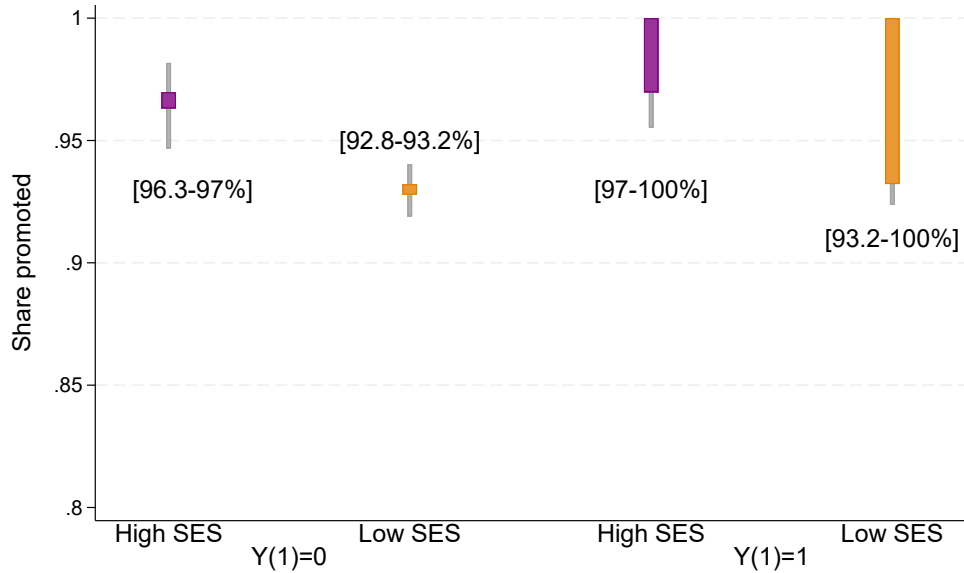
Note: These figures present the distribution of estimates of the discontinuity in treated outcomes obtained by using alternate values of the running variable as the cut-off. We use each value in the $[-1, 85]$ interval as the cut-off, where 0 is the true cut-off, but keep the bandwidth fixed at 15 points on either side of the cut-off, as in the baseline. The vertical lines display the RD estimate with the true cut-off observed in Figure 3. ‘Pr(Greater magnitude estimate)’ computes the share of placebo estimates that are larger than the observed RD estimate. The treated outcome, $Y_i(1)$, is whether a student demonstrated any proficiency on both the Math and ELA M-STEP in 4th grade, as defined in Table A1. The sample and specification details are the same as described in Figure 3.

Figure A6: Average outcomes if promoted ($Y_i(1)$) by compliance group



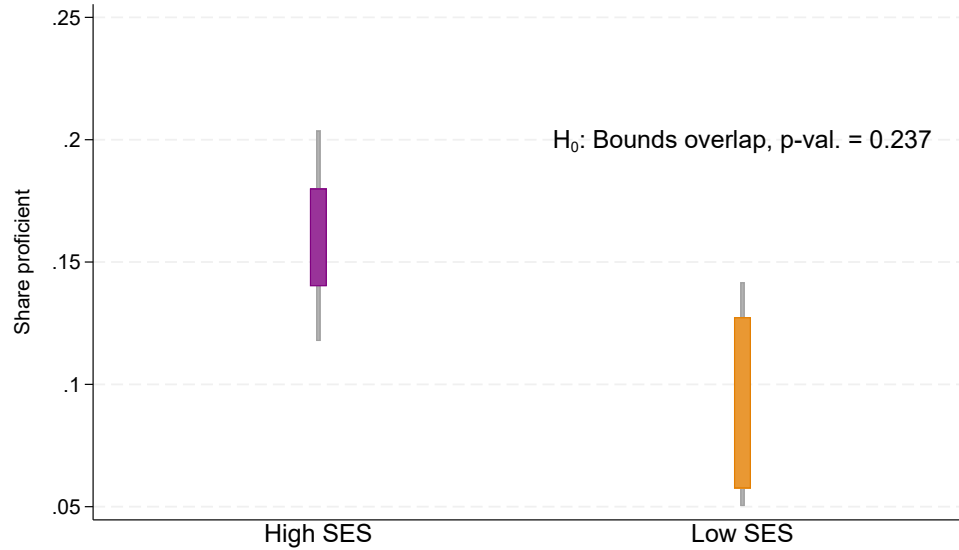
Note: This figure shows the average treated outcomes for always takers ('A'), compliers ('C'), and never takers ('N'). The treatment is promotion and the treated outcome, $Y_i(1)$, is whether a student demonstrated any proficiency on both the Math and ELA M-STEP in 4th grade, as defined in Table A1. The bounds for the treated outcomes for never takers and compliers come from the assumption of weak monotonicity of average treated outcomes across compliance groups, and that $Y_i(1) \in \{0, 1\}$.

Figure A7: Promotion rates at the cut-off, conditional on promoted outcome (π_{0ry})



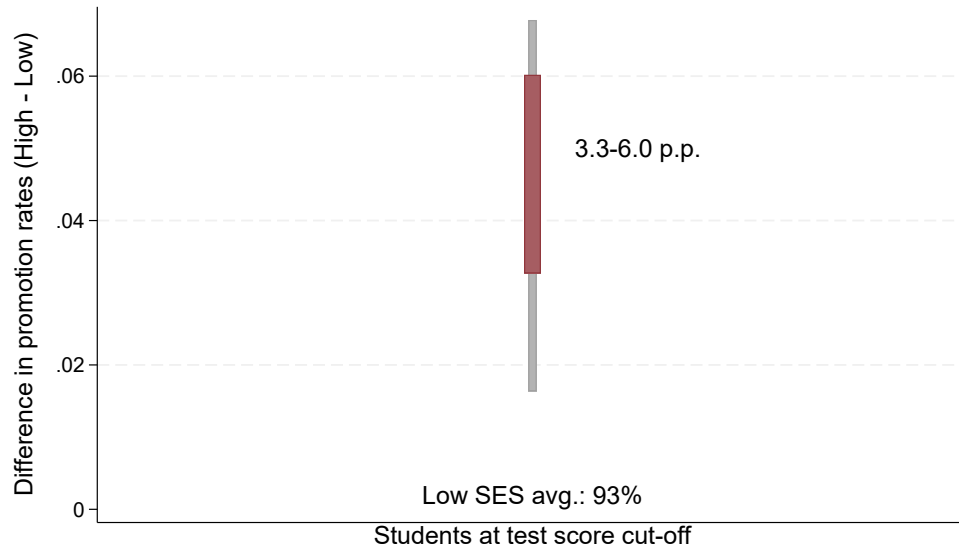
Note: This figure presents bounds on the average prosecution rates for each SES group, conditional on promoted outcomes, $Y_i(1)$, using the approach described in Section 3. $Y_i(1)$ is whether a student demonstrated any proficiency on both the Math and ELA M-STEP in 4th grade, as defined in Table A1. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Figure A8: Average outcomes if promoted: Worst case bounds



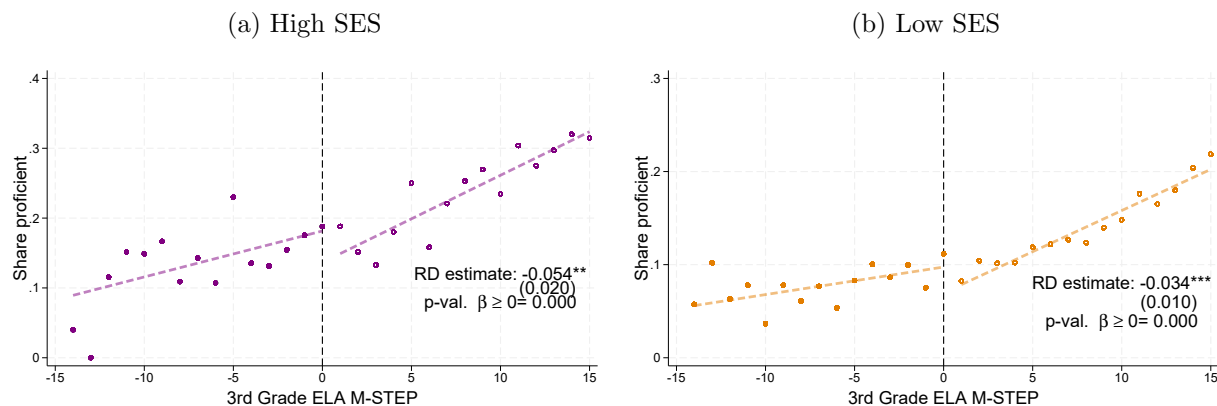
Note: This figure presents bounds on the average treated outcome obtained by modifying the approach described in Section 3 to allow compliers' and never takers' outcomes to lie in $[0, 1]$. The treatment is promotion and the treated outcome, $Y_i(1)$, is whether a student demonstrated any proficiency on both the Math and ELA M-STEP in 4th grade, as defined in Table A1. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap. The p -value is from a formal bootstrapped test of whether the identified sets overlap, described in Appendix C.5.

Figure A9: SES promotion gap: Worst case bounds



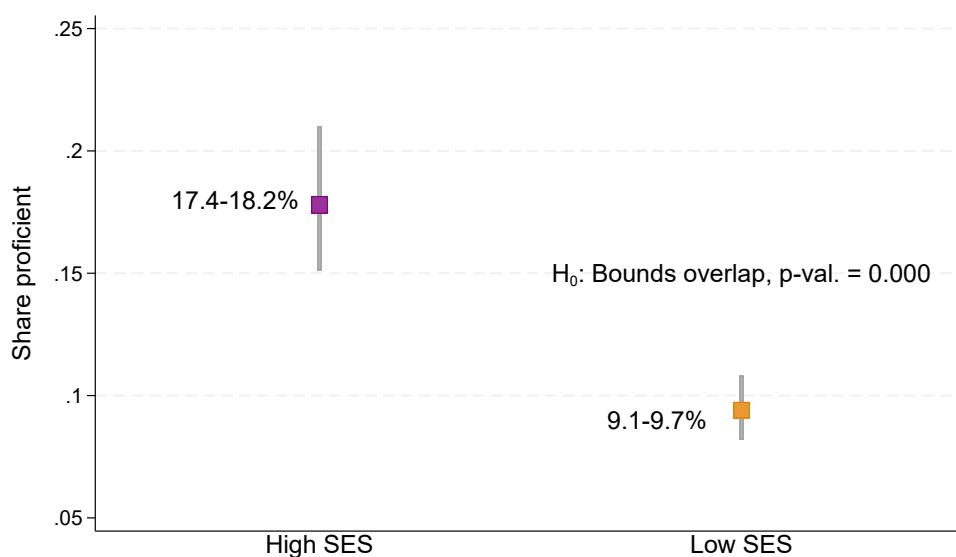
Note: This figure presents bounds on the average difference in promotion rates, conditional on treated potential outcomes, by modifying the approach described in Section 3 to allow compliers' and never takers' outcomes to lie in $[0, 1]$. The treatment is promotion and the treated outcome, denoted by $Y_i(1)$, is whether a student demonstrated any proficiency on both the Math and ELA M-STEP in 4th grade, as defined in Table A1. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Figure A10: Impact of test score cut-off on 4th grade ELA outcomes (Only promoted students)



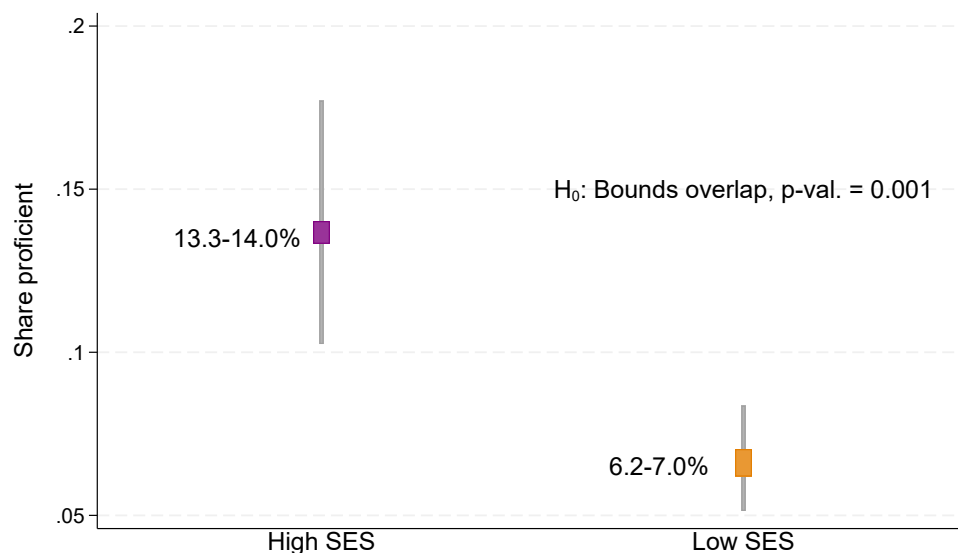
Note: Each Panel presents RD estimates investigating the impact of the RBG3 test score-based promotion policy on 4th grade proficiency rates, using a local linear specification. ‘Share proficient’ represents the share of individuals who demonstrated any proficiency on the ELA M-STEP in 4th grade, as defined in Table A1. The x-axis represents the running variable, the 3rd grade ELA M-STEP, re-centred by the cut-off of 1252. The sample includes first-time 3rd graders during academic years 2020-21 and 2021-22 who scored within 15 points of the ELA M-STEP cut-off, and were promoted to 4th grade. ‘RD estimate’ presents β from Equation 9. The p -value in the second line is a one-sided test of whether the ‘RD estimate’ is weakly positive. Standard errors are clustered at the level of the running variable.

Figure A11: Average outcomes if promoted, only ELA proficiency



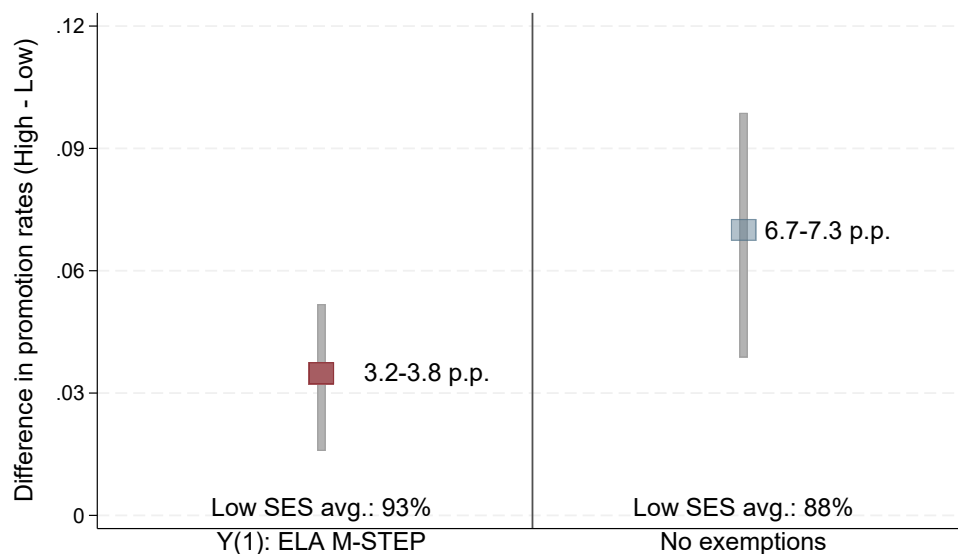
Note: This figure presents bounds on the average treated outcome obtained using the approach described in Section 3. The treatment is promotion and the treated outcome, $Y_i(1)$, is whether a student demonstrated any proficiency on the ELA M-STEP in 4th grade, as defined in Table A1. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap. The p -value is from a formal bootstrapped test of whether the identified sets overlap, described in Appendix C.5.

Figure A12: Average outcomes if promoted, excluding students whose parents requested exemptions



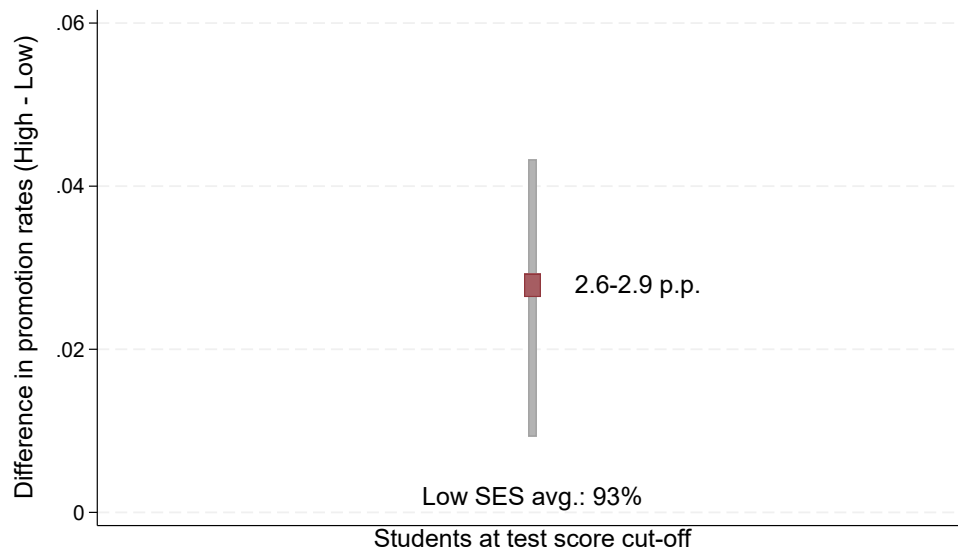
Note: This figure presents bounds on the average treated outcome obtained using the approach described in Section 3. The treatment is promotion and the treated outcome, $Y_i(1)$, is whether a student demonstrated any proficiency on both the Math and ELA M-STEP in 4th grade, as defined in Table A1. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap. The p -value is from a formal bootstrapped test of whether the identified sets overlap, described in Appendix C.5.

Figure A13: SES promotion gap robustness: Alternate $Y_i(1)$; Excluding parent-driven exemptions



Note: This figure presents bounds on the average difference in promotion rates, conditional on treated potential outcomes, using the approach described in Section 3. The treatment is promotion. The treated outcome in the left panel is whether a student demonstrated any proficiency on the 4th grade ELA M-STEP, and the treated outcome in the right panel indicates proficiency on both the 4th grade Math and ELA M-STEP. Proficiency levels are defined in Table A1. The right panel excludes students whose parents requested exemptions. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Figure A14: SES promotion gap, conditional on potential outcome and baseline covariates

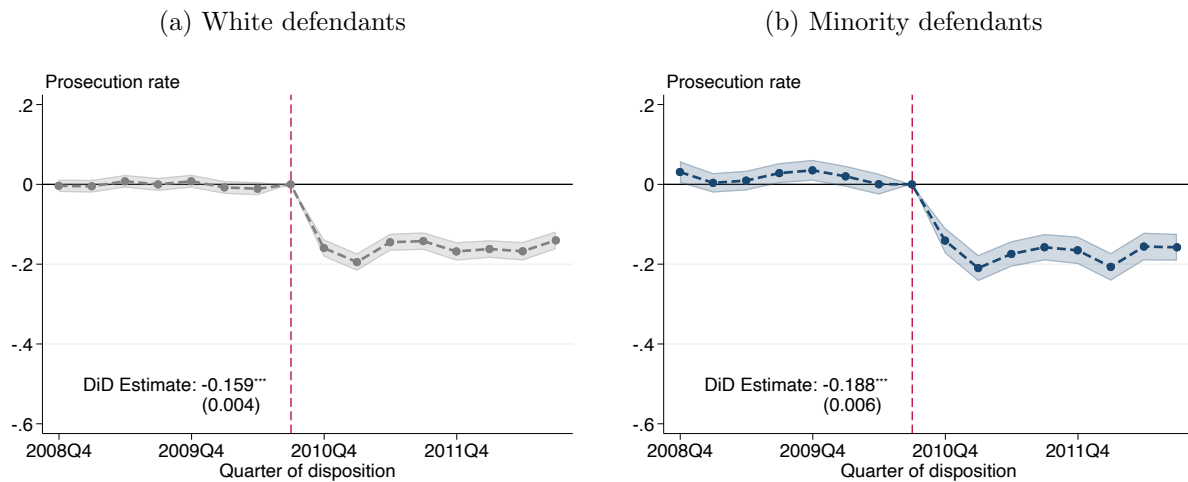


Note: This figure presents bounds on the average difference in promotion rates, conditional on treated potential outcomes and baseline covariates, using the approach described in Section 3. The treatment is promotion and the treated outcome, denoted by $Y_i(1)$, is whether a student demonstrated any proficiency on both the Math and ELA M-STEP in 4th grade, as defined in Table A1. Baseline covariates include gender, special education status, English language learner status, race, school district characteristics, and neighborhood share with at least a BA. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Appendix B Additional results: Misdemeanor prosecution

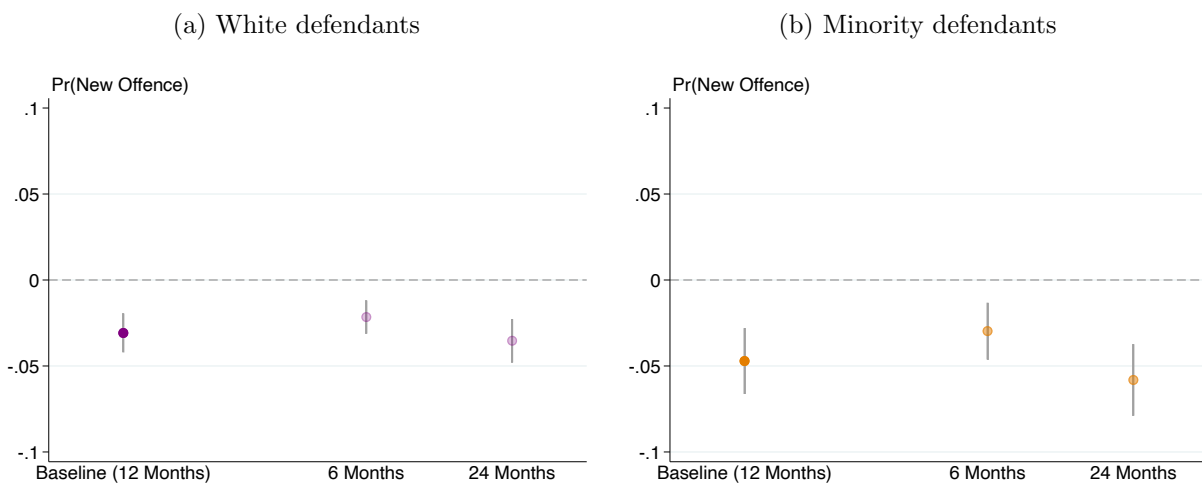
B.1 Robustness checks

Figure B1: Robustness of first stage to broader prosecution decision



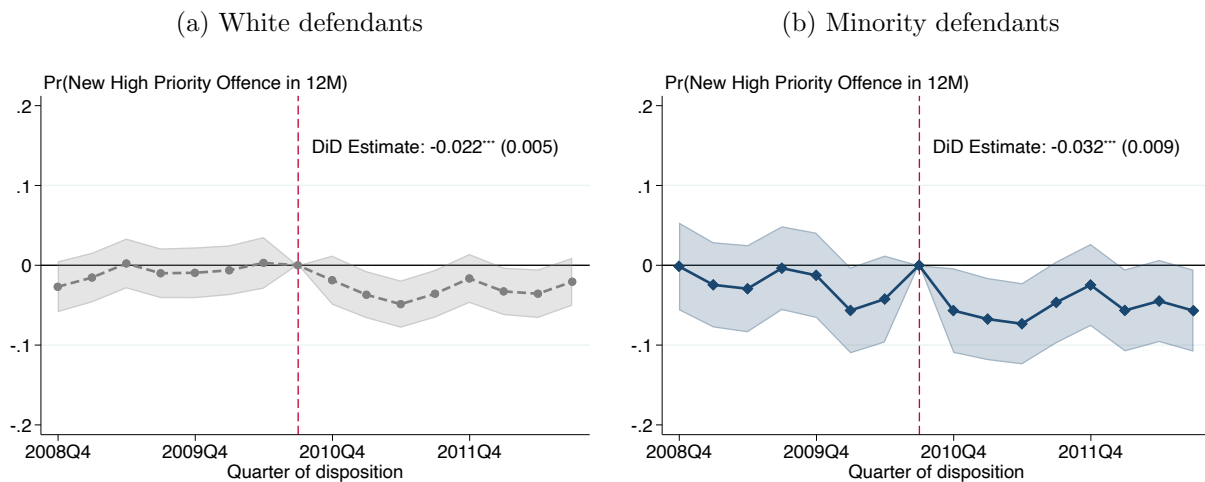
Note: This is identical to [Figure 6](#) except includes cases listed as ‘Dismissed’ but with a fine as part of the sentence as prosecuted.

Figure B2: Impact of King County budget reform on re-offence rates over different time horizons



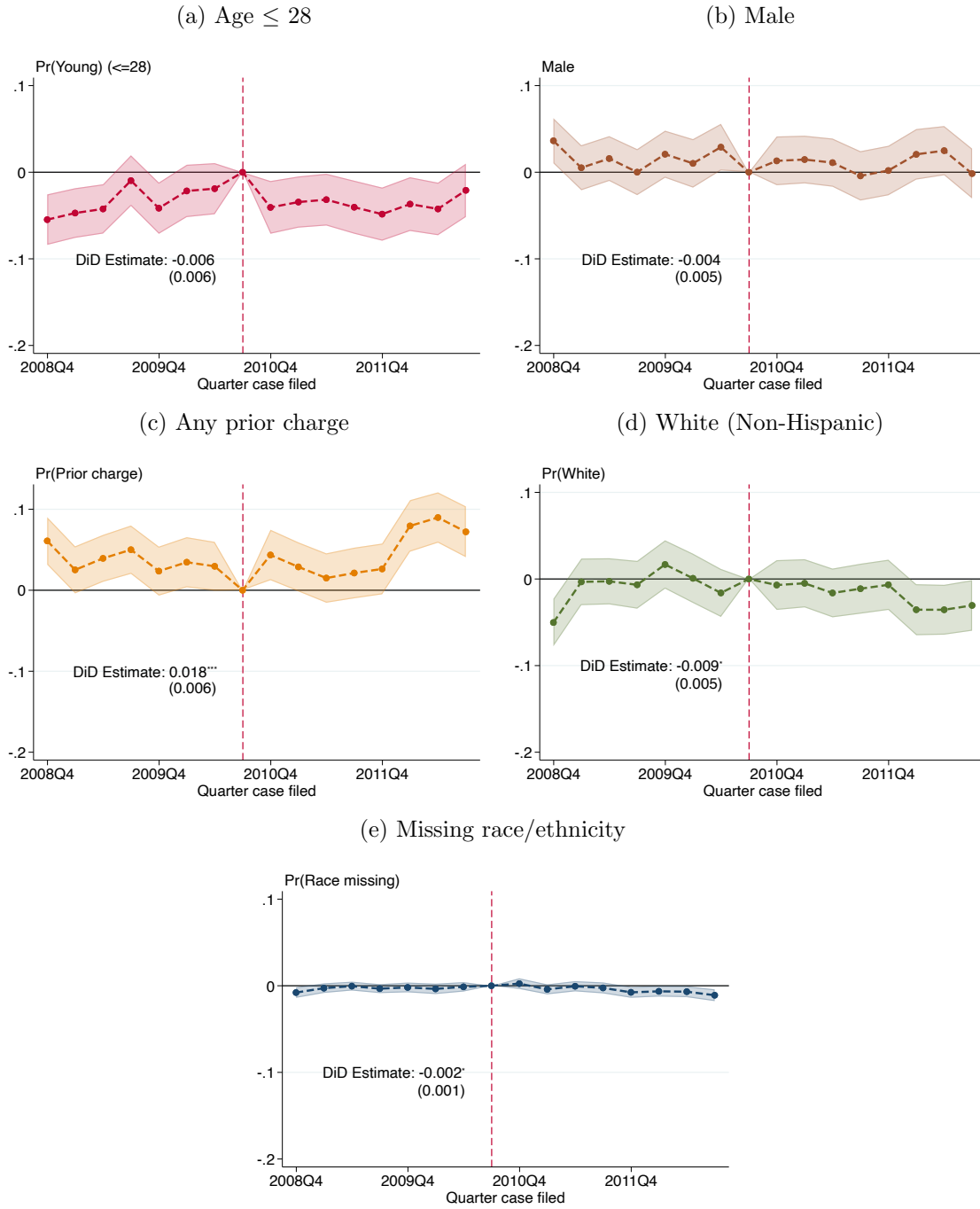
Note: Each Panel presents DiD estimates as reported in [Figure 7](#), but for different amounts of time after disposition.

Figure B3: Impact of King County budget reform on ‘high-priority’ re-offence within one year



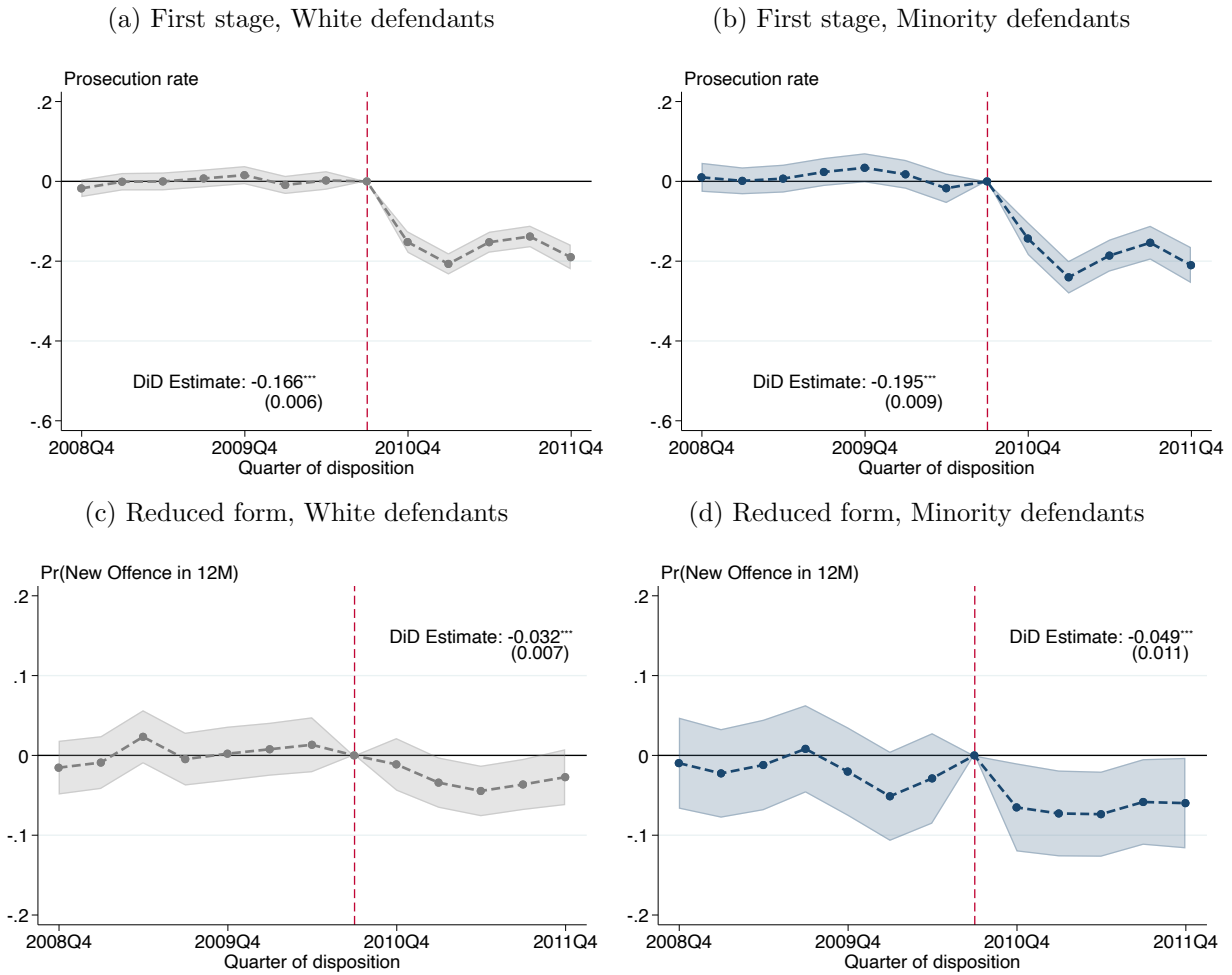
Note: Each Panel presents event study estimates investigating the impact of the King County budget reform. The re-offence outcome includes new ‘high-priority’ offences anywhere in Washington State. ‘High-priority’ offences are those that are not associated with charges that were commonly dismissed in the 2 quarters after the budget reform. Sample includes all misdemeanor defendants, as described in Table 2. ‘DiD Estimate’ pools the coefficients on relative time indicators and estimates $Y_{igt} = \alpha + \delta_1 \mathbb{I}[\text{King County}] + \delta_2 \text{Post}_i + \beta^{DD} \mathbb{I}[\text{King County}] \times \text{Post}_i + \epsilon_{igt}$, where $\text{Post}_i = 1$ if the case is disposed of on or after September 28, 2010, when the budget reform was announced. 95% confidence intervals are constructed using heteroscedasticity-robust standard errors.

Figure B4: Impact of budget reform on caseload characteristics



Note: Each Panel presents event study estimates investigating the impact of the King County budget reform. ‘DiD estimate’ is β^{DD} from $X_{igt} = \alpha + \delta_1 \mathbb{I}[\text{King County}] + \delta_2 Post_i + \beta^{DD} \mathbb{I}[\text{King County}] \times Post_i + \epsilon_{igt}$, where $Post_i = 1$ if the case is filed on or after September 28, 2010, when the budget reform was announced. X_{igt} denotes a baseline characteristic. ‘Young’ includes defendants who less than 29 years old at disposition and ‘Any Prior’ is an indicator for whether an individual has been previously charged with an offence in Washington. 95% confidence intervals constructed with heteroscedasticity-robust standard errors.

Figure B5: Robustness of first stage and reduced form to excluding post-SPD investigation period



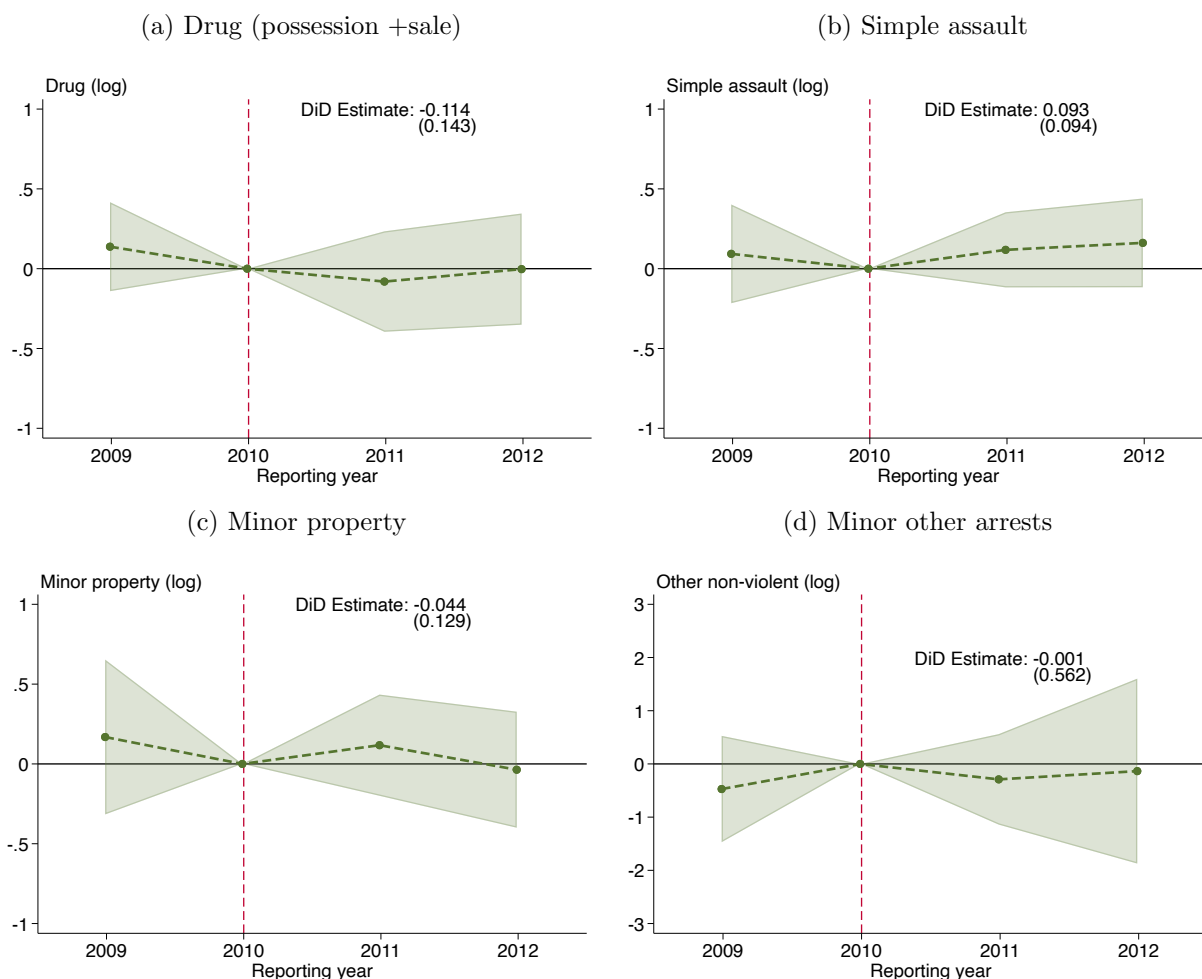
Note: This reproduces Figure 6 (Panels (a) and (b)) and Figure 7 (Panels (c) and (d)), excluding cases disposed after December 11, 2011.

Figure B6: Impact of budget reform on police employment



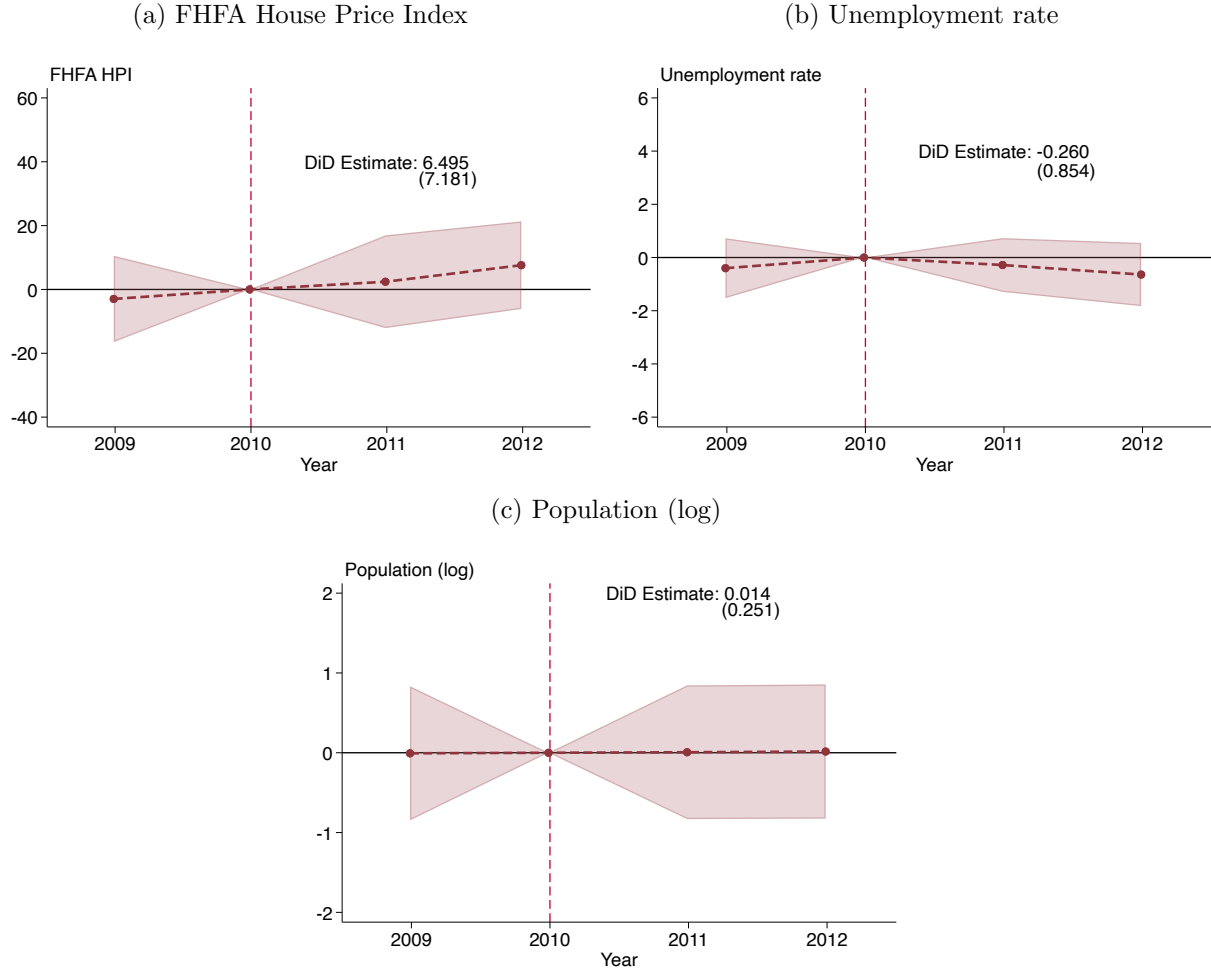
Note: Estimates generated using annual average employment from the Law Enforcement Officers Killed and Assaulted (LEOKA) data (Kaplan, 2023). This specification is similar to those used to estimate the first stage and reduced form except for the inclusion of originating agency (ORI) fixed effects and usage of the average pre-reform county-level population as weights. ORIs in areas with an annual population average less than 1,000 or with zero employment counts throughout the sample are excluded. 95% confidence intervals are constructed using standard errors clustered at the ORI level.

Figure B7: Impact of budget reform on arrests



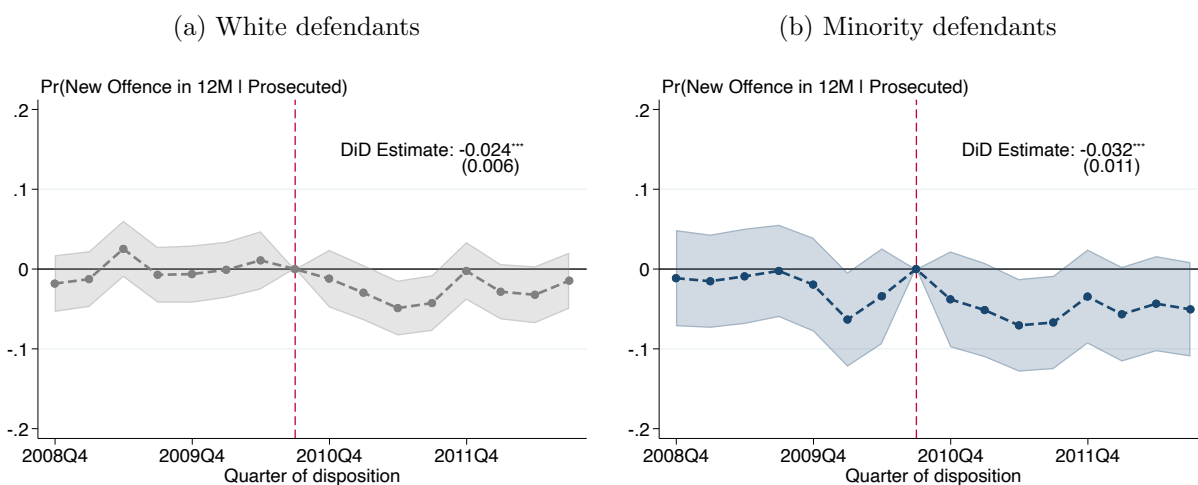
Note: Estimates generated using annual number of arrests from Uniform Crime Report (UCR) data (Kaplan, 2023). ‘Minor property’ arrests include arrests for stolen property, fraud, forgery, and theft. This specification is similar to those used to estimate the first stage and reduced form except for the inclusion of originating agency (ORI) fixed effects and usage of the average pre-reform county-level population as weights. ORIs in areas with an annual population average less than 1,000, with limited time coverage between 2008-13 or with zero arrest counts in a given year are excluded. 95% confidence intervals are constructed using standard errors clustered at the ORI level.

Figure B8: Impact of budget reform on non-crime economic factors



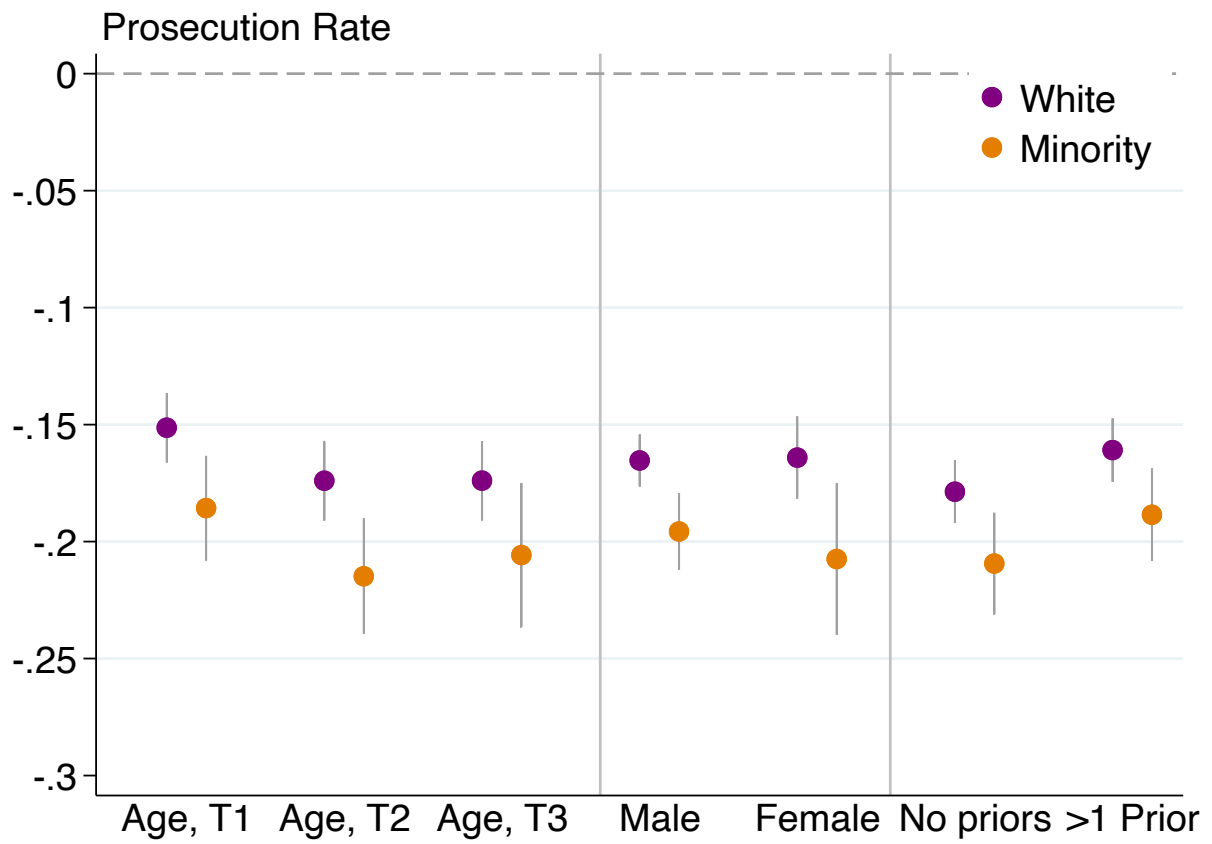
Note: The specification used here is similar to those used to estimate the first stage and reduced form except for the usage of the average pre-reform county-level population as weights in Panels (a) and (b). County-level data on house prices are from the Federal Housing Finance Agency (FHFA), unemployment rates from Local Area Unemployment Statistics and population counts are from the Census Bureau. 95% confidence intervals are constructed using heteroskedasticity-robust standard errors.

Figure B9: Impact of King County budget reform on re-offence within one year, only prosecuted defendants



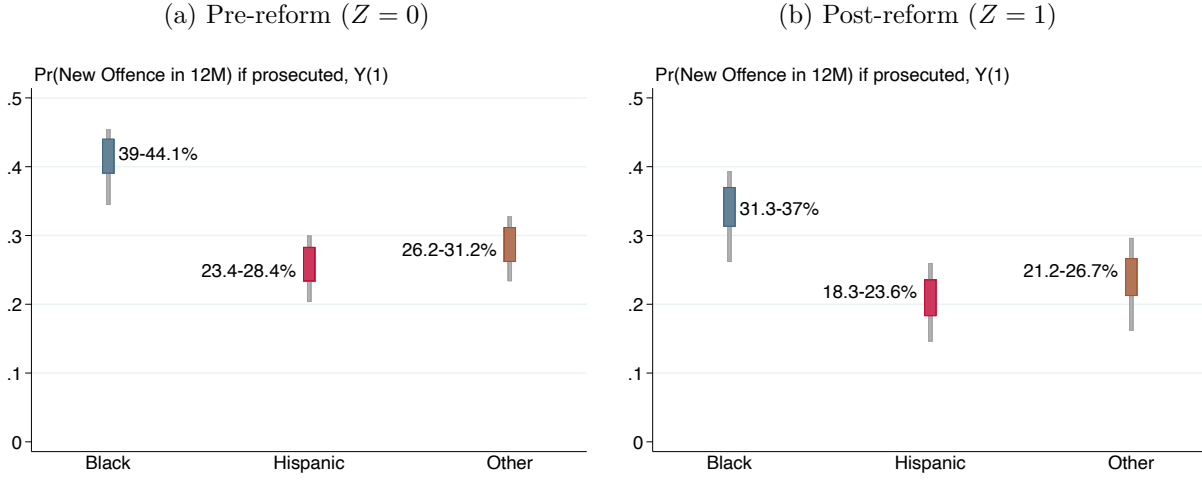
Note: Each Panel is identical to [Figure 7](#), except the sample only includes prosecuted defendants.

Figure B10: Testing for defiers: First stage by subgroup



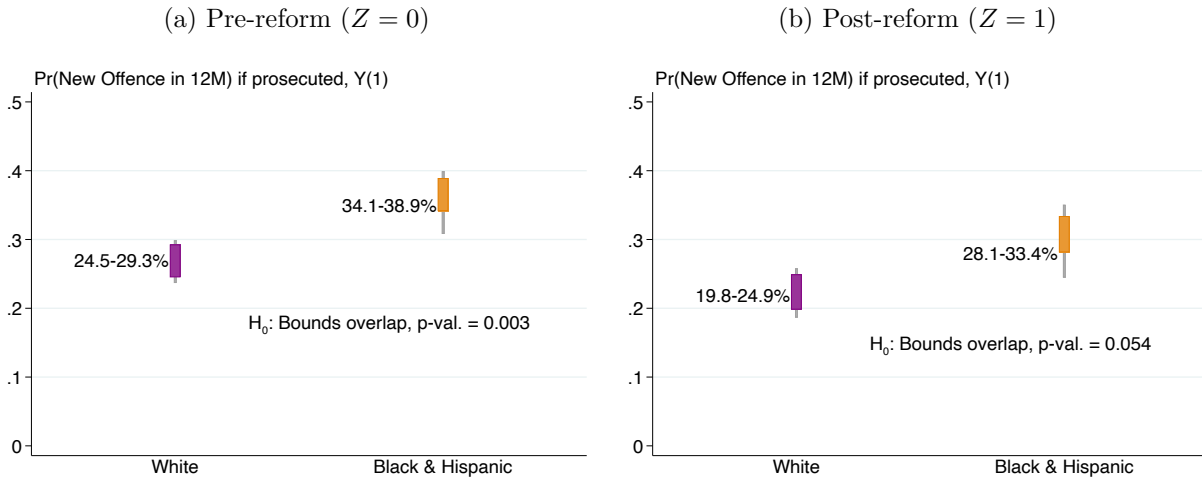
Note: Each Panel presents DiD estimates as reported in [Figure 6](#), but for different covariate subgroups. Age at disposition is split into terciles, represented by T1–T3.

Figure B11: Average outcomes if prosecuted, disaggregated by minority subgroup



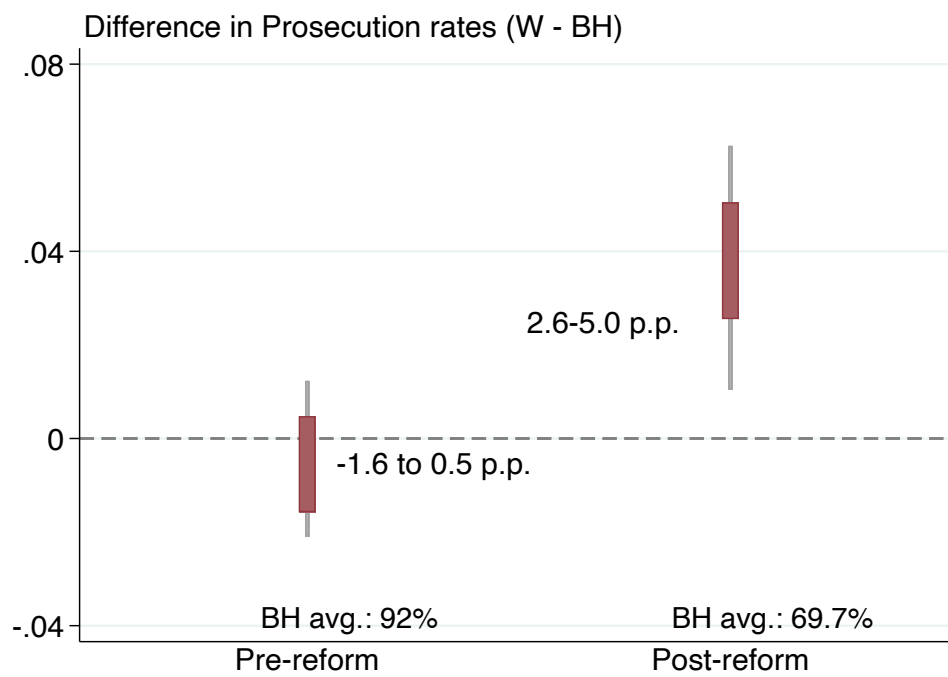
Note: This figure presents bounds on the average treated outcome obtained using the approach described in Section 3, separately by time period and subgroups within minority defendants. The treatment is prosecution and the treated outcome, $Y_i(1)$, is whether an individual re-offends one year after disposition, if prosecuted. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Figure B12: Average outcomes if prosecuted: White vs. Black/Hispanic defendants



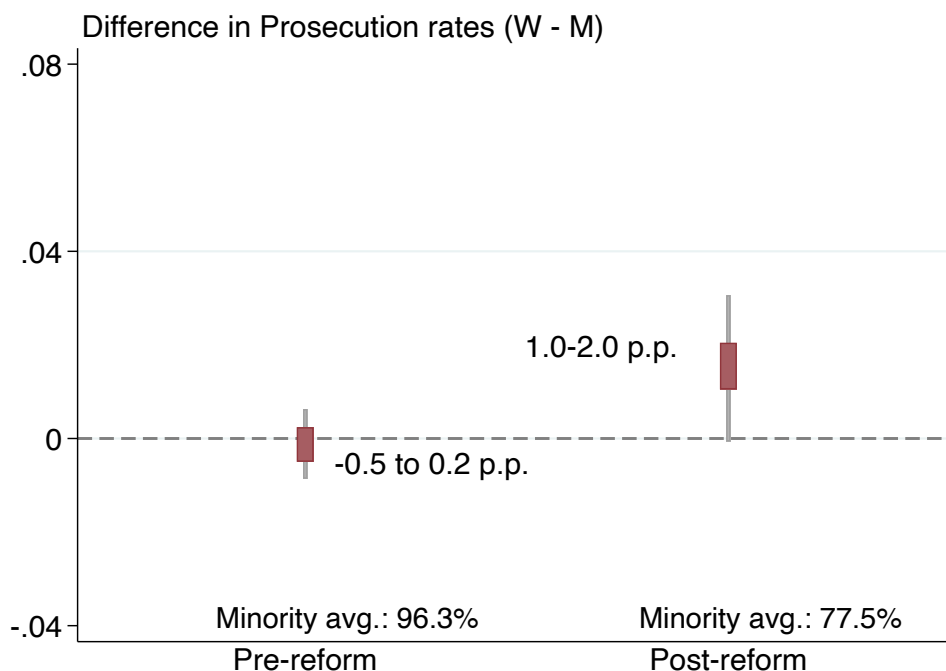
Note: This figure presents bounds on the average treated outcome obtained using the approach described in Section 3, separately by race and time period. The treatment is prosecution and the treated outcome, $Y_i(1)$, is whether an individual re-offends one year after disposition, if prosecuted. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap. The p-value is from a formal bootstrapped test of whether the identified sets overlap, described in Appendix C.5.

Figure B13: Racial prosecution gap conditional on prosecuted outcome: White vs. Black/Hispanic



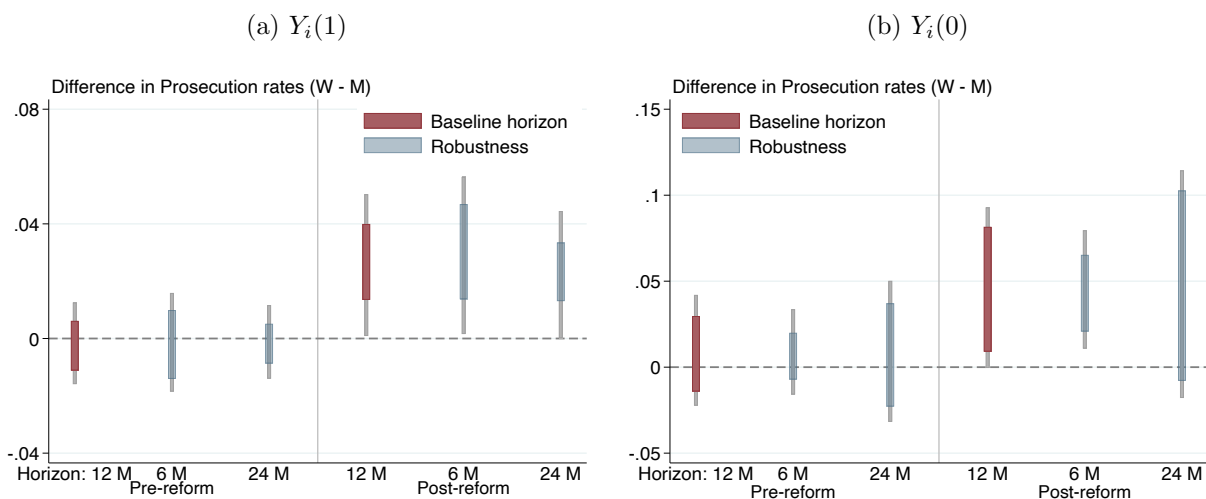
Note: This figure presents bounds on the average difference in prosecution rates in each time period, conditional on prosecuted outcomes, $Y_i(1)$, using the approach described in Section 3. $Y_i(1)$ is whether an individual re-offends one year after disposition, if prosecuted. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Figure B14: Racial prosecution gap cond. on prosecuted outcome: Broad prosecution definition



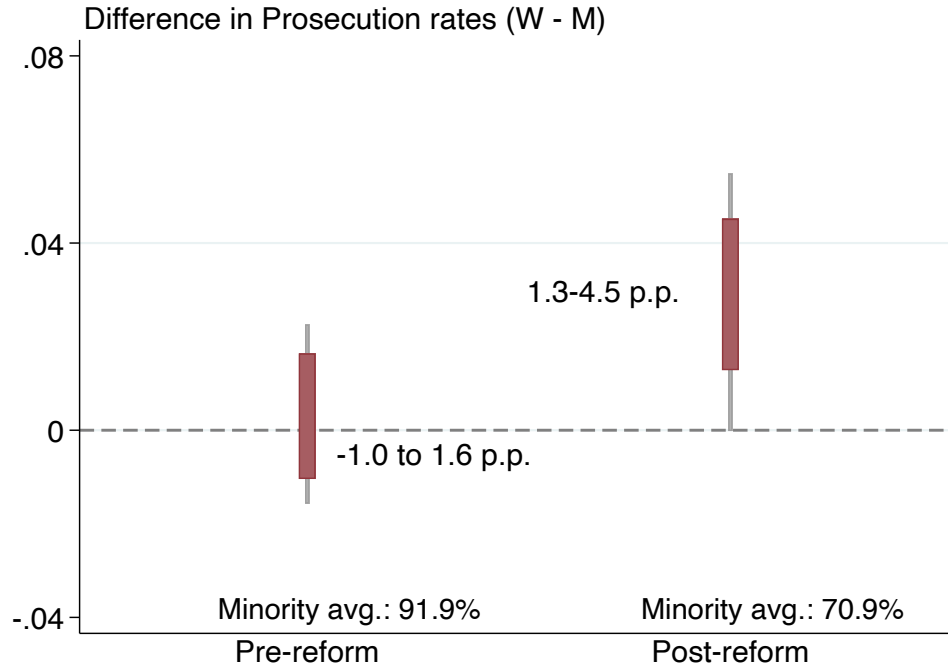
Note: This figure presents bounds on the average difference in prosecution rates in each time period, conditional on prosecuted outcomes, $Y_i(1)$, using the approach described in Section 3. $Y_i(1)$ is whether an individual re-offends one year after disposition, if prosecuted. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Figure B15: Racial prosecution gap conditional on potential outcomes: Variation by outcome horizon



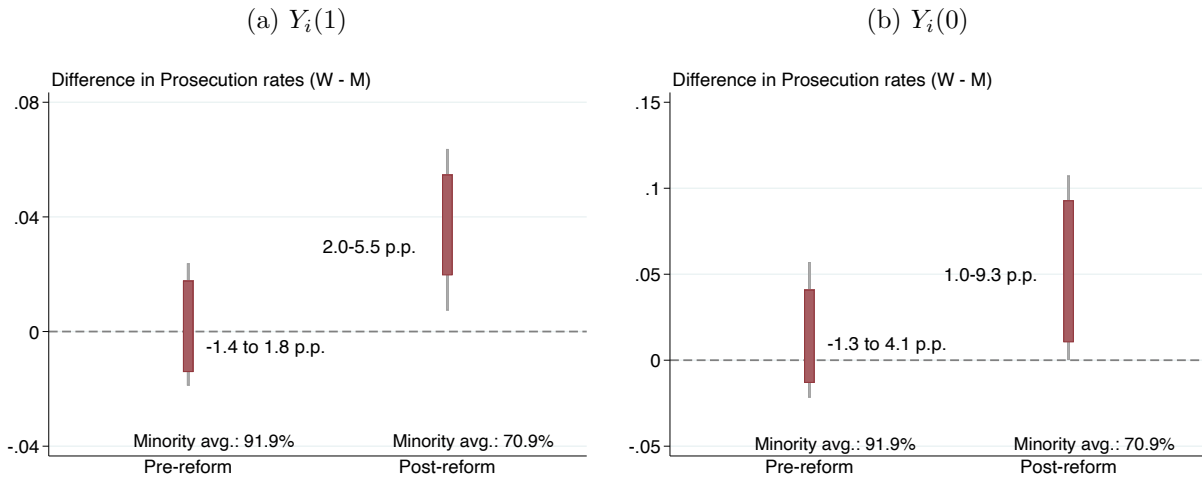
Note: This figure presents bounds on the average difference in prosecution rates in each time period, conditional on potential outcomes, $Y_i(\cdot)$. $Y_i(\cdot)$ is whether an individual re-offends within the amount of time labelled on the x-axis after disposition, if prosecuted (Panel (a)) or if dismissed (Panel (b)). Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Figure B16: Racial prosecution gap conditional on prosecuted outcome: Worst case bounds



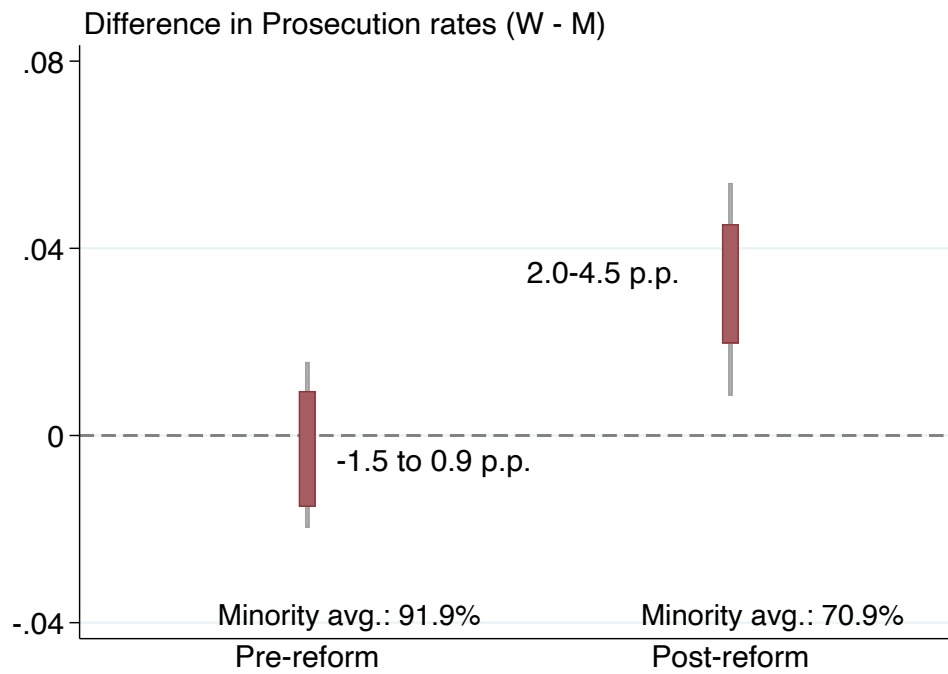
Note: This figure presents bounds on the average difference in prosecution rates in each time period, conditional on prosecuted outcomes, $Y_i(1)$, assuming that never takers' outcomes are bounded between 0 and 1. $Y_i(1)$ is whether an individual re-offends one year after disposition, if prosecuted. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Figure B17: Racial prosecution gap, conditional on potential outcomes and baseline covariates



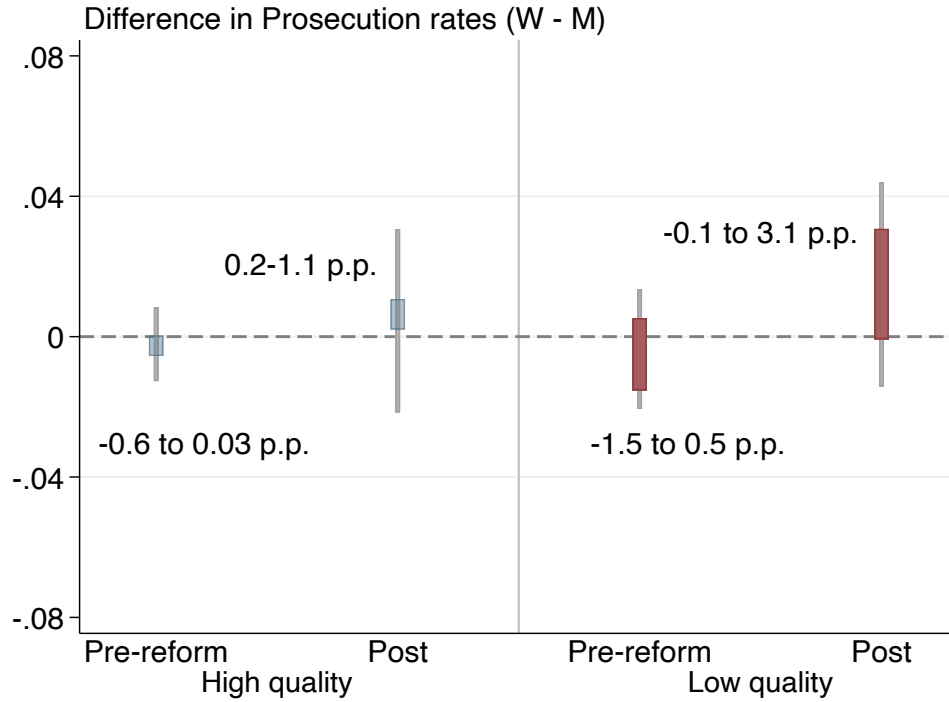
Note: This figure presents bounds on the average difference in prosecution rates in each time period, conditional on potential outcomes, $Y_i(\cdot)$, and baseline covariates. $Y_i(\cdot)$ is whether an individual re-offends one year after disposition if prosecuted (Panel (a)) or if dismissed (Panel (b)). Baseline covariates include age, gender, criminal history (convictions and charges), and court fixed effects. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Figure B18: Racial prosecution gap conditional on receiving any punishment if prosecuted



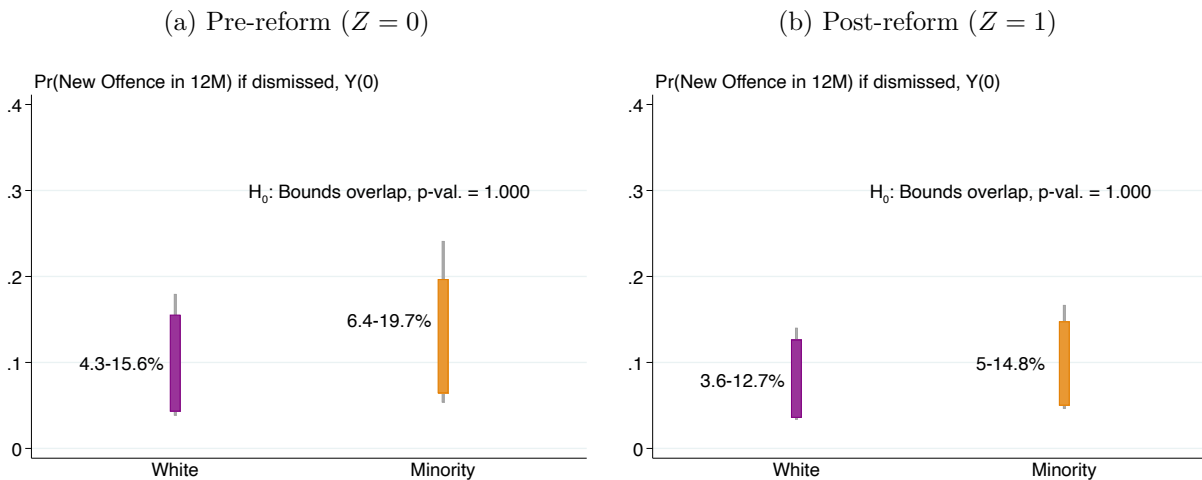
Note: This figure presents bounds on the average difference in prosecution rates in each time period, conditional on case outcome if prosecuted, $Y_i(1)$, using the approach described in Section 3. $Y_i(1)$ is whether an individual is sentenced to any non-fine punishment, if prosecuted. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Figure B19: Racial prosecution gap, by proxy for case quality: excluding cases with missing charges



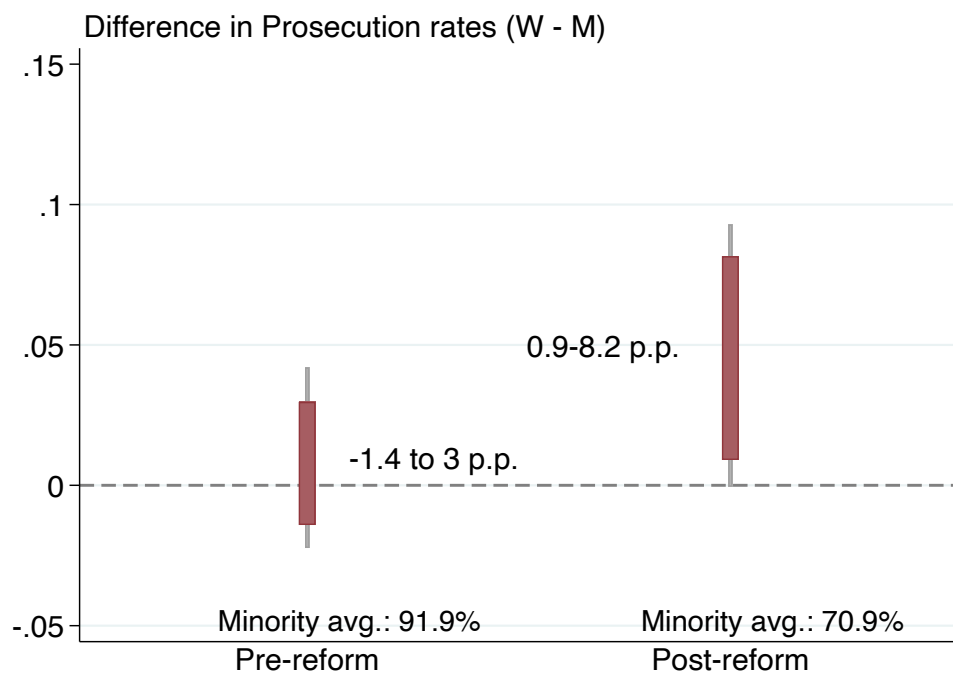
Note: This figure presents bounds on the average difference in prosecution rates in each time period, conditional on prosecuted outcomes, $Y_i(1)$, using the approach described in Section 3. $Y_i(1)$ is whether an individual re-offends one year after disposition, if prosecuted. ‘High quality’/‘Low quality’ offences are those with an above/below median share of charges that result in any punishment using pre-reform data, excluding cases where no charges are listed. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Figure B20: Average outcomes if dismissed, $E[Y_{it}(0)|R_i = r]$



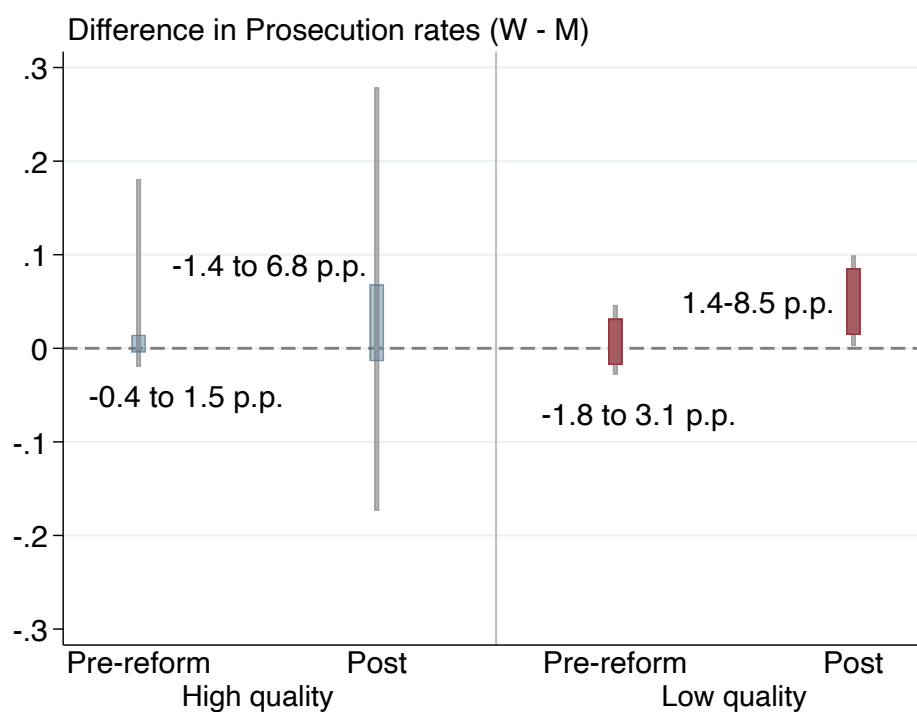
Note: This figure presents bounds on the average untreated outcome obtained using the approach described in Section 3, separately by race and time period. The treatment is prosecution and the untreated outcome, $Y_i(0)$, is whether an individual re-offends one year after disposition, if dismissed. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Figure B21: Racial prosecution gap conditional on dismissed outcome



Note: This figure presents bounds on the average difference in prosecution rates in each time period, conditional on dismissed outcomes, $Y_i(0)$, using the approach described in Section 3. $Y_i(0)$ is whether an individual re-offends one year after disposition, if dismissed. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

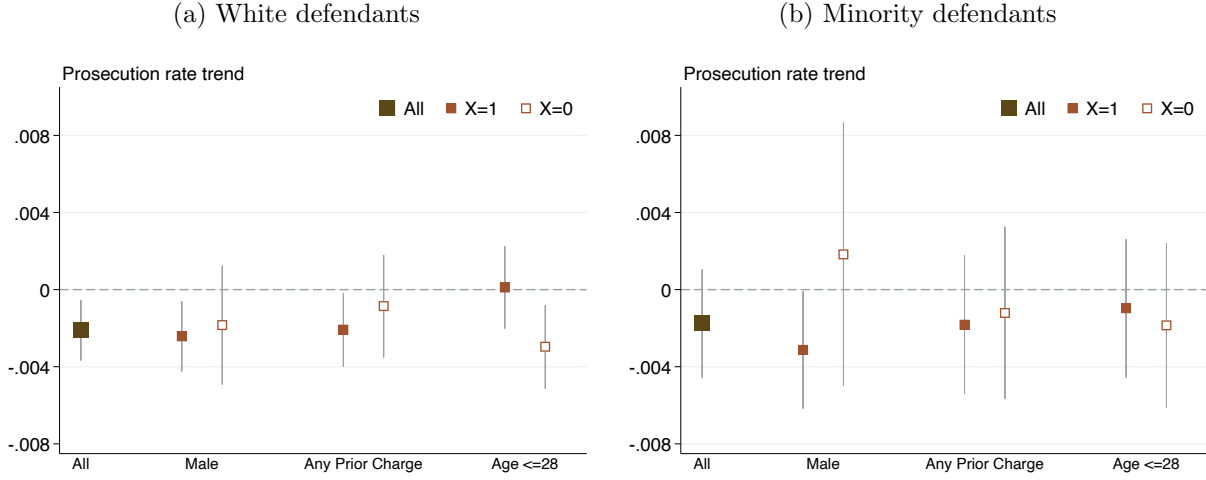
Figure B22: Racial prosecution gap, by proxy for case quality: conditional on dismissed outcome



Note: This figure presents bounds on the average difference in prosecution rates in each time period, conditional on dismissed outcomes, $Y_i(0)$, using the approach described in Section 3. $Y_i(0)$ is whether an individual re-offends one year after disposition, if dismissed. ‘High quality’/‘Low quality’ offences are those with an above/below median share of charges that result in any punishment using pre-reform data. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

B.2 Empirically validating DiD adjustment assumptions

Figure B23: Prosecution trends in adjacent counties, by covariate subgroup



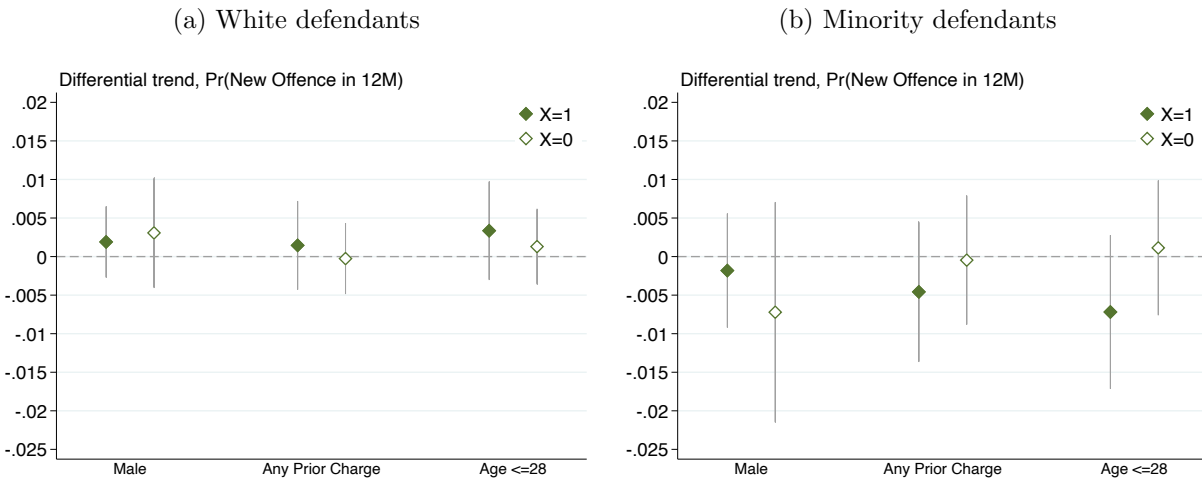
Note: Each coefficient is from estimating a linear regression of prosecution on a linear quarterly trend using pre-period data on individuals from a given subgroup (denoted by X) in the counties adjacent to King County. The estimate labelled ‘All’ reproduces the overall trend estimate from Figure 9. 95% confidence intervals are constructed using heteroscedasticity-robust standard errors.

Figures B24–B27 estimate the following regression; Figure B24 and Figure B26 test hypotheses 1) and 2), while Figure B25 and Figure B27 test hypotheses 3) and 4):

$$Y_{itg} = \beta_1 t + \beta_2 X_i + \beta_3 \text{King County} + \delta_1 X_i \times t + \delta_2 X_i \times \text{King County} + \delta_3 t \times \text{King County} + \delta_4 X_i \times t \times \text{King County} + \varepsilon_{igt}$$

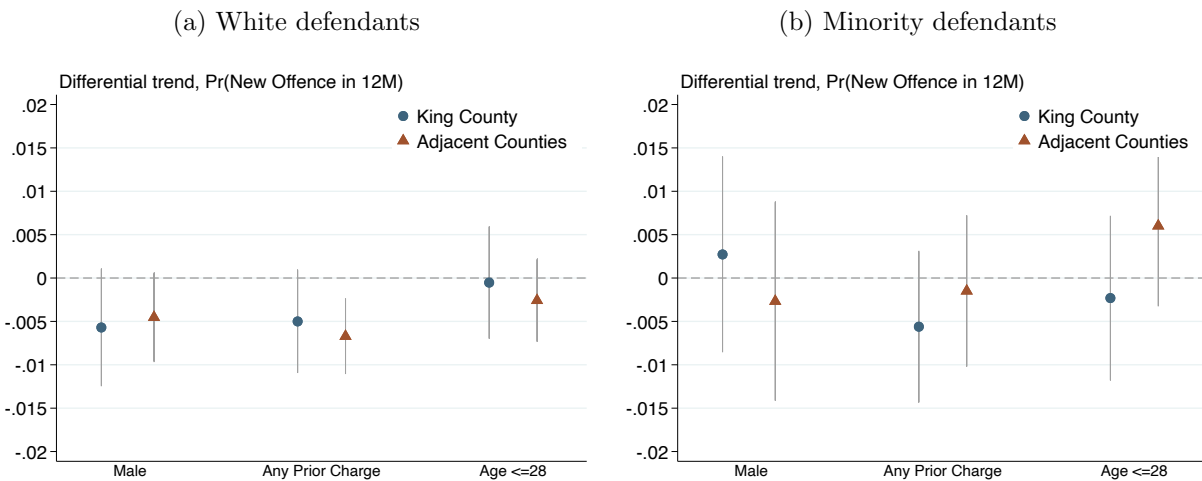
- 1) $X = 0 : H_0 : \delta_3 = 0$
- 2) $X = 1 : H_0 : \delta_3 + \delta_4 = 0$
- 3) King County $H_0 : \delta_1 + \delta_4 = 0$
- 4) Adjacent $H_0 : \delta_1 = 0$

Figure B24: Testing for differential trends in group-specific treated outcomes across counties



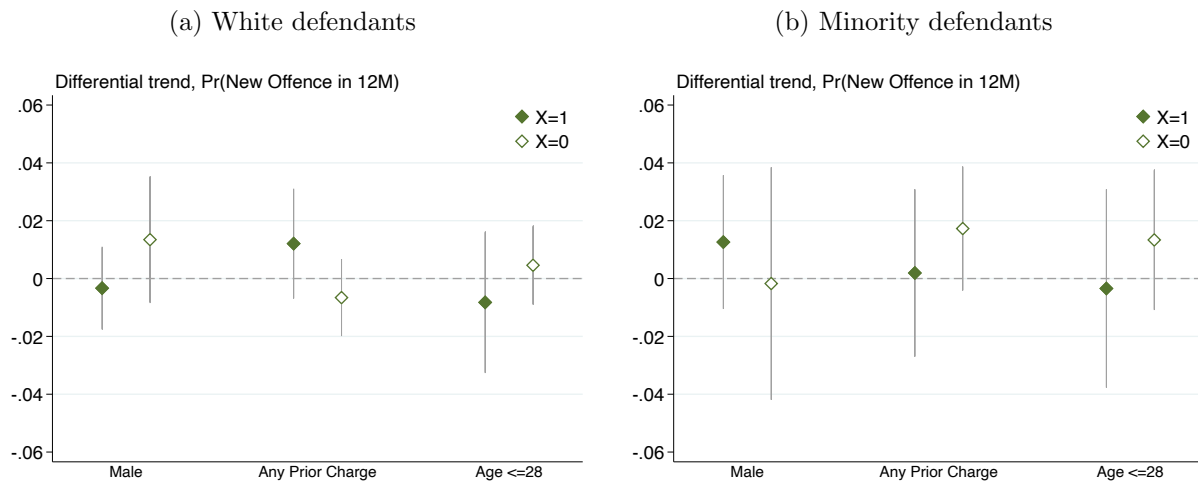
Note: Each coefficient is an estimate of the difference in pre-period trends in treated outcomes (outcomes if prosecuted) across counties, for a given covariate value. The sample only includes individuals who are prosecuted prior to the budget reform. For example, the first solid green diamond in Panel a) represents the difference in trends in treated outcomes for white male defendants between King County and other counties. 95% confidence intervals are constructed using heteroscedasticity-robust standard errors.

Figure B25: Testing for differential trends in treated outcomes between groups, within each county



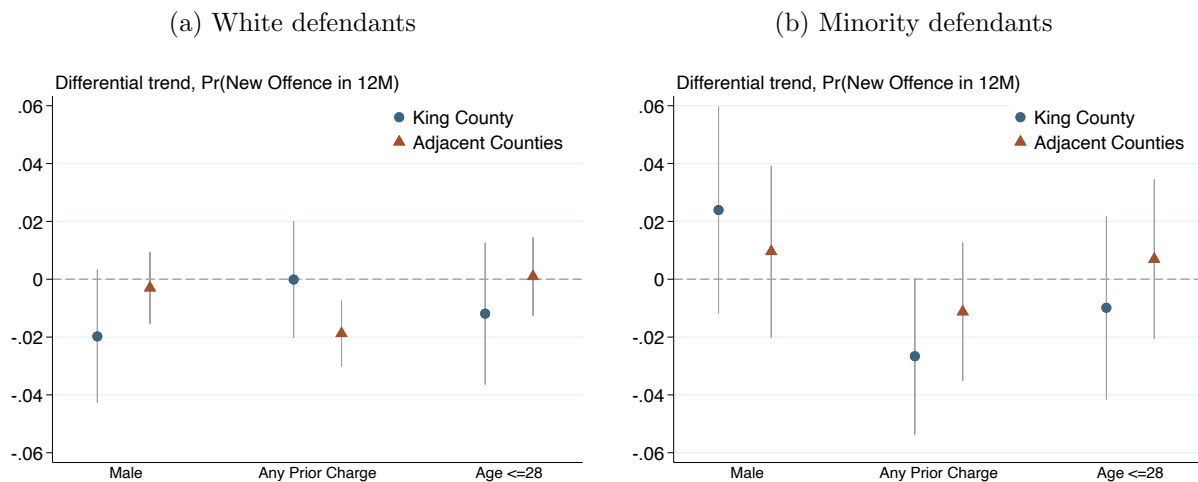
Note: Each coefficient is an estimate of the difference in pre-period trends in treated outcomes (outcomes if prosecuted) across covariate groups, within a given county. The sample only includes individuals who are prosecuted prior to the budget reform. For example, the first blue circle in Panel a) represents the difference in trends in treated outcomes between white male and female defendants in King County. 95% confidence intervals are constructed using heteroscedasticity-robust standard errors.

Figure B26: Testing for differential trends in group-specific untreated outcomes across counties



Note: Each coefficient is an estimate of the difference in pre-period trends in untreated outcomes (outcomes if not prosecuted) across counties, for a given covariate value. The sample only includes individuals who are dismissed prior to the budget reform. For example, the first green diamond in Panel a) represents the difference in trends in untreated outcomes for white male defendants between King County and other counties. 95% confidence intervals are constructed using heteroscedasticity-robust standard errors.

Figure B27: Testing for differential trends in untreated outcomes between groups, within each county

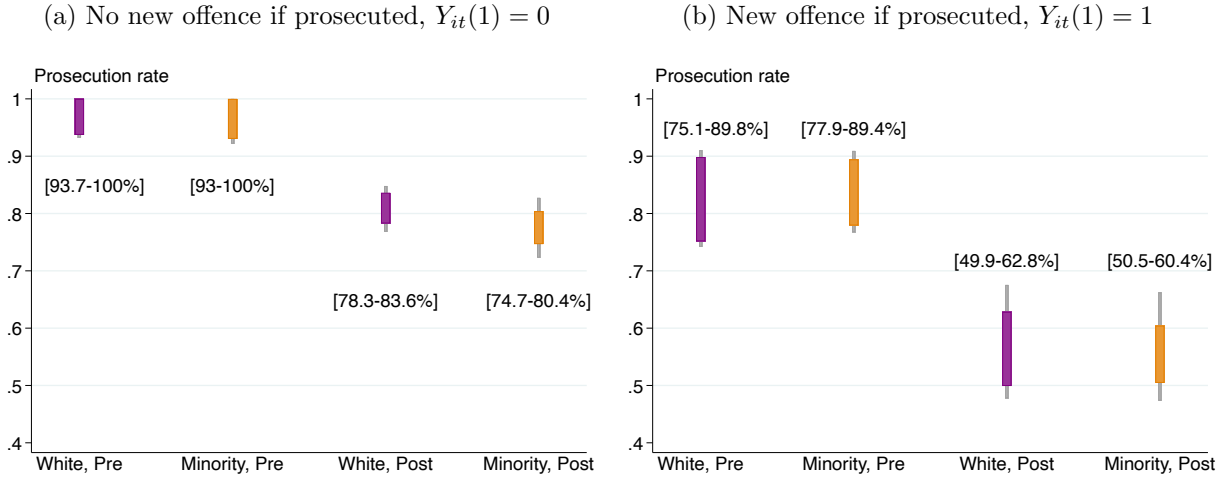


Note: Each coefficient is an estimate of the difference in pre-period trends in untreated outcomes (outcomes if not prosecuted) across covariate groups, within a given County. The sample only includes individuals who are dismissed prior to the budget reform. For example, the first blue circle in Panel a) represents the difference in trends in untreated outcomes between white male and female defendants in King County. 95% confidence intervals are constructed using heteroscedasticity-robust standard errors.

B.3 How prosecution varies by potential outcomes & treatment effects

Here we discuss how the race-, time period- and potential outcome-specific prosecution rates, which are the building blocks of our discrimination estimates (see Equation 3), provide some evidence on prosecutorial decision-making in this setting. We start by conditioning on re-offence outcomes if prosecuted—Figure B28 presents estimates of prosecution rates separately by race, time period, and whether individuals **would not** commit a new offence if prosecuted ($Y_i(1) = 0$) or **would** commit a new offence if prosecuted ($Y_i(1) = 1$).⁵³

Figure B28: Prosecution rates conditional on prosecuted outcome (π_{zry})



Note: This figure presents bounds on the average prosecution rates for each race group and time period, conditional on prosecuted outcomes, $Y_i(1)$, using the approach described in Section 3. $Y_i(1)$ is whether an individual re-offends one year after disposition, if prosecuted. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Panel a) of Figure B28 focuses on prosecution rates for individuals who would not commit a new offence if prosecuted ($Y_i(1) = 0$). The first two estimates are prosecution rates before the reform (“Pre”) and the following two estimates are prosecution rates after the reform (“Post”). We see that among defendants who would not commit a new offence if prosecuted, prosecution rates before the reform are similar across race. However, we see suggestions that the drop after the reform is greater for minority defendants. The prosecution rate for white defendants falls from between 93.7%–100% to between 78.3%–83.6%. The prosecution rate for minority defendants falls from a similar pre-reform level to between 74.7%–80.4%. This is not the case in Panel b), which displays the analogous prosecution rates for individuals who would commit a new offence if they were prosecuted ($Y_i(1) = 1$). Here, prosecution falls evenly by racial group. These patterns suggest

⁵³Using the bounds on prosecution rates in Figure B28 to construct average discrimination conditional on prosecuted outcomes following Equation 3 may not yield the same estimates as the main analysis, which plugs the bounds for the average prosecuted outcomes directly into the expression for our main object of interest, the average period-specific racial gap, Δ_z . The two approaches involve computing averages and minima/maxima in different orders, which will produce numerically different results because minimum/maximum are not linear functions. See Section 3 for additional details.

that the post-reform racial gap that we see in [Figure 12](#) is concentrated among individuals who would not commit an offence if prosecuted.

These patterns also provide suggestive evidence that prosecution is targeted based on defendants' potential outcomes. Prosecutors are less likely to prosecute individuals who **would** commit a new offence if prosecuted. To see this, compare the pre-reform prosecution rates for defendants who **would not** commit an offence if prosecuted (first two bounds in Panel (a)) to the pre-reform prosecution rates for those who **would** commit an offence if prosecuted (first two bounds in Panel (b)). Approximately 93%–100% of defendants of either racial group who would **would not** commit a new offence if prosecuted are being prosecuted, while the same is true for between 75.1%–89.8% of defendants who **would** commit a new offence if prosecuted. This behavior is true in the post-reform period as well. After the reform, between 74.7%–83.6% of defendants of either race who would **would not** commit a new offence if prosecuted are being prosecuted, while the same is true for between 49.9%–62.8% of defendants who **would** commit a new offence if prosecuted.

This pattern is suggestive evidence that prosecutors are less willing to prosecute cases that would result in detrimental outcomes for defendants. To investigate this further, we compare prosecution rates for defendants who would have had the same treatment effect of prosecution, following [Equation 8](#). As discussed in [Section 3](#), this exercise 1) requires restricting the distribution of treatment effects and 2) also bounding the treatment effect on the treated. To address 1), we assume that prosecution can either induce more crime (i.e., be criminogenic) or have no impact.

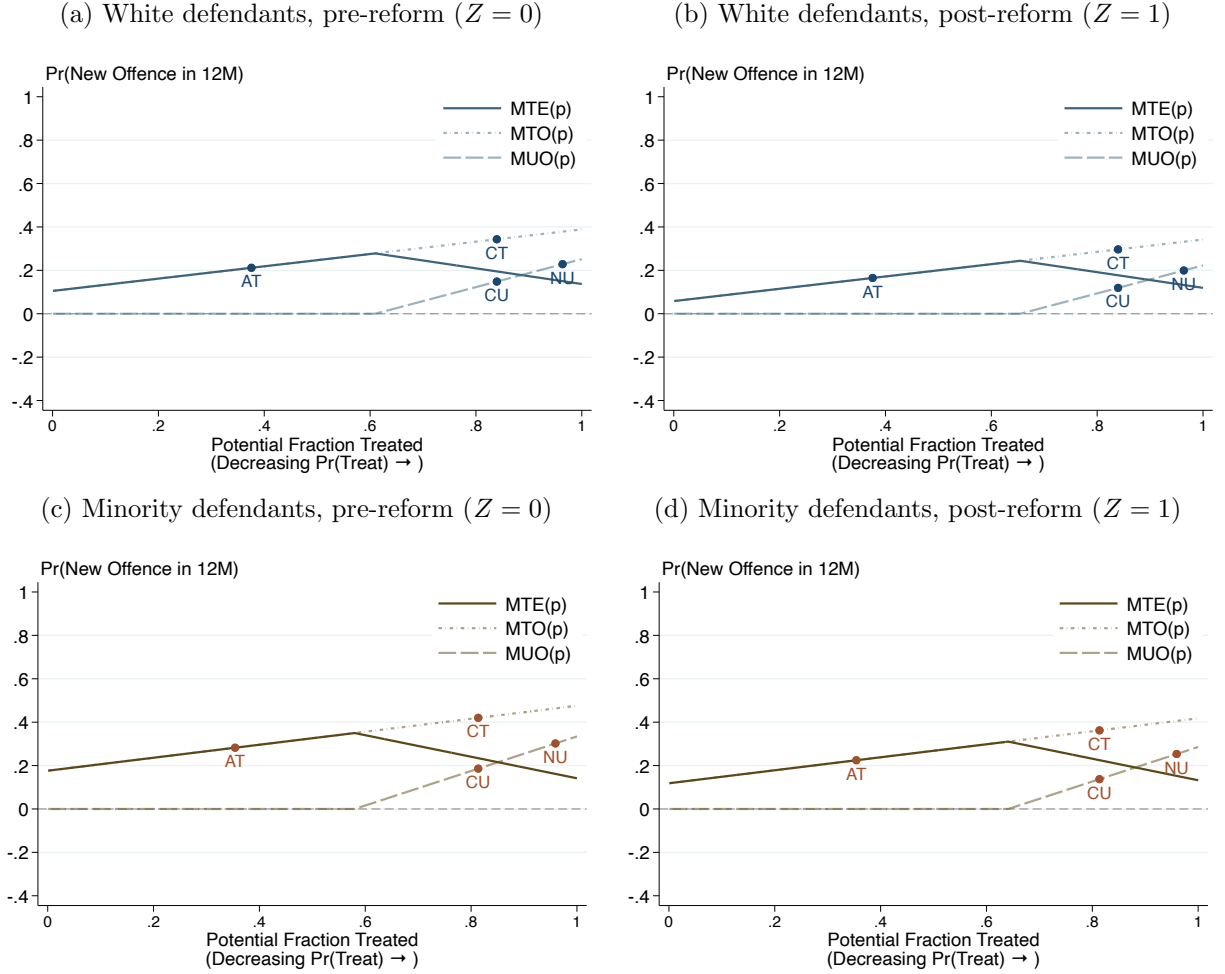
This assumption rules out that prosecution can reduce future criminal legal involvement, which is almost surely not satisfied at the individual-level in the population. However, we think that violations of this assumption are likely small in this context. The literature on the deterrence effects of punishment typically finds small effects that mostly operate through incapacitation (Chalfin and McCrary, 2017), which is uncommon in our context of misdemeanor cases in Washington—only 6% of prosecuted in our sample involve jail time, and average jail sentence of such cases is 40 days.

We also probe this assumption by examining the range of magnitudes that treatment effects in our context take. We do this by identifying the marginal treatment effect (MTE) function under an auxiliary assumption of linearity in the relationship between likelihood of prosecution and potential outcomes.⁵⁴ [Figure B29](#) shows that the MTE function in our context is always positive: i.e., prosecution increases future criminal legal involvement for most people in this setting. Finally, since the treatment effect of prosecution is a function of the re-offence outcomes if dismissed, we need to make the same assumption (**A3**) of parallel trends in re-offence outcomes if dismissed as in [Section 5.4](#) to account for time trends in re-offence outcomes if dismissed.

Under these assumptions, [Figure B30](#) shows bounds of prosecution rates separately where prosecution has no impact (Panel (a)) and where prosecution would be detrimental by inducing future criminal legal involvement (Panel (b)). The bounds are much wider here, given the additional objects that we have to bound. Before the reform, between 90.3%–98.1% of cases where prosecution

⁵⁴As described in the main text, this assumption results in values for the marginal treated and untreated outcomes that lie outside the support of the binary potential outcomes. For this exercise, we constrain values of the marginal treated and untreated outcomes that are negative to be zero.

Figure B29: Marginal treatment response functions for prosecution



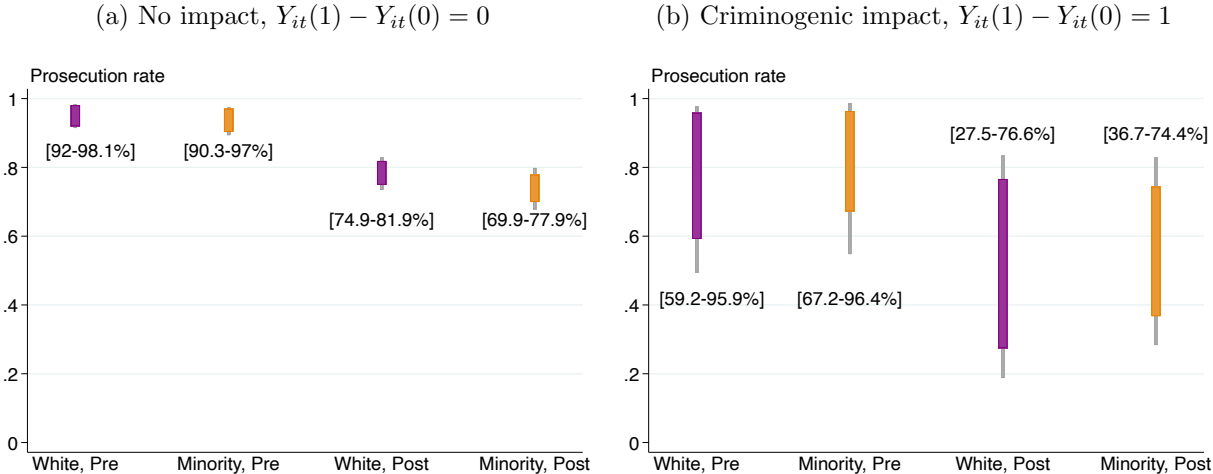
Note: This figure displays the marginal treatment response functions (Mogstad, Santos, and Torgovitsky, 2018) for prosecution. The treatment is prosecution and the outcome is whether an individual re-offends one year after disposition. $MTO(p)$, $MUO(p)$ and $MTE(p)$ represent the marginal treated outcome, marginal untreated outcome, and marginal treatment effect functions respectively. These are identified by assuming a linear relationship between potential outcomes of always takers ('AT'), compliers ('CT'), and never takers ('NT') and their treatment propensities. Since $MTO(p)$ and $MUO(p)$ should lie within the support of the outcome (i.e., between 0 and 1), we constrain values of $MTO(p)$ and $MUO(p)$ that are negative to be zero. Lower values of the x-axis denote individuals who are more likely to be prosecuted.

would have no impact on future criminal legal involvement were being prosecuted, while the same is true for between 59.2%–96.4% of cases where prosecution would be harmful. After the reform, between 69.9%–81.9% of cases where prosecution would have no impact on future criminal legal involvement were being prosecuted, while the same is true for between 27.5%–76.6% of cases where prosecution would be harmful.

Since these bounds overlap, we cannot rule out that the prosecution rates are the same between defendants who would and would not be harmed by prosecution. However, given that the region of overlap is small, we interpret this as suggestive evidence that prosecutors may have been less likely to prosecute cases where it is likely criminogenic. This type of targeting behavior is consistent

with recent work showing that prosecutors have some information about potential outcomes of cases and may choose not to prosecute defendants who may have higher risk of future contact with the criminal legal system (Agan, Doleac, and Harvey, 2023; Harrington, Murdock III, and Shaffer, 2023).

Figure B30: Prosecution rates conditional on treatment effect of prosecution (π_{zry})



Note: This figure presents bounds on the average prosecution rates for each race group and time period, conditional on the treatment effect of prosecution using the approach described in Section 3. The outcome is whether an individual re-offends one year after disposition. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Appendix C Methodological Details

C.1 Comparing change in Δ_z to observed change in disparity

Denote the change in discrimination due to a reform by $\Delta_{\text{change}} = \Delta_1 - \Delta_0$, following the definition of Δ_z in Definition 2. Consider a case with binary potential outcomes. The observational analogue to quantifying how an intervention affected group differences in treatment might be to compare the change in observed treatment rates for each group by estimating the regressions below and constructing the gap defined in Definition 3.

$$\begin{aligned} D_i &= \alpha_w + \beta_w Z_i + \varepsilon_{iw}, \text{ if } R_i = w \\ D_i &= \alpha_m + \beta_m Z_i + \varepsilon_{ib}, \text{ if } R_i = m \end{aligned}$$

Definition 3. Observed change in group treatment gaps due to policy reform (Δ_{obs})

$$\Delta_{\text{obs}} \equiv \beta_w - \beta_m, \text{ where } \beta_r = E[D_i|Z = 1, R_i = r] - E[D_i|Z = 0, R_i = r] \quad (11)$$

We are interested in whether the observed change in treatment responses coincides with the change in discrimination among individuals with identical treated potential outcomes. First, note that each expectation in Definition 3 is a weighted average of treatment rates for individuals with each potential outcome, where the weights are the prevalence of each binary value of the treated outcome. Equation 12 shows this, where $\pi_{zry} \equiv E[D_i|Z = z, R_i = r, Y_i(1) = y]$. The third and fourth lines follow from the fact that $Y_i(1) \perp Z$.

$$\begin{aligned} E[D_i|Z = z, R_i = r] &= E[Y_i(1) = 1|Z = z, R_i = r]E[D_i|Z = z, R_i = r, Y_i(1) = 1] \\ &\quad + E[Y_i(1) = 0|Z = z, R_i = r]E[D_i|Z = z, R_i = r, Y_i(1) = 0] \\ &= E[Y_i(1) = 1|R_i = r]E[D_i|Z = z, R_i = r, Y_i(1) = 1] \\ &\quad + E[Y_i(1) = 0|R_i = r]E[D_i|Z = z, R_i = r, Y_i(1) = 0] \\ \implies E[D_i|Z = z, R_i = r] &= Pr(Y_i(1) = 1|R_i = r)\pi_{zr1} + (1 - Pr(Y_i(1) = 1|R_i = r))\pi_{zr0} \end{aligned} \quad (12)$$

Plugging this into Definition 3, we can rewrite the observed change (Δ_{obs}) in terms of these group- and outcome-specific treatment averages, where $\mu_r \equiv Pr(Y_i(1) = 1|R_i = r)$:

$$\begin{aligned} \Delta_{\text{obs}} &= [(\mu_w\pi_{1w1} + (1 - \mu_w)\pi_{1w0}) - (\mu_w\pi_{0w1} + (1 - \mu_w)\pi_{0w0})] \\ &\quad - [(\mu_m\pi_{1m1} + (1 - \mu_m)\pi_{1m0}) - (\mu_m\pi_{0m1} + (1 - \mu_m)\pi_{0m0})] \end{aligned} \quad (13)$$

Similarly, Equation 14 rewrites the change in discrimination as a function of group- and outcome-specific treatment averages:

$$\begin{aligned}
\Delta_{\text{change}} &= Pr(Y_i(1) = 1) \underbrace{[\Delta_{11} - \Delta_{01}]}_{\text{Change, } Y_i(1) = 1} + (1 - Pr(Y_i(1) = 1)) \underbrace{[\Delta_{10} - \Delta_{00}]}_{\text{Change, } Y_i(1) = 0} \\
&= \underbrace{[Pr(Y_i(1) = 1)(\pi_{1w1} - \pi_{1m1}) + (1 - Pr(Y_i(1) = 1))(\pi_{1w0} - \pi_{1m0})]}_{\text{Gap when } Z=1} \\
&\quad - \underbrace{[Pr(Y_i(1) = 1)(\pi_{0w1} - \pi_{0m1}) + (1 - Pr(Y_i(1) = 1))(\pi_{0w0} - \pi_{0m0})]}_{\text{Gap when } Z=0}
\end{aligned} \tag{14}$$

Comparing Equation 13 and Equation 14, there are only 2 cases in which $\Delta_{\text{change}} = \Delta_{\text{obs}}$:

Case 1. $Y_i(1)$ is similar across groups: $Cov(Y_i(1), R_i) = 0 \implies Pr(Y_i(1) = 1 | R_i = r) = Pr(Y_i(1) = 1) \forall r \in R_i$

Case 2. Group-specific policy-induced responses are constant across $Y_i(1)$: $\pi_{1r1} - \pi_{0r1} = \pi_{1r0} - \pi_{0r0} = \theta_r$

The following shows how $\Delta_{\text{change}} = \Delta_{\text{obs}}$ under the described conditions. We will consider each case in turn.

Case 1

Consider that groups are similar in terms of $Y_i(1)$. Then, substituting $Pr(Y_i(1) = 1)$ for each $Pr(Y_i(1) = 1 | R_i = r)$ in Equation 13, we have:

$$\begin{aligned}
\Delta_{\text{obs}} &= [(Pr(Y_i(1) = 1)\pi_{1w1} + (1 - Pr(Y_i(1) = 1))\pi_{1w0}) - (Pr(Y_i(1) = 1)\pi_{0w1} + (1 - Pr(Y_i(1) = 1))\pi_{0w0})] \\
&\quad - [(Pr(Y_i(1) = 1)\pi_{1m1} + (1 - Pr(Y_i(1) = 1))\pi_{1m0}) - (Pr(Y_i(1) = 1)\pi_{0m1} + (1 - Pr(Y_i(1) = 1))\pi_{0m0})] \\
&= Pr(Y_i(1) = 1)[(\pi_{1w1} - \pi_{0w1}) - (\pi_{1m1} - \pi_{0m1})] + (1 - Pr(Y_i(1) = 1))[(\pi_{1w0} - \pi_{0w0}) - (\pi_{1m0} - \pi_{0m0})] \\
&= \Delta_{\text{change}}
\end{aligned}$$

Case 2

Here, groups differ in terms of $Y_i(1)$ but the policy-induced treatment response for each group is the same, regardless of $Y_i(1)$: $\pi_{1r1} - \pi_{0r1} = \pi_{1r0} - \pi_{0r0} = \theta_r$. Reorganizing Equation 13 to group the policy-induced treatment response terms by race and potential outcome level:

$$\begin{aligned}
\Delta_{\text{obs}} &= [Pr(Y_i(1) = 1 | R_i = w) (\pi_{1w1} - \pi_{0w1}) + (1 - Pr(Y_i(1) = 1 | R_i = w)) (\pi_{1w0} - \pi_{0w0})] \\
&\quad - [Pr(Y_i(1) = 1 | R_i = m) (\pi_{1m1} - \pi_{0m1}) + (1 - Pr(Y_i(1) = 1 | R_i = m)) (\pi_{1m0} - \pi_{0m0})] \\
&= \underbrace{\theta_w - \theta_m}_{\text{Gap in potential outcome-invariant responses}}
\end{aligned}$$

Substituting θ_r for $\pi_{1ry} - \pi_{0ry}$ in Equation 14, we see that $\Delta_{\text{change}} = \Delta_{\text{obs}}$.

$$\begin{aligned}\Delta &= Pr(Y_i(1) = 1)[\theta_w - \theta_m] + (1 - Pr(Y_i(1) = 1))[\theta_w - \theta_m] \\ &= \theta_w - \theta_m = \Delta_{\text{obs}}\end{aligned}$$

C.2 Point identifying discrimination

This section discusses how to implement our discrimination estimation approach to obtain point estimates of discrimination instead of bounds. We first sketch a simplified selection model commonly used to understand treatment non-compliance and treatment effect heterogeneity (Vytlacil, 2002). Individuals are treated if the benefit of treatment, $p_i(Y_i(1), Y_i(0), Z)$, outweighs the cost, where Z is a binary instrument that shifts the benefit of treatment. $p_i(Y_i(1), Y_i(0), Z)$ can also be interpreted as an individual's treatment propensity, and can be normalized such that $p_i(Y_i(1), Y_i(0), Z) \in [0, 1]$. Let $u_i \in U[0, 1]$ be a unidimensional measure summarizing possibly multiple factors that determine an individual's cost of treatment. Given the IV assumptions listed in Section 2, $Y_i(D_i)$ and u_i are unaffected by Z . Consolidating this notation, individual i is treated if $D_i = \mathbb{I}[p_i(Y_i(1), Y_i(0), Z) \geq u_i]$.

Returning to the discussion of always takers (A), compliers (C) and never takers (N) in Section 3, assume that always takers have lower cost of treatment than compliers, who in turn have lower cost of treatment than never takers, $u_A \leq u_C \leq u_N$ (Angrist, Imbens, and Rubin, 1996).⁵⁵ Assuming that the relationship between treatment propensity and treated/untreated potential outcomes is linear, Equation 15 describes the expression for each marginal treatment response function, where $\bar{Y}_{(\cdot)}$ represents the average treated/untreated outcome for always takers, compliers, or never takers (Brinch, Mogstad, and Wiswall, 2017; Mogstad, Santos, and Torgovitsky, 2018; Kowalski, 2023b). The remaining steps to estimate discrimination follows Section 3, except that we estimate average treated or untreated potential outcomes by integrating $MTO(p)$ or $MUO(p)$ over the full support of the treatment propensity. As a result, we obtain point estimates for the average treated/untreated outcomes and discrimination estimands.

$$\begin{aligned}MTO(p) &\equiv E[Y_i(1)|p = u_i] = \bar{Y}_{T,A} - \frac{p_A}{p_C} (\bar{Y}_{T,AC} - \bar{Y}_{T,A}) + \frac{2}{p_C} (\bar{Y}_{T,AC} - \bar{Y}_{T,A}) \times p \\ MUO(p) &\equiv E[Y_i(0)|p = u_i] = \frac{(2 - p_N)\bar{Y}_{U,NC} - (1 + p_A)\bar{Y}_{U,N}}{p_C} + \frac{2}{p_C} (\bar{Y}_{U,N} - \bar{Y}_{U,NC}) \times p \\ MTE(p) &\equiv E[Y_i(1) - Y_i(0)|p = u_i] = MTO(p) - MUO(p)\end{aligned}\tag{15}$$

We briefly demonstrate point identifying discrimination using the context of racial discrimination in incarceration decisions and publicly-available case-level records from Bexar County Criminal District (felony) Courts. We use a large reform meant to reduce overcrowding in Texas jails (SB

⁵⁵Vytlacil (2002) demonstrates how latent index selection models coincide with the local average treatment effect framework (Imbens and Angrist, 1994).

1067 in 1994). The goal of the reform was to reduce the burden on correctional facilities by limiting the incarceration rates for low-level offenders. The reform created a new category of felony: the state jail felony (SB 1067 Article 1, Subchapter C, §12.35) which reduced the punishment associated with a wide range of common offences, including many property and drug crimes. These provisions only applied to offences committed on or after September 1, 1994.⁵⁶

Figure C1 validates this natural experiment separately for white and minority defendants. Panels a) and b) show that the reform resulted in a 6 p.p. (7%) increase in non-incarceration among white defendants and a 14 p.p. (20%) increase among minority defendants. Panels c) and d) demonstrate that future involvement with the criminal legal system falls by 3.1 p.p. for white defendants (−32.6%) and increases by 3.8 p.p. for minority defendants (29%), although the former estimate is imprecise. Panels e) and f) show that a summary measure of the baseline characteristics of defendants is smooth around this date cut-off.

Given that the reform is a valid natural experiment, we apply it to estimate marginal treatment response functions as described above. We define treatment as $D_i = 1$ if an individual is not incarcerated (referred to as ‘released’ henceforth) and $D_i = 0$ if incarcerated. If an individual is released, we observe their treated re-offence outcome $Y_i(1)$. Here, we define $Y_i(1) = 1$ if an individual commits a new offence in Bexar County in the 12 months after they are released. Finally, while we assessed the validity of the natural experiment using regression discontinuity techniques, we parametrize Z as a binary instrument for simplicity: $Z = 1$ if an individual committed an offence after September 1, 1994.⁵⁷

Figure C2 plots the estimated race-specific marginal treated outcome functions, $MTO_r(p)$ and Table C1 integrates these functions to estimate the average outcome that would be realized if everyone were released. The point estimates suggest large differences in underlying potential outcomes. 9.7% of white defendants would re-offend if released, while 17.7% of minority defendants would, although these difference are not significant.

Table C1: Average re-offence estimates (p.p.)

	Average (1)	White (2)	Minority (3)
μ	0.155	0.097	0.177
95% CI	[0.121,0.188]	[0.011,0.180]	[0.143,0.211]

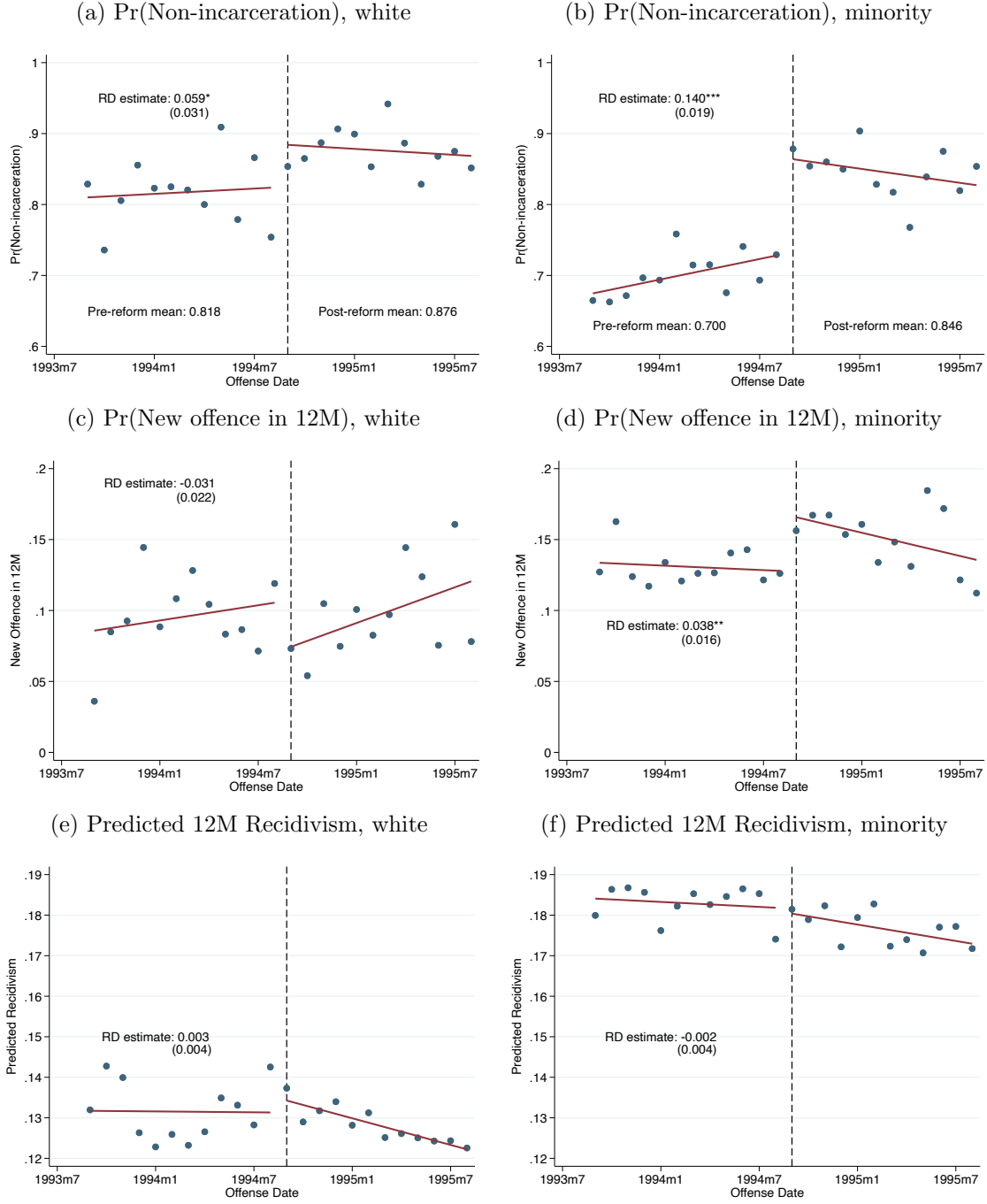
Note: This table presents estimates of the average treated outcome obtained using the approach described in Section 3, separately by race. The treatment is non-incarceration and the treated outcome is whether an individual re-offends within one year after disposition. Confidence intervals are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Following, Equation 3, Table C2 displays point estimates of racial discrimination in non-incarceration decisions that condition on re-offence outcomes if not incarcerated. Prior to the

⁵⁶Mueller-Smith and Schnepel (2021) use data from Harris County, TX to study a related aspect of the same legislation, which changed the incentives to offer deferred adjudication.

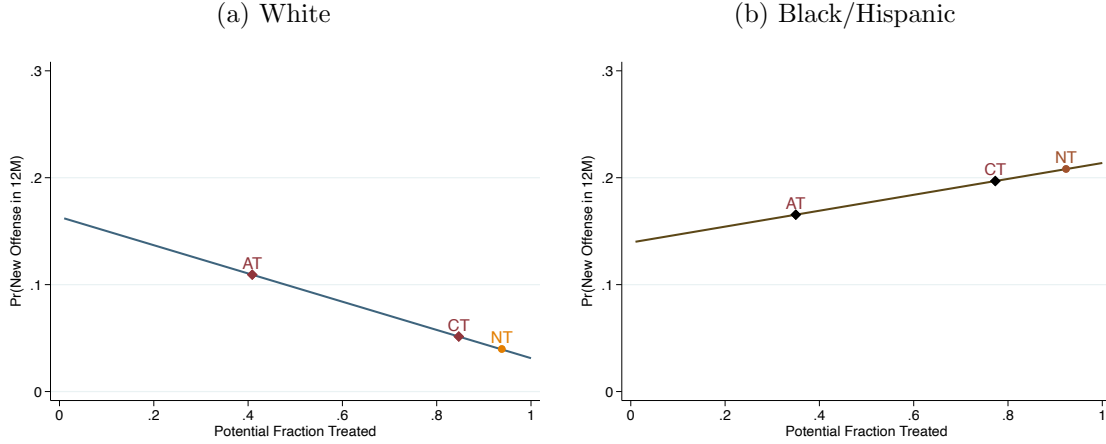
⁵⁷We alternatively could have followed the implementation in Section 4. Given the lack of trends in treatment and in re-offence outcomes here, the implementations are unlikely to meaningfully differ from each other. Hence, we proceed with this binary implementation for simplicity.

Figure C1: Validating the legislative reform



Note: Each Panel presents RD estimates from regressions of the form $Y_i = \alpha + \beta \mathbf{1}(T_i > t) + \delta_1 T_i + \delta_2 \mathbf{1}(T_i > t) \times T_i + \varepsilon_i$, where T_i denotes the running variable, and t denotes the cut-off date of September 1, 1994. Sample includes all felony defendants who committed an offence in a year around the cut-off date. The lines of best fit are estimated on the monthly averages, represented by the blue dots. Incarceration is defined as serving an incarceration sentence. Predicted recidivism is computed by estimating $\mathbf{1}(\text{New Offence})_i = \alpha + \beta X_i + \nu_i$ using pre-reform data, and excluding the RD sample. These coefficients are then used to predict the probability of re-offending for the RD sample. X includes: indicators for race, offence type, felony category, gender, age, criminal history, and neighbourhood characteristics.

Figure C2: Average re-offence extrapolation by race



Note: This figure displays the marginal treated outcome curves ($MTO(p)$) (Mogstad, Santos, and Torgovitsky, 2018) for non-incarceration. The treatment is non-incarceration, or ‘release’, and the outcome is whether an individual commits a new offence within 1 year, if not incarcerated. Each $MTO(p)$ is identified by assuming a linear relationship between potential outcomes of always takers (‘AT’), compliers (‘CT’), and never takers (‘NT’) and their treatment propensities. Lower values of the x-axis denote individuals who are more likely to be released.

reform, white individuals were 12.3pp more likely to be released than minority individuals. The reform significantly narrowed this disparity—after the reform, release rates were 3.4pp higher for White defendants, a 8.9pp reduction.

Table C2: Estimated disparities (p.p.)

	Pre ($Z = 0$) (1)	Post ($Z = 1$) (2)	Change (3)
Δ	0.123	0.034	-0.089
95% CI	[0.084,0.447]	[0.003,0.276]	[-0.170,-0.050]

Note: This figure presents the average disparities in each time period, conditional on treated potential outcomes, using the approach described in Section 3. The treatment is non-incarceration and the treated outcome, denoted by $Y_i(1)$, is whether an individual re-offends within one year after disposition. Confidence intervals are bootstrapped using 1,000 replications and a Bayesian bootstrap.

C.3 Identifying average potential outcomes with difference-in-difference designs

This section describes conditions under which we can non-parametrically bound average potential outcomes and average treatment effects in DiD settings with individual-level treatment non-compliance and heterogeneity in potential outcomes. While ‘treatment’ in a typical DiD might be at an aggregate level, e.g., county-level, we consider a setting where individuals in either county may be treated both before or after some policy. The estimation of causal effects is confounded by the effects of time, which are correlated with the policy adoption. We use changes in the unaffected control county to purge these effects of time. Our approach is similar to “time-corrected” Wald approach to estimate the local average treatment effect (LATE) from recent work (De Chaisemartin

and D’Haultfœuille, 2018). However, we require stronger assumptions to identify average potential outcomes and treatment effects for those who are not compliers.

We start with the following notation:

- $T \in \{0, 1\}$: Denotes periods before (pre) and after (post) a policy
- $G \in \{0, 1\}$: 1 if the county adopts the policy in $T = 1$, 0 if not.
- $Z \equiv T \times G \in \{0, 1\}$: This is the binary instrument
- $D_i(g, z) \in \{0, 1\}$: Whether an **individual** takes up treatment or not.
- $Y_{it}(d, g)$: Potential outcomes for person i , given the time period, treatment status, and the county they are in.
- Refer to always takers, compliers and never takers as “compliance groups”

We make the following assumptions, many of which are common in IV implementations.

Assumption 1. First stage: $Pr(D_i(g, Z = 1)) > Pr(D_i(g, Z = 0)) \forall g$

Assumption 2. Independence and exclusion: $(Y_{it}(1, g), Y_{it}(0, g), D_i(g, 1), D_i(g, 0)) \perp Z | g$.

This implies that within each county, the instrument is random and only affects outcomes via changes in treatment status (Imbens and Angrist, 1994). This allows for a) potential outcomes to differ across counties and b) time-varying factors to directly affect potential outcomes.

Assumption 3. No spillovers: The potential outcomes of individual i are unrelated to the treatment status of other individuals (Angrist, Imbens, and Rubin, 1996).

Assumption 4. IV monotonicity: $D_i(g, 1) \geq D_i(g, 0) \forall g$

This allows the instrument to (weakly) shift individuals in only one direction across treatment contrasts and **does not allow secular trends to change treatment status of individuals**. Together, this means that only the following shifts between treatment contrasts are permitted:

1. $D_i(g, z) = 1 \forall z$: These are always takers in group g
2. $D_i(g, z) = 0 \forall z$: These are never takers in group g
3. $D_i(g, 1) = 1$ and $D_i(g, 0) = 0$: These are compliers in group g , shifted by the instrument

Finally, we make an additional assumption that is in the spirit of parallel trends assumptions, but is not typically made in IV settings or DiD estimation:⁵⁸

⁵⁸This assumption is similar to the assumption underlying the “time-corrected” Wald estimand in De Chaisemartin and D’Haultfœuille (2018). There, the treated (untreated) potential outcomes for those treated (not treated) in the pre-period are the same across group. Their assumption is enough to identify the LATE, but does not allow us to identify the average potential outcomes of each compliance group separately. It pins down time trends in a) treated outcomes for always takers and b) an average of untreated outcomes for both never takers and compliers. This does not pin down time trends for compliers specifically without further assumptions.

Assumption 5. Parallel trends in potential outcomes: This assumes that i) the average change in treated outcomes is the same for always takers and compliers and is independent of county and ii) the average change in untreated outcomes is the same for never takers and compliers and is independent of county. This restricts the effects of time on potential outcomes to be constant across subsets of compliance groups, but not all of them, and does not force the effects of time to be identical across individuals.

$$\begin{aligned} E[Y_{i1}(1, g) - Y_0(1, g)|g, \text{Always taker}] &= E[Y_{i1}(1, g) - Y_0(1, g)|g, \text{Complier}] \text{ and } \perp g \\ E[Y_{i1}(0, g) - Y_0(0, g)|g, \text{Never taker}] &= E[Y_{i1}(0, g) - Y_0(0, g)|g, \text{Complier}] \text{ and } \perp g \end{aligned}$$

We now show how, under these assumptions, we can identify the proportions and average treated/untreated outcomes of always takers (A), never takers (N), and compliers (C) in the $G = 1$ county.

Given Assumption 4, there are no shifts in treatment status due to secular changes. Hence we have that the proportion of always takers (p_A), compliers (p_C) and never takers (p_N) in $G = 1$ are directly observed in the data for $G = 1$.

$$\begin{aligned} p_A &= E[D_i|G = 1, T = 0] \\ p_N &= 1 - (E[D_i|G = 1, T = 1]) \\ p_C &= 1 - (p_A + p_N) \end{aligned} \tag{16}$$

However, since common time trends can affect potential outcomes, the treated and untreated potential outcomes for each of these groups is not directly observed. To see this, recall that in settings where a binary instrument Z increases treatment take-up, the outcomes of individuals who are treated when $Z = 0$ identifies the treated outcomes for always takers. Equation 17 shows that if we try to estimate the treated outcomes for always takers in $G = 1$ using pre-period data (since $Z = 0 \ \& \ G = 1 \implies T = 0$), we only recover treated outcomes for always takers in the pre-period. The difference between this and the treated outcomes for always takers in the post-period is the trend in treated potential outcomes, θ_1 (second line of Equation 17).

$$\begin{aligned} E[Y_i|D_i = 1, G = 1, Z = 0] &= E[Y_i|D_i = 1, G = 1, T = 0] = E[Y_{i0}(1, 1)|A, G = 1] \\ \underbrace{E[Y_{i1}(1, 1)|A, G = 1]}_{\text{Unobserved}} &= \underbrace{E[Y_{i0}(1, 1)|A, G = 1]}_{\text{Observed}} + \underbrace{\theta_1}_{\text{Unobserved}} \end{aligned} \tag{17}$$

Equation 18 makes the same point for the untreated outcomes for never takers. In settings where a binary instrument Z increases treatment take-up, the outcomes of individuals who are not treated when $Z = 1$ identifies the untreated outcomes for never takers. Here, the outcomes of individuals who are not treated in the post-period only identifies the untreated outcomes for

never takers in the post-period. Similarly, the difference between this and the untreated outcomes for never takers in the pre-period is the trend in untreated potential outcomes, θ_0 (second line of Equation 18).

$$\begin{aligned}
E[Y_i|D_i = 0, G = 1, Z = 1] &= E[Y_i|D_i = 0, G = 1, T = 1] = E[Y_{i1}(0, 1)|N, G = 1] \\
\underbrace{E[Y_{i0}(0, 1)|N, G = 1]}_{\text{Unobserved}} &= \underbrace{E[Y_{i1}(0, 1)|N, G = 1]}_{\text{Observed}} - \underbrace{\theta_0}_{\text{Unobserved}}
\end{aligned} \tag{18}$$

Assumption 5 lets us identify the time trends in treated and untreated outcomes in $G = 1$ (θ_1 & θ_0) using the change over time in $G = 0$. Starting with treated outcomes, note that the only individuals who would be treated in the control county are always takers. The average change in treated outcomes in $G = 0$ identifies the time trend for always takers in $G = 1$, since Assumption 5 states that the trend in treated potential outcomes for always takers is identical across counties and is equal to the trend for compliers (see Equation 19).

$$\begin{aligned}
\theta_1 &= E[Y_{i1}(1, 1) - Y_{i0}(1, 1)|G = 1, A] = E[Y_{i1}(1, 1) - Y_{i0}(1, 1)|G = 1, C] \\
&= E[Y_{i1}(1, 0) - Y_{i0}(1, 0)|G = 0]
\end{aligned} \tag{19}$$

Similarly, the individuals who are untreated in the control county, $G = 0$, consist of compliers and never takers. From Assumption 5, the change in untreated outcomes for never takers and compliers are identical to each other, which means the average change in untreated outcomes in $G = 0$ equals the average change in untreated outcomes for never takers in $G = 0$. Additionally, Assumption 5 states that the time trend in untreated outcomes for never takers is identical across counties, which allows us to identify the time trend in untreated outcomes for never takers in $G = 1$ (see Equation 20).

$$\begin{aligned}
\theta_0 &= E[Y_{i1}(0, 1) - Y_{i0}(0, 1)|G = 1, N] = E[Y_{i1}(0, 1) - Y_{i0}(0, 1)|G = 1, C] \\
&= E[Y_{i1}(0, 0) - Y_{i0}(0, 0)|G = 0]
\end{aligned} \tag{20}$$

Equation 21 restates this by combining this with Equations 17 and 18, to show how the time trend in the treated outcomes of always takers and untreated potential outcomes of never takers in $G = 1$ can be identified using the aggregate changes in treated and untreated potential outcomes in the control county, $G = 0$.

$$\begin{aligned}
E[Y_{i1}(1, 1)|A, G = 1] &= E[Y_{i0}(1, 1)|A, G = 1] + \theta_1 \\
&= E[Y_{i0}(1, 1)|A, G = 1] + E[Y_{i1}(1, 0) - Y_{i0}(1, 0)|G = 0]
\end{aligned} \tag{21}$$

$$\begin{aligned}
E[Y_{i0}(0, 1)|N, G = 1] &= E[Y_{i1}(0, 1)|N, G = 1] - \theta_0 \\
&= E[Y_{i1}(0, 1)|N, G = 1] - E[Y_{i1}(0, 0) - Y_{i0}(0, 0)|G = 0]
\end{aligned}$$

We now have the treated outcomes for always takers and the untreated outcomes for never takers from $G = 1$ in both periods. We can use this information with other observed moments in the data to estimate the treated and untreated outcomes for compliers (Imbens and Rubin, 1997).

Starting with treated outcomes, the first line of Equation 22 notes that the observed outcomes among treated individuals in period $T = 1$ is a weighted average of treated outcomes for always takers and compliers in $T = 1$. Rearranging this expression, the second line shows that the treated outcomes for compliers in $T = 1$, $E[Y_{i1}(1, 1)|C, G = 1]$, is a function of moments that we can estimate. $E[Y_{i1}(1, 1)|A, G = 1]$ is obtained from Equation 21, $E[Y_i|D_i = 1, G = 1, T = 1]$ is a sample average, and each of the proportions, $p(\cdot)$, is obtained using Equation 16.

$$\begin{aligned}
E[Y_i|D_i = 1, G = 1, T = 1] &= \frac{p_A E[Y_{i1}(1, 1)|A, G = 1] + p_C E[Y_{i1}(1, 1)|C, G = 1]}{p_A + p_C} \\
E[Y_{i1}(1, 1)|C, G = 1] &= \frac{(p_A + p_C) E[Y_i|D_i = 1, G = 1, T = 1] - p_A E[Y_{i1}(1, 1)|A, G = 1]}{p_C}
\end{aligned} \tag{22}$$

We can estimate untreated outcomes for compliers in a similar way. The first line of Equation 23 shows that the observed outcomes among untreated individuals in period $T = 0$ is a weighted average of untreated outcomes for never takers and compliers in $T = 0$. Rearranging this expression, the second line shows that the untreated outcomes for compliers in $T = 0$, $E[Y_{i0}(0, 1)|C, G = 1]$, is a function of moments that we can estimate. $E[Y_{i0}(1, 1)|N, G = 1]$ is obtained from Equation 21, $E[Y_i|D_i = 0, G = 1, T = 0]$ is a sample average, and each of the proportions, $p(\cdot)$, is obtained using Equation 16.

$$\begin{aligned}
E[Y_i|D_i = 0, G = 1, T = 0] &= \frac{p_N E[Y_{i0}(0, 1)|N, G = 1] + p_C E[Y_{i0}(0, 1)|C, G = 1]}{p_N + p_C} \\
E[Y_{i0}(0, 1)|C, G = 1] &= \frac{(p_N + p_C) E[Y_i|D_i = 0, G = 1, T = 0] - p_N E[Y_{i0}(0, 1)|N, G = 1]}{p_C}
\end{aligned} \tag{23}$$

As a result, we have identified the following objects:

- Treated outcomes in each period for always takers

- Untreated outcomes in each period for never takers
- Treated outcomes in $T = 1$ for compliers
- Untreated outcomes in $T = 0$ for compliers

We are missing 2 objects: Treated outcomes in $T = 0$ for compliers and untreated outcomes in $T = 1$ for compliers. Assumption 5 allows us to recover this, since it implies the time trend in treated/untreated potential outcomes of each compliance group in $G = 1$ can be identified using the aggregate changes in treated/untreated potential outcomes in the control county, $G = 0$. We now have average treated and untreated outcomes for each compliance group and in each period.

$$\begin{aligned} E[Y_{i0}(1, 1)|C, G = 1] &= E[Y_{i1}(1, 1)|C, G = 1] - \theta_1 \\ E[Y_{i1}(0, 1)|C, G = 1] &= E[Y_{i0}(0, 1)|C, G = 1] + \theta_0 \end{aligned} \tag{24}$$

C.4 Bounds when conditioning on $Y_i(0)$ or $Y_i(1) - Y_i(0)$

Conditioning on $Y_i(0)$

Equation 25 shows how the share of individuals **not treated**, with a given untreated outcome (e.g., the dismissal rate for people with a specific outcome if dismissed) can be used to quantify the share of **treated** individuals with a specific untreated outcome, abstracting away from the period-specific notation Z . The first line subtracts the share of individuals not treated with a given untreated outcome from 1, where $\bar{D} = E[D_i|R_i = r]$, and where $E[D_i = 0|R_i = r, Y_i(0) = y]$ is rewritten in the same way as described in Equation 2. The third line uses the fact that the average untreated outcome observed if everyone were untreated is a weighted average of the untreated outcomes for the treated individuals and the untreated individuals: $E[Y_i(0) = y|R_i = r] = E[Y_i(0) = y|R_i = r, D_i = 1]\bar{D} + E[Y_i(0) = y|R_i = r, D_i = 0](1 - \bar{D})$. This recovers the share of treated individuals with a specific untreated outcome, described in Equation 7.

$$\begin{aligned} 1 - E[D_i = 0|R_i = r, Y_i(0) = y] &= 1 - \frac{E[Y_i(0) = y|R_i = r, D_i = 0] \times (1 - \bar{D})}{E[Y_i(0) = y|R_i = r]} \\ &= \frac{E[Y_i(0) = y|R_i = r] - E[Y_i(0) = y|R_i = r, D_i = 0] \times (1 - \bar{D})}{E[Y_i(0) = y|R_i = r]} \\ &= \frac{E[Y_i(0) = y|R_i = r, D_i = 1] \times \bar{D}}{E[Y_i(0) = y|R_i = r]} \\ &= E[D_i|R_i = r, Y_i(0) = y] \end{aligned} \tag{25}$$

Conditioning on $Y_i(1) - Y_i(0)$

The equation below reproduces Equation 8, where $\tau_i \equiv Y_i(1) - Y_i(0)$. The right hand side is a function of two partially identified objects (first term in numerator and denominator) and a point estimate (second term in numerator). The two partially identified objects are dependent—the denominator is a function of the first term in the numerator.

$$E[D_i|Z = z, R_i = r, \tau_i = y] = \frac{E[\tau_i = y|Z = z, R_i = r, D_i = 1] \times E[D_i|Z = z, R_i = r]}{E[\tau_i = y|R_i = r]}$$

We rewrite the treatment rate conditional on the treatment effect, omitting group conditioning for brevity and considering the case where the policy reform has taken effect, i.e. $Z = 1$. Here, we assume that $\tau_i \in \{0, 1\}$, because this assumption is required to identify this type of treatment rate, as described in Section 3.

$$\begin{aligned} E[D_i|Z = 1, \tau_i = 1] &= \frac{E[\tau_i|Z = z, D_i = 1] \times E[D_i|Z = z]}{E[\tau_i]} \\ &= \frac{(p_A E[\tau_i|A] + p_C E[\tau_i|C]) / (p_A + p_C)}{p_A E[\tau_i|A] + p_C E[\tau_i|C] + p_N E[\tau_i|N]} \times E[D_i|Z = 1] \end{aligned} \quad (26)$$

The average treatment effect on the treated is a function of the bounds on the treatment effect for always takers and the point estimated treatment effect for compliers. The denominator ($E[\tau_i]$) is a function of two partially identified objects, $E[\tau_i|A]$ and $E[\tau_i|N]$, since we never observe untreated outcomes for always takers or treated outcomes for never takers.

This implies that the treatment rate conditional on the treatment effect is of the form $y = f(p, q) = \frac{p \times r}{p + q}$, where r is a point estimate and the other quantities are bounds. We compute bounds for y by holding one partially identified object fixed at a time, evaluating the function at the extremes of the other partially identified object. That is, we take the minimum and maximum values of y over the following cases: $f(\underline{p}, \underline{q})$, $f(\underline{p}, \bar{q})$, $f(\bar{p}, \underline{q})$, $f(\bar{p}, \bar{q})$.

C.5 Inference for tests of overlapping bounds

Here we discuss how we test whether the average potential outcome bounds estimated for each group overlap. Let the true parameter of interest for each group be μ_r , where $r \in \{m, w\}$ denotes the group. Let the estimated bounds be $[\mu_{r,L}, \mu_{r,U}]$. The goal is to test whether $[\mu_{m,L}, \mu_{m,U}]$ and $[\mu_{w,L}, \mu_{w,U}]$ overlap.

We construct a set that denotes the difference between the upper bound for one group and the lower bound for another: $\mathbf{M}_d \equiv [\mu_{m,L} - \mu_{w,U}, \mu_{m,U} - \mu_{w,L}] = [\tilde{\mu}_L, \tilde{\mu}_U]$. Note that $0 \in \mathbf{M}_d$ only if the bounds for each group are overlapping. To see this, consider the following three cases:⁵⁹

⁵⁹There are 3 more cases if you switch m and w , but they yield the same conclusions. These conditions also hold if the intervals themselves contain 0.

Case 1. $[\mu_{m,L}, \mu_{m,U}]$ and $[\mu_{w,L}, \mu_{w,U}]$ are disjoint. E.g., $[\mu_{m,L}, \mu_{m,U}] = [0.5, 0.6]$ and $[\mu_{w,L}, \mu_{w,U}] = [0.2, 0.3]$. Then, $\mathbf{M}_d = [0.2, 0.4]$

Case 2. $[\mu_{m,L}, \mu_{m,U}]$ and $[\mu_{w,L}, \mu_{w,U}]$ overlap but one is not a subset of the other. E.g., $[\mu_{m,L}, \mu_{m,U}] = [0.25, 0.6]$ and $[\mu_{w,L}, \mu_{w,U}] = [0.2, 0.3]$. Then, $\mathbf{M}_d = [-0.05, 0.4]$

Case 3. $[\mu_{m,L}, \mu_{m,U}]$ is contained within $[\mu_{w,L}, \mu_{w,U}]$. E.g., $[\mu_{m,L}, \mu_{m,U}] = [0.2, 0.8]$ and $[\mu_{w,L}, \mu_{w,U}] = [0.3, 0.4]$. Then, $\mathbf{M}_d = [-0.2, 0.5]$

Our goal is to test the following null hypothesis: $H_0 : 0 \in \mathbf{M}_d$. We bootstrap the estimation of the bounds using a Bayesian bootstrap (Rubin, 1981). For each bootstrap replication, we construct the interval $\mathbf{M}_d \equiv [\mu_{m,L} - \mu_{w,U}, \mu_{m,U} - \mu_{w,L}] = [\tilde{\mu}_L, \tilde{\mu}_U]$. We then calculate a p -value as the share of the bootstrap replications in which $0 \in \mathbf{M}_d$, i.e., in which the bounds overlap.

C.6 Extrapolation discrimination using averages identified at RD cut-off

Here we discuss the assumptions needed to apply the average potential outcome estimates derived from information at the cut-off to adjust treatment rates away from the cut-off. Let the analysis sample include the following values of the running variable (e.g., a test score): $s \in [\underline{s}, \bar{s}]$ and let s^* be the cut-off. Let D_i denote the treatment decision (e.g., whether student i is promoted) and Z denote whether a student is above or below the test-score cut-off. Consider that we want to condition on the treated potential outcome, $Y_i(1)$. Finally, any assumptions discussed below will need to hold by subgroup, although we omit subgroup notation for brevity.

Assume that the following assumptions hold:

RD1: $D_i \perp s$ and $E[D_i|Z = 1, s^*] - E[D_i|Z = 0, s^*] = E[D_i|Z = 1, s] - E[D_i|Z = 0, s] \forall s \in [\underline{s}, \bar{s}]$

RD2: $E[Y_i(1)|s = s^*] = E[Y_i(1)|s < s^*] = E[Y_i(1)|s > s^*]$

To understand the plausibility of these assumptions, consider them in the context of the student grade promotion application in Section 4. RD1 states that within the analysis window, the test score does not influence promotion decisions by itself—only the cut-off does. RD1 also implies that the size of the first-stage would be the same in counterfactuals where the test score cut-off was elsewhere in the window. This ensures that the proportions of always takers, compliers, and never takers identified at the cut-off are applicable to the wider window. RD2 assumes that the average promoted outcomes that would be realized if all students **at the cut-off** were promoted is equal to the average promoted outcomes if students elsewhere in the window were all promoted. In the simplest case, this would be satisfied if $Y_i(1)$ did not vary across s .

While RD1 may hold in some settings, RD2 is a strong assumption that may not be satisfied in many contexts, including student grade promotion. To see this, note that we can split $E[Y_i(1)|s = s^*]$ into average treated outcomes for always takers, compliers, and never takers at the cut-off s^* . Let $p_{(\cdot)}$ denote the proportion of always takers (A), compliers (C), or never takers (N). Consider

the first part of the equality in RD2, which compares students at the cut-off to those below. RD2 implies that the equality in [Equation 27](#) must hold.

$$\begin{aligned}
E[Y_i(1)|s = s^*] &= E[Y_i(1)|s < s^*] \\
\implies p_A E[Y_i(1)|A, s = s^*] + p_C E[Y_i(1)|C, s = s^*] + p_N E[Y_i(1)|N, s = s^*] &= \\
p_A E[Y_i(1)|A, s < s^*] + p_C E[Y_i(1)|C, s < s^*] + p_N E[Y_i(1)|N, s < s^*] & \\
\implies p_C (E[Y_i(1)|C, s = s^*] - E[Y_i(1)|C, s < s^*]) + p_N (E[Y_i(1)|N, s = s^*] - E[Y_i(1)|N, s < s^*]) &= \\
= p_A (E[Y_i(1)|A, s < s^*] - E[Y_i(1)|A, s = s^*]) & \\
\end{aligned} \tag{27}$$

We can map the implications of [Equation 27](#) to how promoted outcomes vary with the running variable in the data. In [Figure 3](#), all individuals below the cut-off are always takers for promotion (they are being promoted despite being below the cut-off). Here, always takers below the cut-off clearly have lower treated outcomes than those at the cut-off, implying that always takers **below the cut-off** are less likely to be prepared for 4th grade than always takers **at the cut-off**. Mathematically, this means that the right hand side of final expression of [Equation 27](#) is negative. For this assumption to hold with equality, the left hand side needs to be sufficiently negative to offset this. However, that would imply that compliers and never takers **at the cut-off** are **less prepared** for 4th grade than compliers and never takers **below the cut-off**, which is at odds with reasonable models of selection into promotion. Comparing the average treated outcomes at the cut-off to that above the cut-off yields a similar conclusion.